Topics in Mathematics Science VII

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Introduction to Koszul Duality

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Convention

Throughout the course, $\Bbbk$ is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as $\Bbbk$-vector spaces.

There are a lot of contents not covered in lecture, which will be written in coloured blocks:

> This is supplementary material: proofs and routine verifications of statements made in the main text, often 'unnecessarily long or pedantic check of definitions' – things that are only there for rigour.

Supplementary material are often (but not always) QA'ed.

> This is advanced material; things that I personally would like to add, from personal experience in the theory, but definitely not necessary for lecture purpose.

Advanced material is mostly produced by AI and most of them has not been rigorously checked, so the reader should read it with extra care.

The main content of the notes are drafted by AI and QA'ed (heavily edited) by myself.

1 Graded algebras, modules, and duals

1.1 Algebras and modules

Definition 1.1. *A $\Bbbk$-algebra is a $\Bbbk$-vector space with an associative unital multiplication that is bilinear, meaning linear in each variable.*

A right A-module is understood to be a $\Bbbk$-vector space M with a bilinear action satisfying $(ma)b = m(ab)$ and $m1_A = m$ for $m \in M$ and $a, b \in A$. A module homomorphism $f : M \to N$ is linear and satisfies $f(ma) = f(m)a$. A homomorphism of $\Bbbk$-algebras is a linear map preserving multiplication and the identity.

Example 1.2. *(1) The matrix ring $\mathrm{Mat}_n(\Bbbk)$ is a $\Bbbk$-algebra. The space of $1 \times n$ row vectors is a right $\mathrm{Mat}_n(\Bbbk)$-module by matrix multiplication.*

(2) Let $\Bbbk A_n$ be the upper triangular matrix ring, which is also a $\Bbbk$-algebra.

$$\Bbbk A_n = \begin{pmatrix} \Bbbk & \Bbbk & \cdots & \Bbbk \\ 0 & \Bbbk & \cdots & \Bbbk \\ 0 & 0 & \ddots & \vdots \\ 0 & \cdots & 0 & \Bbbk \end{pmatrix}$$

For those familiar with path algebras, $\Bbbk A_n$ is the path algebra of a linearly oriented A_n quiver, which explains the notation. We use matrix calculations, so no knowledge of quivers is required. Write e_{ij} for the matrix with 1 in entry (i, j) and zero elsewhere. Our convention composes paths from left to right: e_{ij} represents $i \to j$, and $e_{ij}e_{j\ell} = e_{i\ell}$.

1.2 Categories and functors

We organise modules together with their homomorphisms. A category records both: specifying the objects alone does not specify a category.

Definition 1.3. *A category $\mathcal{C}$ consists of objects, a set $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ of morphisms $f : X \to Y$ for each pair of objects, and composition sending $f : X \to Y$ and $g : Y \to Z$ to $g \circ f : X \to Z$. Each object X has an identity $\mathrm{id}_X : X \to X$, and*

$$h \circ (g \circ f) = (h \circ g) \circ f, \qquad \mathrm{id}_Y \circ f = f = f \circ \mathrm{id}_X \quad \text{for all composable } f, g, h.$$

A morphism $f : X \to Y$ is an isomorphism if some $g : Y \to X$ satisfies $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$; we then write $X \cong Y$. A full subcategory of $\mathcal{C}$ consists of chosen objects of $\mathcal{C}$, with all morphisms between them and the same composition.

Example 1.4. *For a $\Bbbk$-algebra A, we use the notation*

$$\begin{aligned} \mathrm{Mod}\, A &:= \textit{category of right } A\textit{-modules and } A\textit{-linear maps}, \\ \mathrm{mod}\, A &:= \textit{full subcategory of finitely generated right } A\textit{-modules}. \end{aligned}$$

In either category, $\mathrm{Hom}_A(M, N)$ is the space of A-linear maps $M \to N$; composition is the usual composition of maps. For $A = \Bbbk$, these are the categories of all and of finite-dimensional vector spaces, respectively.

Note that, if $\dim_{\Bbbk} A < \infty$, then $M \in \mathrm{mod}\, A \Leftrightarrow M$ is finite-dimensional ($\because$ finite generation implies that M is a quotient of $A^{\oplus n}$ some n, and a finite $\Bbbk$-basis of M is clearly a finite generating set.)

Example 1.5. *For $M \in \operatorname{mod} A$, let $\operatorname{add} M := \operatorname{add}_{\operatorname{mod} A}(M)$ be the full subcategory with*

$$N \in \operatorname{add} M \iff N \oplus N' \cong M^{\oplus r} \quad \text{for some } N' \in \operatorname{mod} A \text{ and integer } r \geq 0.$$

Thus $\operatorname{add} M$ consists of direct summands of finite direct sums of M.

In modern representation theory, remembering A and $\operatorname{add} A$ holds the same categorical information is crucial in utilising cateogircal technique in representation theory; this allows one to push many results to a more abstract framework.

Definition 1.6. *A functor $F : \mathcal{C} \to \mathcal{D}$ assigns an object $F(X)$ to each object X and a morphism $F(f) : F(X) \to F(Y)$ to each $f : X \to Y$, such that*

$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)}, \qquad F(g \circ f) = F(g) \circ F(f) \quad \text{for all composable } f, g.$$

For example, forgetting the A-action gives a functor $\operatorname{Mod} A \to \operatorname{Mod} \Bbbk$: it keeps the underlying vector spaces and linear maps. The *identity functor* $\mathrm{id}_{\mathcal{C}}$ leaves every object and morphism unchanged; functors compose by applying one after the other. Every functor preserves isomorphisms, since $F(f^{-1})$ is inverse to $F(f)$.

The *opposite category* $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$, with $\operatorname{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, Y) = \operatorname{Hom}_{\mathcal{C}}(Y, X)$ and composition reversed. A *contravariant functor* from $\mathcal{C}$ to $\mathcal{D}$ means a functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Thus it sends $f : X \to Y$ in $\mathcal{C}$ to $F(f) : F(Y) \to F(X)$ and satisfies $F(g \circ f) = F(f) \circ F(g)$. Duals will provide our main examples.

Definition 1.7. *For functors $F, G : \mathcal{C} \to \mathcal{D}$, a natural transformation $\eta : F \to G$ is a family of morphisms $\eta_X : F(X) \to G(X)$ such that*

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array} \qquad G(f) \circ \eta_X = \eta_Y \circ F(f) \quad \text{for every } f : X \to Y. \tag{1.1}$$

If every η_X is an isomorphism, η is a natural isomorphism, written $F \cong G$.

The square in (1.1) is said to *commute*: its two routes from $F(X)$ to $G(Y)$ give the same morphism. We often say a family of morphisms (resp. isomorphisms) in a category $\mathcal{C}$ is *natural in $M \in \mathcal{C}$* to mean that it defines a natural transformation (resp. isomorphism).

For example, for fixed $c \in \Bbbk$, the maps $c\,\mathrm{id}_M : M \to M$ define a natural transformation $\mathrm{id}_{\operatorname{Mod} A} \to \mathrm{id}_{\operatorname{Mod} A}$ because $f(cm) = cf(m)$ for every module homomorphism $f : M \to N$. It is a natural isomorphism when $c \neq 0$.

A functor $F : \mathcal{C} \to \mathcal{D}$ is an *equivalence* if there is a functor $G : \mathcal{D} \to \mathcal{C}$ with natural isomorphisms $GF \cong \mathrm{id}_{\mathcal{C}}$ and $FG \cong \mathrm{id}_{\mathcal{D}}$. A *duality* between $\mathcal{C}$ and $\mathcal{D}$ is an equivalence $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Thus an equivalence recovers objects and maps up to natural isomorphism; a duality also reverses the direction of maps.

Definition 1.8. *Let A be a $\Bbbk$-algebra, and let $L, M, N \in \operatorname{Mod} A$, with $i : L \to M$ and $p : M \to N$ module homomorphisms. The sequence $L \xrightarrow{i} M \xrightarrow{p} N$ is exact at M if $\operatorname{Im} i = \operatorname{Ker} p$. A sequence of module homomorphisms is exact if it is exact at every term with an incoming and an outgoing map. A short exact sequence is an exact sequence*

$$0 \longrightarrow L \xrightarrow{i} M \xrightarrow{p} N \longrightarrow 0, \qquad \operatorname{Ker} i = 0, \quad \operatorname{Im} i = \operatorname{Ker} p, \quad \operatorname{Im} p = N. \tag{1.2}$$

Definition 1.9. *Let A, B be $\Bbbk$-algebras. A functor $F : \operatorname{Mod} A \to \operatorname{Mod} B$ is exact if $F(0) = 0$ and every short exact sequence (1.2) gives a short exact sequence*

$$0 \longrightarrow F(L) \xrightarrow{F(i)} F(M) \xrightarrow{F(p)} F(N) \longrightarrow 0.$$

An exact functor preserves injective and surjective maps: apply it to $0 \to L \xrightarrow{i} M \to M/i(L) \to 0$ and $0 \to \operatorname{Ker} p \to M \xrightarrow{p} N \to 0$, respectively. For a functor $F : (\operatorname{Mod} A)^{\mathrm{op}} \to \operatorname{Mod} B$, exactness instead means that (1.2) gives

$$0 \longrightarrow F(N) \xrightarrow{F(p)} F(M) \xrightarrow{F(i)} F(L) \longrightarrow 0.$$

Thus an exact contravariant functor exchanges injective and surjective maps.

Example 1.10 (𝕜-linear duality). *For a vector space V, define its 𝕜-linear dual by $D(V) := \operatorname{Hom}_{\Bbbk}(V, \Bbbk)$. For a linear map $u : V \to W$, set*

$$D(u) : D(W) \longrightarrow D(V), \qquad \lambda \longmapsto \lambda \circ u.$$

The identities $D(\mathrm{id}_V) = \mathrm{id}_{D(V)}$ and $D(v \circ u) = D(u) \circ D(v)$ make $D : (\operatorname{Mod} \Bbbk)^{\mathrm{op}} \to \operatorname{Mod} \Bbbk$ a functor. The functor D is exact.

> *For a short exact sequence (1.2) of vector spaces, $D(p)$ is injective because p is surjective, and $D(i)$ is surjective because a functional on a subspace extends to the whole vector space. If $\lambda \in D(M)$ satisfies $\lambda \circ i = 0$, then*
>
> $$\mu : N \longrightarrow \Bbbk, \qquad p(m) \longmapsto \lambda(m)$$
>
> *is well-defined since $\operatorname{Ker} p = i(L)$, and $\lambda = \mu \circ p = D(p)(\mu)$. Conversely, $p \circ i = 0$ gives $D(i) \circ D(p) = 0$, so $\operatorname{Ker} D(i) = \operatorname{Im} D(p)$. Thus D exchanges injective and surjective linear maps.*

For every vector space V, evaluation defines a natural injective map

$$\mathrm{ev}_V : V \longrightarrow D(D(V)), \qquad v \longmapsto (\lambda \mapsto \lambda(v)). \tag{1.3}$$

If V is finite-dimensional, choose a basis $v_1, \dots, v_n$ and its dual basis $v_1^*, \dots, v_n^*$, defined by $v_i^*(v_j) = \delta_{ij}$. Then

$$\mathrm{ev}_V^{-1} : D(D(V)) \longrightarrow V, \qquad \Phi \longmapsto \sum_{i=1}^{n} \Phi(v_i^*) v_i.$$

Consequently, $D : (\operatorname{mod} \Bbbk)^{\mathrm{op}} \to \operatorname{mod} \Bbbk$ is a duality.

> *If $v \neq 0$, choose $\lambda \in D(V)$ with $\lambda(v) \neq 0$; hence ev_V is injective. For $u : V \to W$, $v \in V$, and $\lambda \in D(W)$,*
>
> $$D(D(u))(\mathrm{ev}_V(v))(\lambda) = \lambda(u(v)) = \mathrm{ev}_W(u(v))(\lambda),$$
>
> *which proves naturality.*
>
> *In finite dimension, writing $\lambda = \sum_{i=1}^n \lambda(v_i) v_i^*$ gives $\Phi(\lambda) = \lambda(\sum_{i=1}^n \Phi(v_i^*) v_i)$. Thus the displayed formula is the inverse of (1.3), and is independent of the chosen basis.*

1.3 Semisimple algebras

Definition 1.11. *A module L is simple if $L \neq 0$ and its only submodules are 0 and L. A module is semisimple if it is a direct sum of simple modules. An algebra B is semisimple if it is semisimple as a right module B_B over itself. The Jacobson radical $\operatorname{rad} B$ is the intersection of all maximal right ideals of B. For a right B-module M, the radical $\operatorname{rad} M$ is the intersection of all maximal submodules of M, with $\operatorname{rad} M = M$ if there are none.*

We use the following standard ring-theoretic facts without proof. For a finite-dimensional $\Bbbk$-algebra B and any right B-module M,

$$\operatorname{rad} M = M \operatorname{rad} B.$$

Recall that a *division $\Bbbk$-algebra* is a non-zero $\Bbbk$-algebra whose non-zero elements are invertible.

Theorem 1.12 (Artin–Wedderburn). *For a non-zero finite-dimensional $\Bbbk$-algebra B,*

$$B \text{ is semisimple} \iff \operatorname{rad} B = 0 \iff B \cong \prod_{i=1}^{n} \operatorname{Mat}_{n_i}(D_i),$$

where $n, n_i \geq 1$ and each D_i is a finite-dimensional division $\Bbbk$-algebra. If $\Bbbk$ is algebraically closed, then $D_i = \Bbbk$ for every i. For semisimple B, every left or right B-module is semisimple, and every submodule has a complement. In particular, the definition is independent of the chosen side.

> For proofs of $\operatorname{rad} M = M \operatorname{rad} B$ and of Theorem 1.12, see for example [DK94]; [Bez23] gives a short account of semisimple modules and Wedderburn's theorem.

Corollary 1.13. *Every finite-dimensional left or right module over a finite-diemnsional semisimple algebra S is a direct summand of $S^{\oplus n}$ for some n, i.e. $M \in \operatorname{add}(S)$ for all $M \in \operatorname{mod} S$.*

Proof Choose a surjection $p : S^{\oplus n} \twoheadrightarrow M$. By Theorem 1.12, there is a submodule L such that $S^{\oplus n} = \operatorname{Ker} p \oplus L$. The restriction $p|_L : L \to M$ has kernel $L \cap \operatorname{Ker} p = 0$ and image $p(L) = p(S^{\oplus n}) = M$. Hence $L \cong M$, so $S^{\oplus n} \cong \operatorname{Ker} p \oplus M$. □

Example 1.14. *The algebras $\Bbbk^{\oplus n}$ and $\operatorname{Mat}_n(\Bbbk)$ are semisimple by Theorem 1.12. Upper triangular matrices need not form a semisimple algebra: for $A = \Bbbk A_2$,*

$$\operatorname{rad} A = \Bbbk e_{12} \neq 0, \qquad A/\operatorname{rad} A \cong \Bbbk^{\oplus 2}.$$

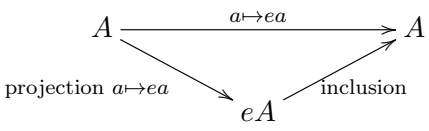
> *Put $J = \Bbbk e_{12}$, so $J^2 = 0$. For a simple right module L, simplicity gives $LJ = 0$ or $LJ = L$; the second alternative implies $L = LJ^2 = 0$, a contradiction. Hence J is contained in every maximal right ideal. The quotient $A/J \cong \Bbbk^{\oplus 2}$ has exactly two maximal right ideals, whose inverse images have intersection*
>
> $$(\Bbbk e_{11} \oplus \Bbbk e_{12}) \cap (\Bbbk e_{12} \oplus \Bbbk e_{22}) = J.$$

Definition 1.15. *An element $e \in A$ is idempotent if $e^2 = e$. Idempotents e, f are orthogonal if $ef = fe = 0$. A non-zero idempotent is primitive if it is not the sum of two non-zero orthogonal idempotents. A finite family $e_1, \ldots, e_r$ of pairwise orthogonal idempotents is complete if $\sum_{i=1}^{r} e_i = 1_A$.*

Remark 1.16. For orthogonal idempotents $e, f \in A$,

$$(e + f)A = eA \oplus fA.$$

In particular, we always have $A \cong eA \oplus (1 - e)A$. This strengthen the viewpoint of idempotent-as-local-identity: left-multiplying e defines a canonical projection onto the direct summand eA of A, and yields a commutative diagram



So multiplying e means we work inside the direct summand eA. The *corner subalgebra* $eAe = \{eae \mid a \in A\}$ has identity e.[1] For $A = \prod_{i=1}^{r} A_i$, the *coordinate idempotents* $e_i = (0, \dots, 1_{A_i}, \dots, 0)$ are a complete orthogonal family and are the identities of the individual factors. In $A = \Bbbk A_n$, the diagonal matrix units $e_{11}, \dots, e_{nn}$ are primitive, since $e_{ii}Ae_{ii} = \Bbbk e_{ii}$. For $n \geq 2$, the sum $e = e_{11} + \cdots + e_{n-1,n-1}$ gives the corner

$$eAe = \left\{ \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \;\middle|\; B \in \Bbbk A_{n-1} \right\} \cong \Bbbk A_{n-1}.$$

Idempotents also describe maps out of the summands of A_A.

Lemma 1.17 (Yoneda lemma). *For an idempotent $e \in A$ and a right A-module M, evaluation at e gives a natural (in M) vector space isomorphism*

$$\mathrm{Hom}_A(eA, M) \xrightarrow{\sim} Me, \qquad f \longmapsto f(e), \quad \text{with inverse} \quad m \longmapsto (ea \mapsto ma).$$

Proof We have $f(e) = f(e)e$ and $f(ea) = f(e)a$. If $m = me$, the map $ea \mapsto ma$ is well-defined because $ea = ea'$ implies $ma = mea = mea' = ma'$. These formulas prove that the maps are inverse; postcomposition by a module map $M \to N$ proves naturality. □

1.4 Graded algebras

Definition 1.18. *Let G be a group with identity e. A G-graded $\Bbbk$-algebra is a $\Bbbk$-algebra A with $1_A \in A_e$ and a decomposition*

$$A = \bigoplus_{g \in G} A_g \quad \text{such that} \quad A_g A_h \subset A_{gh} \quad \text{for all } g, h \in G.$$

An element of A_g is homogeneous of degree g.

Convention. *Unless stated otherwise, a graded algebra is $\mathbb{Z}$-graded and concentrated in non-negative degrees.*

Note that the decomposition is into a direct sum, so every element of a graded algebra A can always be written as a *finite* sum of homogeneous elements.

Example 1.19. *For the upper triangular matrix ring $\Bbbk A_n$ of Example 1.2, the decomposition*

$$(\Bbbk A_n)_d = \mathrm{span}_{\Bbbk}\{e_{ij} \mid j - i = d\}$$

makes $\Bbbk A_n$ a graded algebra.

Example 1.20. *The algebra $\Bbbk A_3$ also has a $\mathbb{Z}^{\oplus 2}$-grading with*

$$\deg e_{ii} = (0,0), \qquad \deg e_{12} = (1,0), \qquad \deg e_{23} = (0,1), \qquad \deg e_{13} = (1,1).$$

The homomorphism $\mathbb{Z}^{\oplus 2} \to \mathbb{Z}$, $(p,q) \mapsto p+q$, gives the grading of Example 1.19. Alternatively, setting $\deg e_{ij} = 2(j-i)$ gives a $\mathbb{Z}$-grading with $A_1 = 0$ but $A_2 \neq 0$.

Definition 1.21. *A homomorphism of graded algebras $f : A \to B$ is a $\Bbbk$-algebra homomorphism with $f(A_i) \subset B_i$ for all $i \in \mathbb{Z}$.*

Definition 1.22. *A graded algebra A is locally finite-dimensional (lfd) if $\dim_{\Bbbk} A_i < \infty$ for every $i \in \mathbb{Z}$.*

[1] note that this is not a unital subalgebra since the identity element changes; a more proper way to think about this is as a subcategory $\mathrm{add}_{\mathrm{mod}\, A}(eA) \simeq \mathrm{add}_{\mathrm{mod}\, eAe}(eAe)$.

1.5 Graded modules

Definition 1.23. *Let $A = \bigoplus_{g\in G} A_g$ be a G-graded algebra. A G-graded right A-module is a right A-module M with a direct sum decomposition*

$$M = \bigoplus_{g\in G} M_g \quad \text{such that} \quad M_g A_h \subset M_{gh} \quad \text{for all } g, h \in G.$$

An element of M_g is called homogeneous of degree g.

A G-graded A-module $M = \bigoplus_{g\in G} M_g$ is locally finite-dimensional (lfd) if $\dim_{\Bbbk} M_g < \infty$ for all $g \in G$.

Unless specified otherwise, graded modules are $\mathbb{Z}$-graded; their degrees need not be non-negative.

For an algebra A, the opposite algebra A^{op} has the same vector space as A, with multiplication $a^{\mathrm{op}}b^{\mathrm{op}} = (ba)^{\mathrm{op}}$. A left A-module is a right A^{op}-module by $m \cdot a^{\mathrm{op}} = am$. Thus $\mathrm{Mod}(A^{\mathrm{op}})$ is the category of left A-modules and left A-linear maps; not to be confused with the opposite category $(\mathrm{Mod}\, A)^{\mathrm{op}}$ though. For a $\mathbb{Z}$-graded algebra $A = \bigoplus_{i\geq 0} A_i$,

$$\text{the opposite algebra is also graded with } (A^{\mathrm{op}})_i = A_i.$$

Definition 1.24. *Let M be a graded A-module and let $d \in \mathbb{Z}$. The degree shift, or grading shift, $M(d)$ is the graded A-module whose underlying A-module is M, but whose grading is*

$$M(d)_j = M_{j+d}.$$

By Definition 1.24, $1 \in A_0 = A(-d)_d$, so $A(-d)$ has a generator in degree d. If M is concentrated in degree 0, then $M(d)$ is concentrated in degree $-d$.

Example 1.25. *For $A = \Bbbk A_3$ with the grading of Example 1.19, the module $P_1 := e_{11}A$ consists of matrices supported on the first row. Its basis e_{11}, e_{12}, e_{13} has degrees $0, 1, 2$, respectively; in $P_1(-2)$, the same basis has degrees $2, 3, 4$.*

Definition 1.26. *Let M, N be graded A-modules. A homomorphism $f : M \to N$ is homogeneous of degree d if $f(M_j) \subset N_{j+d}$ for all $j \in \mathbb{Z}$. A homomorphism of graded modules means a homogeneous A-module homomorphism of degree 0, also said to preserve degrees.*

Example 1.27. *For $A = \Bbbk A_3$, left multiplication by e_{12} yields a map*

$$\begin{aligned} e_{22}A &\longrightarrow e_{11}A, \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & u & v \\ 0 & 0 & 0 \end{pmatrix} &\longmapsto \begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad (u, v \in \Bbbk). \end{aligned}$$

The u and v entries have degrees $0, 1$ in the source and $1, 2$ in the target, and so the map has degree 1.

Notation. *For graded right A-modules M, N, the space of homomorphisms preserving degrees is*

$$\mathrm{Hom}_A^{\mathbb{Z}}(M, N) = \{f \in \mathrm{Hom}_A(M, N) \mid f(M_j) \subset N_j \text{ for all } j \in \mathbb{Z}\}. \tag{1.4}$$

Thus $\mathrm{Hom}_A^{\mathbb{Z}}(M, N(d))$ consists of homomorphisms $M \to N$ of degree d, since $N(d)_j = N_{j+d}$. Note that, like the grading shift functor, different literature has different notation for this (there does not seem to be any widely agreed notation).

We write

$$\begin{aligned} \mathrm{GrMod}\, A &:= \text{category of graded right } A\text{-modules with morphism spaces } \mathrm{Hom}_A^{\mathbb{Z}}(M, N), \\ \mathrm{grmod}\, A &:= \text{full subcategory of } \mathrm{GrMod}\, A \text{ consisting of locally finite-dimensional modules.} \end{aligned}$$

Thus $\mathrm{GrMod}(A^{\mathrm{op}})$ is the category of graded left A-modules and degree-preserving maps. Forgetting the grading gives a functor $\mathrm{GrMod}\, A \to \mathrm{Mod}\, A$.

Remark 1.28. grmod A is somewhat analogous to mod A but not really the same: If A is locally finite-dimensional, every finitely generated graded A-module M is locally finite-dimensional. ($\because$ a finite homogeneous generating set of a graded A-module M gives a surjection from a finite direct sum of shifts of A to M, so every M_i has to be finite-dimensional.) However, a locally finite-dimensional M need not be finitely generated.

For $d \in \mathbb{Z}$, the shift of Definition 1.24 is a functor

$$(d) : \operatorname{GrMod} A \to \operatorname{GrMod} A.$$

It sends $f : M \to N$ to the same underlying map $f(d) : M(d) \to N(d)$. Since $M(d)(-d) = M$ and $f(d)(-d) = f$, the shift is an equivalence with inverse $(-d)$; it restricts to an equivalence $\operatorname{grmod} A \to \operatorname{grmod} A$.

Remark 1.29. The notion of exactness makes sense for graded module categories – a sequence in $\operatorname{GrMod} A$ is exact if the underlying sequence in $\operatorname{Mod} A$ is exact. In particular, we talk about functors invovle such categories being exact.

> For $L, M, N \in \operatorname{GrMod} A$ with maps $i : L \to M$ and $p : M \to N$, the sequence $0 \to L \xrightarrow{i} M \xrightarrow{p} N \to 0$ is exact if and only if
>
> $$0 \longrightarrow L_j \xrightarrow{i|_{L_j}} M_j \xrightarrow{p|_{M_j}} N_j \longrightarrow 0 \quad \text{is exact for every } j \in \mathbb{Z}.$$
>
> Indeed, i and p preserve the direct sum decompositions, so their kernels and images decompose into the kernels and images of the restrictions.

Example 1.30. *Let $A = \Bbbk A_2$. Consider $P_1 := e_{11}A$ which has $\operatorname{rad} P_1 = \Bbbk e_{12}$ and $L_1 := P_1/\operatorname{rad} P_1$. Then we have a short exact sequence in* $\operatorname{GrMod} A$*:*

$$0 \longrightarrow \operatorname{rad} P_1 \xrightarrow{i} P_1 \xrightarrow{p} P_1/\operatorname{rad} P_1 \longrightarrow 0, \qquad i(ce_{12}) = ce_{12}, \quad p(m) = \overline{m}.$$

Note that $\deg \operatorname{rad} P_1 = 1$ *and* $\deg L_1 = 0$*. Let $L_2 := \Bbbk e_{22} = e_{22}A$, which we treat as a graded A-module concentrated in degree 0. Note that $L_2 \cong \operatorname{rad} P_1$ as ungraded modules but we only have $L_2 \cong (\operatorname{rad} P_1)(1)$, which yields the following short exact sequence in* $\operatorname{GrMod} A$*:*

$$0 \longrightarrow L_2(-1) \to P_1 \to P_1/\operatorname{rad} P_1 \longrightarrow 0,$$

Lemma 1.31. *For graded right A-modules M, N, there is an inclusion*

$$\bigoplus_{d \in \mathbb{Z}} \operatorname{Hom}_A^{\mathbb{Z}}(M, N(d)) \subset \operatorname{Hom}_A(M, N), \text{ which is equality when } M \text{ is finitely generated.}$$

Proof For $f \in \operatorname{Hom}_A(M, N)$, define $f_d(m) = f(m)_{j+d}$ when $m \in M_j$, and extend linearly. For $a \in A_i$, we have $f_d(ma) = f(ma)_{j+i+d} = f(m)_{j+d}a$, so f_d is A-linear of degree d. The sum of the homogeneous Hom spaces is direct: on M_j, distinct degrees have values in distinct summands of N.

If M is finitely generated, the homogeneous components of a finite generating set form a finite homogeneous generating set. Only finitely many f_d are non-zero on these homogeneous generators, hence on M, giving $f = \sum_d f_d$. □

1.6 Homogeneous ideals and quotients

Definition 1.32. *Let $A = \bigoplus_{i \in \mathbb{Z}} A_i$ be a graded algebra. A right ideal $I \subset A$ is homogeneous if*

$$I = \bigoplus_{i \in \mathbb{Z}} (I \cap A_i).$$

Equivalently, whenever $a \in I$ and $a = a_0 + \cdots + a_N$ is its decomposition into homogeneous components, each a_i also belongs to I.

For quotient algebras we need two-sided ideals. If I is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i \in \mathbb{Z}} A_i/(I \cap A_i)$$

is again a graded algebra.

Example 1.33. *In $A = \Bbbk A_3$ with the grading of Example 1.19, the two-sided ideal $I = (e_{13}) = \Bbbk e_{13}$ is homogeneous. Writing $\overline{e}_{ij}$ for the image of e_{ij},*

$$(A/I)_0 \cong \Bbbk^{\oplus 3}, \qquad (A/I)_1 = \Bbbk\overline{e}_{12} \oplus \Bbbk\overline{e}_{23}, \qquad (A/I)_d = 0 \quad (d \geq 2).$$

In particular, $\overline{e}_{12}\overline{e}_{23} = 0$.

Example 1.34. *Let $A = \Bbbk[x]$ with $\deg x = 1$, and let $I = (x - x^2)$. The ideal I is two-sided because A is commutative. Every element of I vanishes at $x = 1$, whereas neither x nor x^2 does. Thus $x - x^2 \in I$ but $x, x^2 \notin I$, so I is not homogeneous. Writing $\overline{f} = f + I$, we have $0 \neq \overline{x} = \overline{x^2}$. The images of A_1 and A_2 therefore intersect non-trivially, so A/I does not inherit the quotient grading. As an ungraded algebra, the quotient is semisimple:*

$$\Bbbk[x]/(x - x^2) \xrightarrow{\sim} \Bbbk \oplus \Bbbk, \qquad \overline{f} \longmapsto (f(0), f(1)).$$

The inverse sends (a, b) to $a + (b - a)\overline{x}$.

Definition 1.35. *The positive degree ideal of a graded algebra A is $A_+ := \bigoplus_{i>0} A_i$.*[2]

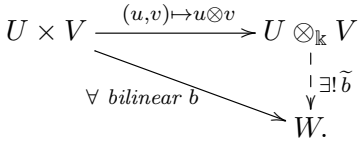
The subspace A_+ is a homogeneous two-sided ideal.

1.7 Tensor products of vector spaces

Definition 1.36. *Choose a basis $\{u_i \mid i \in I\}$ of U and a basis $\{v_j \mid j \in J\}$ of V. The tensor product $U \otimes_\Bbbk V$ is the vector space with basis $\{u_i \otimes v_j \mid i \in I,\ j \in J\}$.*

The assignment $(u_i, v_j) \mapsto u_i \otimes v_j$ extends uniquely to a bilinear map $U \times V \to U \otimes_\Bbbk V$ sending $(u, v) \mapsto u \otimes v$. Note that this means that elements like $(au_1 + bu_2) \otimes (cv_1 + dv_2)$ in $U \otimes_\Bbbk V$ is by definition the linear combination $ac(u_1 \otimes v_1) + ad(u_1 \otimes v_2) + bc(u_2 \otimes v_1) + bd(u_2 \otimes v_2)$.

Proposition 1.37 (Universal property). *Every bilinear map $b : U \times V \to W$ has a unique linear factorisation (commutative diagram)*



Consequently, $U \otimes_\Bbbk V$ is independent of the chosen bases up to a unique isomorphism preserving the bilinear map.

Proof The formula $\widetilde{b}(u_i \otimes v_j) = b(u_i, v_j)$ determines a unique linear map $\widetilde{b} : U \otimes_\Bbbk V \to W$. Bilinearity gives $\widetilde{b}(u \otimes v) = b(u, v)$ for all $u \in U$ and $v \in V$.

Suppose $t : U \times V \to T$ and $t' : U \times V \to T'$ both satisfy the universal property. They give

[2] It is also called the augmentation ideal; we avoid that terminology.

unique linear maps $a : T \to T'$ and $b : T' \to T$ with $a \circ t = t'$ and $b \circ t' = t$. Both $b \circ a : T \to T$ and $\mathrm{id}_T : T \to T$ factor t through itself, since

$$(b \circ a) \circ t = b \circ t' = t = \mathrm{id}_T \circ t.$$

Uniqueness for (T, t) gives $b \circ a = \mathrm{id}_T$. Likewise, $(a \circ b) \circ t' = a \circ t = t' = \mathrm{id}_{T'} \circ t'$, so uniqueness for (T', t') gives $a \circ b = \mathrm{id}_{T'}$. □

For finite bases with $|I| = r$ and $|J| = s$, we obtain $\dim_{\Bbbk}(U \otimes_{\Bbbk} V) = rs$. We use the canonical identifications

$$(U \otimes_{\Bbbk} V) \otimes_{\Bbbk} W \xrightarrow{\sim} U \otimes_{\Bbbk} (V \otimes_{\Bbbk} W), \quad (u \otimes v) \otimes w \longmapsto u \otimes (v \otimes w),$$
$$\Bbbk \otimes_{\Bbbk} U \xrightarrow{\sim} U, \qquad\qquad c \otimes u \longmapsto cu,$$

and set $U^{\otimes_{\Bbbk} 0} := \Bbbk$.

1.8 Direct sums and direct products

For an arbitrary family $(M_i)_{i \in I}$ of right A-modules, indexed by a set I, the *direct product* and *direct sum* are

$$\prod_{i \in I} M_i = \{(m_i)_{i \in I} \mid m_i \in M_i \text{ for all } i \in I\},$$
$$\bigoplus_{i \in I} M_i = \{(m_i)_{i \in I} \in \prod_{i \in I} M_i \mid m_i = 0 \text{ for all but finitely many } i\}.$$

Addition and the A-action are componentwise; the same definitions apply to vector spaces by taking $A = \Bbbk$. Thus $\bigoplus_{i \in I} M_i \subset \prod_{i \in I} M_i$, with equality for finite I. For example, $(1, 1, \ldots) \in \prod_{i \geq 0} \Bbbk$ does not belong to $\bigoplus_{i \geq 0} \Bbbk$. In a direct sum, an element is a finite sum of its components; no notion of convergence is involved. For instance, ordinary polynomials use non-negative powers, whereas Laurent polynomials allow integer powers, but both are finite sums:

$$\Bbbk[x] = \bigoplus_{n \geq 0} \Bbbk x^n, \qquad \Bbbk[x, x^{-1}] = \bigoplus_{n \in \mathbb{Z}} \Bbbk x^n.$$

The *formal power series ring* allows arbitrary coefficients:

$$\Bbbk[[x]] = \left\{ \sum_{n \geq 0} a_n x^n \;\middle|\; a_n \in \Bbbk \text{ for every } n \geq 0 \right\} \cong \prod_{n \geq 0} \Bbbk x^n \quad \text{as vector spaces.}$$

Addition is coefficientwise, and multiplication is

$$\Big(\sum_{n \geq 0} a_n x^n\Big)\Big(\sum_{n \geq 0} b_n x^n\Big) = \sum_{n \geq 0} \Big(\sum_{j=0}^{n} a_j b_{n-j}\Big) x^n.$$

Tensor products distribute over direct sums in either variable. For vector spaces U, V and arbitrary families $(U_i)_{i \in I}$, $(V_j)_{j \in J}$ indexed by sets I, J, there are natural linear isomorphisms

$$\Big(\bigoplus_{i \in I} U_i\Big) \otimes_{\Bbbk} V \xrightarrow{\sim} \bigoplus_{i \in I} (U_i \otimes_{\Bbbk} V), \quad (u_i)_{i \in I} \otimes v \longmapsto (u_i \otimes v)_{i \in I},$$
$$U \otimes_{\Bbbk} \Big(\bigoplus_{j \in J} V_j\Big) \xrightarrow{\sim} \bigoplus_{j \in J} (U \otimes_{\Bbbk} V_j), \quad u \otimes (v_j)_{j \in J} \longmapsto (u \otimes v_j)_{j \in J}. \tag{1.5}$$

The displayed bilinear assignments induce these linear maps by Proposition 1.37. On the ith summand, the inverse of the first map sends $u_i \otimes v$ to the element with u_i in coordinate i, tensored with v. On the jth summand, the inverse of the second sends $u \otimes v_j$ to u tensored with the element having v_j in coordinate j. Extending over the direct sums gives the two inverses.

For direct products, the corresponding map need not be an isomorphism.

Lemma 1.38. *For an arbitrary family $(U_i)_{i\in I}$ of vector spaces, indexed by a set I, and a vector space V, the coordinate projections define a linear map*

$$\Big(\prod_{i\in I} U_i\Big) \otimes_{\Bbbk} V \longrightarrow \prod_{i\in I}(U_i \otimes_{\Bbbk} V), \qquad (u_i)_i \otimes v \longmapsto (u_i \otimes v)_i. \tag{1.6}$$

This map is injective, and is an isomorphism if I is finite or V is finite-dimensional. The same statements hold with the two tensor factors reversed.

Proof The bilinear map $\big(\prod_{i\in I} U_i\big) \times V \to \prod_{i\in I}(U_i \otimes_{\Bbbk} V)$, $((u_i)_i, v) \mapsto (u_i \otimes v)_i$, induces (1.6) by Proposition 1.37. Every element can be written $\sum_{\ell=1}^{r} u^{(\ell)} \otimes v_\ell$ with the v_ℓ linearly independent: choose a basis of the span of its finitely many second factors. If its image is zero, then $\sum_{\ell=1}^{r} u_i^{(\ell)} \otimes v_\ell = 0$ for every i. The tensor-product basis construction gives $u_i^{(\ell)} = 0$ for all i, ℓ, proving injectivity. If $v_1, \dots, v_r$ is a basis of V, the inverse is

$$\prod_{i\in I}(U_i \otimes_{\Bbbk} V) \longrightarrow \Big(\prod_{i\in I} U_i\Big) \otimes_{\Bbbk} V, \qquad \Big(\sum_{\ell=1}^{r} u_i^{(\ell)} \otimes v_\ell\Big)_i \longmapsto \sum_{\ell=1}^{r} (u_i^{(\ell)})_i \otimes v_\ell.$$

For finite I, products equal sums. Reversing the tensor factors gives the other statements, since $U \otimes_{\Bbbk} V \to V \otimes_{\Bbbk} U$, $u \otimes v \mapsto v \otimes u$, is an isomorphism. □

Example 1.39. *The comparison in* (1.6) *need not be surjective when V is infinite-dimensional. Take $U_i = \Bbbk$ for $i \geq 1$ and $V = \bigoplus_{j\geq 1} \Bbbk v_j$. Under $\Bbbk \otimes_{\Bbbk} V \cong V$, the image of the product comparison in* (1.6) *is*

$$\left\{ (w_i)_i \in \prod_{i\geq 1} V \;\middle|\; \dim_{\Bbbk} \operatorname{span}_{\Bbbk}\{w_i \mid i \geq 1\} < \infty \right\}.$$

Indeed, the image of $\sum_{\ell=1}^{r} (c_i^{(\ell)})_i \otimes w_\ell$ has every entry in $\operatorname{span}_{\Bbbk}\{w_1, \dots, w_r\}$. Conversely, expand all entries in a basis of their common finite-dimensional span. Since the v_i are linearly independent, $(v_i)_i$ is not in the image.

1.9 Bimodules and enveloping algebras

Definition 1.40. *For $\Bbbk$-algebras B, C, a bimodule ${}_BM_C$ is a vector space with $\Bbbk$-bilinear left B- and right C-module actions satisfying $(bm)c = b(mc)$ for all $b \in B$, $c \in C$, and $m \in M$. The subscripts indicate the left and right acting algebras; when $B = C = A$, we call M an A-bimodule. A bimodule homomorphism $f : {}_BM_C \to {}_BN_C$ is a linear map satisfying $f(bmc) = bf(m)c$ for all $b \in B$, $c \in C$, and $m \in M$. For graded B, C, a bimodule M is graded if $M = \bigoplus_{i\in\mathbb{Z}} M_i$ and $B_iM_jC_k \subset M_{i+j+k}$ for all $i, j, k \in \mathbb{Z}$. A graded bimodule homomorphism is a bimodule homomorphism $f : M \to N$ satisfying $f(M_i) \subset N_i$ for all $i \in \mathbb{Z}$.*

Convention.

- *Bilinearity implies that $\Bbbk$ acts centrally: $(c1_B)m = cm = m(c1_C)$ for $c \in \Bbbk$. From the perspective of general module theory over general rings (algebras over non-field commutative rings), we should have said that our bimodules are $\Bbbk$-central bimodule. Since we alawys work*

over a field, all bimodules we considered are always $\Bbbk$*-central and we will not write such adjective every time.*

- *Instead of saying* (B,C)*-bimodule, we usually just write* ${}_BM_C$*. It would be natural to write* $B\operatorname{Mod} C$ *for the category of* (B,C)*-bimodules, but since* $\Bbbk$*-centrality comes for free from the assumption of* $\Bbbk$ *being a field, we can use* $\operatorname{Mod} C \otimes_{\Bbbk} B^{\mathrm{op}}$ *instead.*

Example 1.41.

- *Multiplication makes* A_A *a graded right module and* ${}_AA_A$ *a graded bimodule; each degree component is a bimodule* ${}_{A_0}(A_i)_{A_0}$*.*
- *For a bimodule* ${}_AM_A$ *and idempotents* $e, f \in A$*, the space* $eMf := \{enf \mid n \in M\}$ *is a bimodule* ${}_{eAe}(eMf)_{fAf}$*.*

Definition 1.42. *For a* $\Bbbk$*-algebra* A*, its enveloping algebra is* $A^{\mathrm{en}} := A \otimes_{\Bbbk} A^{\mathrm{op}}$*, with multiplication extended bilinearly from*

$$(a \otimes b^{\mathrm{op}})(a' \otimes (b')^{\mathrm{op}}) = aa' \otimes (b'b)^{\mathrm{op}}, \qquad 1_{A^{\mathrm{en}}} = 1_A \otimes 1_A^{\mathrm{op}} \quad (a,b,a',b' \in A).$$

A bimodule ${}_AM_A$ becomes a right A^{en}-module by

$$m \cdot (a \otimes b^{\mathrm{op}}) = bma \quad (a,b \in A,\ m \in M). \tag{1.7}$$

The first tensor factor records the right action, and the second the left action:

$$\big(m \cdot (a \otimes b^{\mathrm{op}})\big) \cdot (a' \otimes (b')^{\mathrm{op}}) = b'bmaa' = m \cdot \big(aa' \otimes (b'b)^{\mathrm{op}}\big).$$

Conversely, for $M \in \operatorname{Mod} A^{\mathrm{en}}$, define $ma = m \cdot (a \otimes 1)$ and $bm = m \cdot (1 \otimes b^{\mathrm{op}})$. The two actions commute because $a \otimes 1$ and $1 \otimes b^{\mathrm{op}}$ commute in A^{en}. For bimodules ${}_AM_A$, ${}_AN_A$, a linear map $u : M \to N$ satisfies

$$u(bma) = bu(m)a \quad (a,b \in A,\ m \in M) \quad \Longleftrightarrow \quad u \in \operatorname{Hom}_{A^{\mathrm{en}}}(M,N).$$

Thus $A\operatorname{Mod} A$ is identified with $\operatorname{Mod} A^{\mathrm{en}}$. More generally, for $\Bbbk$-algebras B, C, the vector space $C \otimes_{\Bbbk} B^{\mathrm{op}}$ is an algebra with

$$(c \otimes b^{\mathrm{op}})(c' \otimes (b')^{\mathrm{op}}) = cc' \otimes (b'b)^{\mathrm{op}}, \qquad 1 = 1_C \otimes 1_B^{\mathrm{op}}.$$

The same construction identifies the categories

$$B\operatorname{Mod} C \simeq \operatorname{Mod}(C \otimes_{\Bbbk} B^{\mathrm{op}}), \qquad m \cdot (c \otimes b^{\mathrm{op}}) = bmc.$$

Grant–Iyama [GI20, §2, p. 2591] identify bimodules with left modules over A^{en}, using $(a \otimes b^{\mathrm{op}})m = amb$. Our right module convention is (1.7): $m \cdot (a \otimes b^{\mathrm{op}}) = bma$. Both describe the same bimodules; the side on which A^{en} acts determines the formula.

Remark 1.43. For a finite-dimensional semisimple $\Bbbk$-algebra S, semisimplicity of S^{en} requires an additional condition. For a field extension $K/\Bbbk$, give $K \otimes_{\Bbbk} S$ the multiplication $(c \otimes s)(d \otimes t) = cd \otimes st$. The algebra S is called *separable over* $\Bbbk$ if $K \otimes_{\Bbbk} S$ is semisimple for every field extension $K/\Bbbk$. By [DK94, Theorem 6.1.2, p. 105],

$$S^{\mathrm{en}} \text{ is semisimple} \quad \Longleftrightarrow \quad S \text{ is separable over } \Bbbk.$$

For a product of matrix algebras, every field extension $K/\Bbbk$ gives

$$K \otimes_{\Bbbk} \prod_{j=1}^{n} \mathrm{Mat}_{n_j}(\Bbbk) \xrightarrow{\sim} \prod_{j=1}^{n} \mathrm{Mat}_{n_j}(K), \qquad c \otimes (a_j)_j \longmapsto (ca_j)_j.$$

The target is semisimple by Theorem 1.12. Thus products of matrix algebras over $\Bbbk$ are separable; in particular, every finite-dimensional semisimple algebra over an algebraically closed field is separable. Semisimplicity of S alone does not imply semisimplicity of S^{en} over an arbitrary field.

Example 1.44. *Let $\Bbbk = \mathbb{F}_2(t)$, where t is an indeterminate, and let $S = \Bbbk[u]/(u^2 - t)$, also writing u for its residue class. The element t is not a square in $\Bbbk$: an equality $p(t)^2 = tq(t)^2$ for non-zero polynomials would give $2\deg p = 1 + 2\deg q$. Thus $u^2 - t$ is irreducible and S is a field, hence semisimple. However, since S is commutative and has characteristic 2,*

$$S^{\mathrm{en}} = S \otimes_{\Bbbk} S \cong S[v]/(v^2 - t) = S[v]/((v-u)^2) \cong S[\varepsilon]/(\varepsilon^2).$$

Here the first tensor factor supplies the coefficients, v corresponds to $1 \otimes u$, and $\varepsilon = 1 \otimes u - u \otimes 1$. The ideal $S\varepsilon$ is non-zero, has square zero, and is the unique maximal ideal, so $\mathrm{rad}(S^{\mathrm{en}}) = S\varepsilon \neq 0$. Thus S^{en} is not semisimple, and the field extension $S/\Bbbk$ already shows that S is not separable over $\Bbbk$.

1.10 Graded $\Bbbk$-linear duals

A linear functional on a graded vector space can be non-zero on infinitely many degrees. To obtain a graded space, retain only finite sums of homogeneous functionals.

Definition 1.45. *For a graded vector space $V = \bigoplus_{i \in \mathbb{Z}} V_i$, the graded $\Bbbk$-linear dual is the graded vector space*

$$\mathbb{D}(V) = \bigoplus_{i \in \mathbb{Z}} \mathbb{D}(V)_i, \qquad \mathbb{D}(V)_i = D(V_{-i}) = \mathrm{Hom}_{\Bbbk}^{\mathbb{Z}}(V, \Bbbk(i)).$$

Note that here $\Bbbk$ is regarded as a graded module concentrated in degree 0.

Linear duality reverses the side of a module action. For a right A-module M and a left A-module N, set

$$(af)(m) = f(ma), \qquad (ga)(n) = g(an) \quad \text{for } a \in A,\ f \in D(M),\ g \in D(N),\ m \in M,\ n \in N. \tag{1.8}$$

For graded modules these actions restrict to the graded linear duals: if $a \in A_p$ and $f \in \mathbb{D}(M)_q$, then af vanishes outside M_{-p-q}.

Example 1.46. *Let $A = \Bbbk A_3$ and $P_1 = e_{11}A$ as in Example 1.25. The functionals $\varepsilon_{1j}(e_{1\ell}) = \delta_{j\ell}$ for $1 \leq j, \ell \leq 3$ form a basis of $\mathbb{D}(P_1)$, with*

$$\deg \varepsilon_{1j} = 1 - j, \qquad e_{12}\varepsilon_{12} = \varepsilon_{11}, \qquad e_{23}\varepsilon_{13} = \varepsilon_{12}, \qquad e_{jj}\varepsilon_{1j} = \varepsilon_{1j}.$$

The actions follow from (1.8).

For any $u \in \mathrm{Hom}_A(M, N)$, we have a dual map

$$D(u) : D(N) \to D(M), \qquad f \mapsto f \circ u,$$

which is a left A-linear map, i.e. $D(u) \in \mathrm{Hom}_{A^{\mathrm{op}}}(D(N), D(M))$. This gives rise to four exact (contravariant) functors (without any finiteness assumption):

$$\begin{aligned} D &: (\mathrm{Mod}\, A)^{\mathrm{op}} \longrightarrow \mathrm{Mod}(A^{\mathrm{op}}), & D &: (\mathrm{Mod}(A^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \mathrm{Mod}\, A, \\ \mathbb{D} &: (\mathrm{GrMod}\, A)^{\mathrm{op}} \longrightarrow \mathrm{GrMod}(A^{\mathrm{op}}), & \mathbb{D} &: (\mathrm{GrMod}(A^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \mathrm{GrMod}\, A. \end{aligned} \tag{1.9}$$

Exactness follows from Example 1.10, degree by degree for $\mathbb{D}$.

For these to be duality, we need to recover a module from its bidual; this requires some finiteness conditions.

Lemma 1.47. *Let A be a $\Bbbk$-algebra.*

(1) *The functor D defines a duality between finite-dimensional right and left A-modules. In particular, if A is finite-dimensional, then we have a duality on finitely generated modules*

$$D : (\operatorname{mod} A)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod}(A^{\mathrm{op}}), \qquad D : (\operatorname{mod}(A^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod} A.$$

(2) *Suppose that A is graded. For locally finite-dimensional $M \in \operatorname{grmod} A$, evaluation is a natural graded module isomorphism*

$$\mathrm{ev}_M : M \xrightarrow{\sim} \mathbb{D}(\mathbb{D}(M)), \qquad m \longmapsto (f \mapsto f(m)). \tag{1.10}$$

In particular, we have a duality between locally finite-dimensional graded right and left A-modules:

$$\mathbb{D} : (\operatorname{grmod} A)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod}(A^{\mathrm{op}}), \qquad \mathbb{D} : (\operatorname{grmod}(A^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod} A. \tag{1.11}$$

Proof First we show that ev_M is A-linear. Indeed,

$$\mathrm{ev}_M(ma)(f) = f(ma) = \mathrm{ev}_M(m)(af) = (\mathrm{ev}_M(m)a)(f) \quad (m \in M,\ a \in A,\ f \in D(M)).$$

For (1), the claim follows from Example 1.10. For (2), apply (1) degree-wise. □

For a locally finite-dimensional graded vector space V, restriction gives

$$\mathbb{D}(V) = \bigoplus_{i \in \mathbb{Z}} D(V_{-i}) \subset D(V) \cong \prod_{i \in \mathbb{Z}} D(V_i).$$

The inclusion is equality exactly when only finitely many V_i are non-zero. For an arbitrary vector space, evaluation $V \to D(D(V))$ is injective but need not be surjective; hence the functors in (1.9) are not dualities on all modules.

Example 1.48. *Graded linear duality need not preserve finite generation, even for a polynomial ring. Let $A = \Bbbk[x]$ with $\deg x = 1$ and let $\varepsilon_n(x^m) = \delta_{nm}$ for $n, m \geq 0$. Then $\deg \varepsilon_n = -n$, $x\varepsilon_0 = 0$, and $x\varepsilon_n = \varepsilon_{n-1}$ for $n \geq 1$. Any finite set of elements of $\mathbb{D}(A)$ lies in the span of $\varepsilon_0, \ldots, \varepsilon_N$ for some N, and multiplication by x preserves that span. Thus $\mathbb{D}(A)$ is locally finite-dimensional but not finitely generated.*

For vector spaces V, W, a functional $f \in D(V)$ and a vector $w \in W$ define a linear map $V \to W$, $v \mapsto f(v)w$. Finite sums of these maps give the following description of Hom.

Proposition 1.49. *For vector spaces V, W, there is a natural linear map*

$$W \otimes_{\Bbbk} D(V) \longrightarrow \operatorname{Hom}_{\Bbbk}(V, W), \qquad w \otimes f \longmapsto \big(v \mapsto f(v)w\big). \tag{1.12}$$

It is an isomorphism if either V or W is finite-dimensional.

Proof If V has a finite basis $\{v_i \mid 1 \leq i \leq n\}$ with dual basis $\{v_i^* \mid 1 \leq i \leq n\}$, the inverse of (1.12) is

$$\operatorname{Hom}_{\Bbbk}(V, W) \longrightarrow W \otimes_{\Bbbk} D(V), \qquad F \longmapsto \sum_{i=1}^{n} F(v_i) \otimes v_i^*,$$

since $v = \sum_{i=1}^{n} v_i^*(v) v_i$. If instead W has a finite basis $\{w_j \mid 1 \le j \le m\}$, write $F(v) = \sum_{j=1}^{m} f_j(v) w_j$; the uniquely determined coordinate functionals $f_j \in D(V)$ give the inverse $F \mapsto \sum_{j=1}^{m} w_j \otimes f_j$. The formulas commute with precomposition and postcomposition, so the comparison is natural. □

Assumption 1.50. *For simplicity, from now on and unless stated otherwise, we assume that all graded algebras are $\mathbb{Z}$-graded with $A_{<0} = 0$, and the degree 0 component is a product of fields:*

$$A_0 = S = \bigoplus_{i=1}^{n} \Bbbk e_i.$$

That is S is semisimple with coordinate idempotents $e_1, \dots, e_n$.

This simplifies many cumbersome check when dealing with dual modules – the general setting requires using S-dual, which is naturally isomorphic to $\Bbbk$-dual via a choice of 'symmetrising form'; we will try to deal with this general setting in the advanced sections. We note as a fact that such form of S also guarantees S^{en} to be a semisimple algebra with primitive idempotents $e_i \otimes e_j$ over all $1 \le i, j \le n$.

It is enough to show that S is separable. We use the fact that S is separable over $\Bbbk$ if the multiplication map $\mu : S \otimes_{\Bbbk} S \to A$ splits as a map of A-bimodules. Equivalently, there is an element $p \in S \otimes_{\Bbbk} S$ with $\mu(p) = 1$ and $sp = ps$ for all $s \in S$. Take

$$p = \sum_{i=1}^{n} e_i \otimes e_i.$$

$\mu(p) = \sum e_i^2 = \sum e_i = 1$. For each j: $(e_j \otimes 1)p = e_j \otimes e_j = p\,(1 \otimes e_j)$, and since $e_j \otimes 1$ and $1 \otimes e_j$ both act on p the same way on either side, $(e_j \otimes 1)p = p(e_j \otimes 1) = e_j \otimes e_j$ (because $e_i e_j = \delta_{ij} e_j$). Hence $sp = ps$ for all s, by linearity.

For a bimodule ${}_S M_S$, the $\Bbbk$-linear dual retains both actions:

$$(s\lambda t)(m) = \lambda(tms) \qquad (s, t \in S,\ \lambda \in D(M),\ m \in M). \tag{1.13}$$

With precomposition on morphisms, we obtain functors

$$D : (\operatorname{Mod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{Mod} S^{\mathrm{en}}, \qquad \mathbb{D} : (\operatorname{GrMod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{GrMod} S^{\mathrm{en}},$$

whose restrictions to finite-dimensional bimodules, and to locally finite-dimensional graded bimodules, are dualities by Lemma 1.47.

Evaluation preserves both actions: for $m \in M$, $\lambda \in D(M)$, and $s, t \in S$,

$$(s \operatorname{ev}_M(m)\, t)(\lambda) = \operatorname{ev}_M(m)(t\lambda s) = (t\lambda s)(m) = \lambda(smt) = \operatorname{ev}_M(smt)(\lambda).$$

Applying this in each degree gives the graded statement.

In particular, dualising reverses the two idempotents: for $1 \le i, j \le n$,

$$e_j D(M) e_i = \{\lambda \in D(M) \mid \lambda(e_k M e_\ell) = 0 \text{ unless } (k, \ell) = (i, j)\} \cong D(e_i M e_j), \tag{1.14}$$

since $(e_j \lambda e_i)(m) = \lambda(e_i m e_j)$ by (1.13).

1.11 S-dual

Under Assumption 1.50, the $\Bbbk$-linear dual is all we need. For an arbitrary finite-dimensional semisimple S, the choice-free substitutes are the one-sided S-duals below; they are the duals used in [BGS96, §2.7]. Under Assumption 1.50 they are canonically isomorphic to D; see Remark 1.58.

One-sided S-duals.

Definition 1.51. *For a right S-module M and a left S-module N, define their S-duals by*

$$M^* := \operatorname{Hom}_S(M, S) \in \operatorname{Mod}(S^{\mathrm{op}}), \quad (sf)(m) := sf(m),$$
$$^*N := \operatorname{Hom}_{S^{\mathrm{op}}}(N, S) \in \operatorname{Mod} S, \qquad (gs)(n) := g(n)s.$$

Here $s \in S$, $m \in M$, $n \in N$, $f \in M^*$, and $g \in {}^*N$.

Similar to $\Bbbk$-lienar duality, this definition extends to exact contravariant functors: for a right module map $u : M \to N$, define $u^* : N^* \to M^*$ by precomposition $u^*(f) = f \circ u$; for a left module map $u : M \to N$, the same recipe defines ${}^*u : {}^*N \to {}^*M$. This yields the desired functors

$$(-)^* : (\operatorname{Mod} S)^{\mathrm{op}} \longrightarrow \operatorname{Mod}(S^{\mathrm{op}}), \qquad {}^*(-) : (\operatorname{Mod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{Mod} S.$$

We also have graded analogue similar to Definition 1.45.

Definition 1.52. *For graded right and left S-modules M, respectively, define their graded S-duals by*

$$M^{(*)} := \bigoplus_{i\in\mathbb{Z}} (M_{-i})^* = \bigoplus_{i\in\mathbb{Z}} \operatorname{Hom}^{\mathbb{Z}}_S(M, S(i)), \qquad {}^{(*)}M := \bigoplus_{i\in\mathbb{Z}} {}^*(M_{-i}) = \bigoplus_{i\in\mathbb{Z}} \operatorname{Hom}^{\mathbb{Z}}_{S^{\mathrm{op}}}(M, S(i)),$$

where each indicated summand has degree i.

This also extends to a functor: for a degree 0 right S-module map $u : M \to N$, define $u^{(*)} : N^{(*)} \to M^{(*)}$ by

$$(u^{(*)})_i : (N_{-i})^* \longrightarrow (M_{-i})^*, \qquad f \longmapsto f \circ u|_{M_{-i}} \quad (i \in \mathbb{Z}).$$

For a degree 0 left module map $u : M \to N$, the same formula defines ${}^{(*)}u : {}^{(*)}N \to {}^{(*)}M$. With the actions from Definition 1.51, we obtain functors

$$(-)^{(*)} : (\operatorname{GrMod} S)^{\mathrm{op}} \longrightarrow \operatorname{GrMod}(S^{\mathrm{op}}), \qquad {}^{(*)}(-) : (\operatorname{GrMod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{GrMod} S. \tag{1.15}$$

Each functor in (1.15) restricts to locally finite-dimensional modules: replace GrMod by grmod in its source and target. Indeed, for $R = S$ or S^{op} and $M \in \operatorname{grmod} R$,

$$\dim_{\Bbbk} \operatorname{Hom}_R(M_{-i}, R) \le (\dim_{\Bbbk} M_{-i})(\dim_{\Bbbk} R) < \infty \quad (i \in \mathbb{Z}). \tag{1.16}$$

Lemma 1.53. *We have dualities*

$$(-)^* : (\operatorname{mod} S)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod}(S^{\mathrm{op}}), \qquad {}^*(-) : (\operatorname{mod}(S^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod} S,$$
$$(-)^{(*)} : (\operatorname{grmod} S)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod}(S^{\mathrm{op}}), \quad {}^{(*)}(-) : (\operatorname{grmod}(S^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod} S.$$

Proof We show that evaluation gives natural isomorphisms of functors

$$\mathrm{id}_{\mathrm{mod}\, S} \xrightarrow{\sim} {}^*((-)^*), \qquad \mathrm{id}_{\mathrm{mod}(S^{\mathrm{op}})} \xrightarrow{\sim} ({}^*(-))^*,$$

with components

$$\begin{aligned} \mathrm{ev}_M : M \xrightarrow{\sim} {}^*(M^*), \quad m \longmapsto (f \mapsto f(m)) \quad (M \in \mathrm{mod}\, S), \\ \mathrm{ev}_N : N \xrightarrow{\sim} ({}^*N)^*, \quad n \longmapsto (g \mapsto g(n)) \quad (N \in \mathrm{mod}(S^{\mathrm{op}})). \end{aligned}$$

For $u : M \to N$ in $\mathrm{mod}\, S$, $m \in M$, and $f \in N^*$,

$$\big({}^*(u^*)(\mathrm{ev}_M(m))\big)(f) = (u^* f)(m) = f(u(m)) = \mathrm{ev}_N(u(m))(f).$$

Hence ${}^*(u^*) \circ \mathrm{ev}_M = \mathrm{ev}_N \circ u$. For $M = S$, evaluation is an isomorphism: its inverse sends $F \in {}^*(S^*)$ to $F(\mathrm{id}_S)$. Indeed, every $f \in S^*$ satisfies $f(s) = f(1)s$, so $F(f) = f(1)F(\mathrm{id}_S) = f(F(\mathrm{id}_S))$. Restriction to summands identifies the dual of a finite direct sum with the direct sum of the duals; evaluation therefore is an isomorphism for $S^{\oplus n}$. By Corollary 1.13, $M \oplus L \cong S^{\oplus n}$ for some L. Under the direct sum identifications, $\mathrm{ev}_{M\oplus L} = \mathrm{ev}_M \oplus \mathrm{ev}_L$, so ev_M is an isomorphism. Replacing S by S^{op} proves the left module statement.

For the graded cases, apply the ungraded case degree-wise. Note that since S is finite-dimensional, finitely generated S-module is the same as finite-dimensional S-modules. □

For a bimodule ${}_S M_S$, the linear and one-sided duals retain both actions:

$$(s\lambda t)(m) = \lambda(tms), \qquad (sft)(m) = sf(tm), \qquad (sgt)(m) = g(ms)t, \tag{1.17}$$

where $s, t \in S$, $m \in M$, $\lambda \in D(M)$, $f \in \mathrm{Hom}_S(M, S)$, and $g \in \mathrm{Hom}_{S^{\mathrm{op}}}(M, S)$. The functionals still take values in $\Bbbk$ or S. With precomposition on morphisms, we obtain

$$\begin{aligned} D,\ (-)^*,\ {}^*(-) : (\mathrm{Mod}\, S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \mathrm{Mod}\, S^{\mathrm{en}}, \\ \mathbb{D},\ (-)^{(*)},\ {}^{(*)}(-) : (\mathrm{GrMod}\, S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \mathrm{GrMod}\, S^{\mathrm{en}}. \end{aligned} \tag{1.18}$$

The restrictions to finite-dimensional modules, and to locally finite-dimensional graded modules, are dualities by Lemmas 1.47 and 1.53. Evaluation preserves both actions: for $m \in M$, $f \in M^*$, and $s, t \in S$,

$$(s\, \mathrm{ev}_M(m)\, t)(f) = f(sm)t = f(smt) = \mathrm{ev}_M(smt)(f).$$

The corresponding checks for D and the left dual use the same evaluation formula; applying them in each degree gives the graded statements.

Recall from Definition 1.42 and (1.7) that $S^{\mathrm{en}} = S \otimes_{\Bbbk} S^{\mathrm{op}}$ and S-bimodules are the same thing as S^{en}-modules via $m \cdot (s \otimes t^{\mathrm{op}}) = tms$.

Definition 1.54. *The two-sided dual of a bimodule ${}_S M_S$ is*

$$M^{*_{\mathrm{en}}} := \mathrm{Hom}_{S^{\mathrm{en}}}(M, S^{\mathrm{en}}), \qquad (sFt)(m) := (s \otimes t^{\mathrm{op}})F(m) \quad (s, t \in S,\ F \in M^{*_{\mathrm{en}}},\ m \in M).$$

In Definition 1.54, M has the right action (1.7), so $F(tms) = F(m)(s \otimes t^{\mathrm{op}})$. The output is a left S^{en}-module by multiplication on the values, which gives the displayed bimodule action. The two-sided dual takes values in S^{en}, not in S. On bimodule morphisms it acts by precomposition, as do D, $(-)^*$, and ${}^*(-)$; each is a contravariant functor on the category of S-bimodules. The two-sided dual records both module actions in the target algebra S^{en}.

For a graded bimodule ${}_SM_S$, define its *graded two-sided dual* by

$$M^{(*_{\mathrm{en}})} := \bigoplus_{i\in\mathbb{Z}} (M_{-i})^{*_{\mathrm{en}}}, \qquad (M^{(*_{\mathrm{en}})})_i = (M_{-i})^{*_{\mathrm{en}}}.$$

Definition 1.54 gives a functor

$$(-)^{(*_{\mathrm{en}})} : (\mathrm{GrMod}\, S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \mathrm{GrMod}\, S^{\mathrm{en}}. \tag{1.19}$$

On a degree 0 bimodule map $u : M \to N$, its degree i component is

$$(u^{(*_{\mathrm{en}})})_i : (N_{-i})^{*_{\mathrm{en}}} \longrightarrow (M_{-i})^{*_{\mathrm{en}}}, \qquad F \longmapsto F \circ u|_{M_{-i}} \quad (i \in \mathbb{Z}).$$

The functor restricts to locally finite-dimensional bimodules by (1.16) with $R = S^{\mathrm{en}}$.

The four duals above follow the distinction in Grant–Iyama [GI20, §2.1, pp. 2592–2593], with the following notation changes:

construction	Grant–Iyama	these notes
$\Bbbk$-linear dual	M^*	$D(M)$
left S-linear dual	M^{*_ℓ}	${}^*M$
right S-linear dual	M^{*_r}	M^*
two-sided dual	$M^\vee$	$M^{*_{\mathrm{en}}}$

The last row also changes the convention for enveloping modules. Their $F : M \to S^{\mathrm{en}}$ is left S^{en}-linear. Define

$$\sigma : S^{\mathrm{en}} \longrightarrow S^{\mathrm{en}}, \qquad a \otimes b^{\mathrm{op}} \longmapsto b \otimes a^{\mathrm{op}}.$$

Since $\sigma(xy) = \sigma(y)\sigma(x)$ for $x, y \in S^{\mathrm{en}}$, the map $\sigma \circ F : M \to S^{\mathrm{en}}$ satisfies

$$(\sigma \circ F)(tms) = (\sigma \circ F)(m)(s \otimes t^{\mathrm{op}}) \quad (s, t \in S,\ m \in M),$$

and belongs to our $M^{*_{\mathrm{en}}}$. Thus $M^\vee \to M^{*_{\mathrm{en}}}$, $F \mapsto \sigma \circ F$, translates their two-sided dual into our convention. In particular, their star denotes the linear dual; our star denotes the right S-dual.

1.12 Comparing the duals

The one-sided S-duals and the $\Bbbk$-linear dual are naturally isomorphic after choosing a symmetrising form on S.

Definition 1.55. *A symmetrising form on a finite-dimensional $\Bbbk$-algebra S is a linear map $\tau : S \to \Bbbk$ satisfying*

$$\tau(st) = \tau(ts) \quad \text{for all } s, t \in S, \qquad \big(\tau(st) = 0 \text{ for all } t \in S\big) \implies s = 0.$$

An algebra admitting such a form is symmetric.

The second condition in Definition 1.55 says that the bilinear form $(s, t) \mapsto \tau(st)$ is *non-degenerate*: every non-zero $s \in S$ pairs non-trivially with some $t \in S$.

Every finite-dimensional semisimple $\Bbbk$-algebra admits a symmetrising form. For products of matrix algebras over $\Bbbk$, the forms are explicit:

Example 1.56. *The following maps are symmetrising forms:*

$$\tau : \Bbbk^{\oplus n} \longrightarrow \Bbbk, \quad (a_1, \dots, a_n) \longmapsto \sum_{j=1}^{n} a_j,$$

$$\tau : \mathrm{Mat}_n(\Bbbk) \longrightarrow \Bbbk, \qquad a \longmapsto \mathrm{Tr}(a) := \sum_{i=1}^{n} a_{ii},$$

and more generally, $\tau : \prod_{j=1}^{n} \mathrm{Mat}_{n_j}(\Bbbk) \longrightarrow \Bbbk, \quad (a_1, \dots, a_n) \longmapsto \sum_{j=1}^{n} \mathrm{Tr}(a_j).$

Generally, any finite-dimensional semisimple algebra has a symmetrising form, by Eilenberg–Nakayama.[a] For the graded analogue, see [DNN17]. For example, let $\lambda_j : \Delta_j \to \Bbbk$ be a symmetrising form on a finite-dimensional division $\Bbbk$-algebra Δ_j for $j = 1, \dots, n$. Then the semisimple algebra $S = \prod_{j=1}^{n} \mathrm{Mat}_{n_j}(\Delta_j)$ has a symmetrising form given by

$$\tau : S \longrightarrow \Bbbk, \qquad (a_1, \dots, a_n) \longmapsto \sum_{j=1}^{n} \sum_{i=1}^{n_j} \lambda_j((a_j)_{ii}).$$

Theorem 1.57 (Comparison of duals). *Choose a symmetrising form $\tau : S \to \Bbbk$. For $M \in \operatorname{mod} S$ and $N \in \operatorname{mod}(S^{\mathrm{op}})$, postcomposition with τ gives S-module isomorphisms*

$$\alpha_M : M^* \longrightarrow D(M), \quad f \longmapsto \tau \circ f, \qquad \beta_N : {}^*N \longrightarrow D(N), \quad g \longmapsto \tau \circ g, \tag{1.20}$$

natural in M and N. Applied in each degree, the same maps give natural isomorphisms $M^{(*)} \cong \mathbb{D}(M)$ and ${}^{(*)}N \cong \mathbb{D}(N)$ for $M \in \operatorname{grmod} S$ and $N \in \operatorname{grmod}(S^{\mathrm{op}})$; we again write α_M and β_N.

We can spell this out more precisely:

Ungraded duals. The isomorphisms (1.20) give natural isomorphisms

$$\begin{aligned} \alpha &: (-)^* \xrightarrow{\sim} D \quad \text{of functors } (\operatorname{mod} S)^{\mathrm{op}} \longrightarrow \operatorname{mod}(S^{\mathrm{op}}), \\ \beta &: {}^*(-) \xrightarrow{\sim} D \quad \text{of functors } (\operatorname{mod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{mod} S. \end{aligned} \tag{1.21}$$

Graded duals. Composition with τ in each degree gives natural isomorphisms

$$\begin{aligned} \alpha &: (-)^{(*)} \xrightarrow{\sim} \mathbb{D} \quad \text{of functors } (\operatorname{grmod} S)^{\mathrm{op}} \longrightarrow \operatorname{grmod}(S^{\mathrm{op}}), \\ \beta &: {}^{(*)}(-) \xrightarrow{\sim} \mathbb{D} \quad \text{of functors } (\operatorname{grmod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{grmod} S. \end{aligned} \tag{1.22}$$

For $M \in \operatorname{grmod} S$ and $N \in \operatorname{grmod}(S^{\mathrm{op}})$, the degree i components are $\alpha_{M_{-i}}$ and $\beta_{N_{-i}}$, respectively.

Proof We prove the right module statement, i.e. taking $\alpha_M(f) := \tau \circ f$ yields

$$\text{ungraded case } \alpha : (-)^* \xrightarrow{\sim} D, \quad \text{and} \quad \text{graded case } \alpha : (-)^{(*)} \xrightarrow{\sim} \mathbb{D}.$$

First, non-degeneracy of τ gives an isomorphism $S \to D(S)$, $a \mapsto (s \mapsto \tau(as))$. Hence, for $\lambda \in D(M)$, there is a unique linear map $f : M \to S$ satisfying

$$\tau(f(m)s) = \lambda(ms) \quad \text{for all } m \in M,\ s \in S. \tag{1.23}$$

For $m \in M$ and $s, t \in S$,

$$\tau(f(mt)s) = \lambda(mts) = \tau(f(m)ts),$$

so non-degeneracy implies $f(mt) = f(m)t$. Taking $s = 1$ in (1.23) gives $\tau \circ f = \lambda$; conversely, every right S-linear preimage of λ must satisfy (1.23). Thus α_M is bijective.

Both M^* and $D(M)$ are left S-modules, so the component $\alpha_M : M^* \to D(M)$ must preserve the left action:

$$\alpha_M(sf)(m) = \tau(sf(m)) = \tau(f(m)s) = \tau(f(ms)) = (s\alpha_M(f))(m) \quad (s \in S,\ f \in M^*,\ m \in M).$$

For a right module map $u : M \to N$ and $f \in N^*$, associativity of composition gives $\alpha_M(f \circ u) = \alpha_N(f) \circ u$, proving naturality.

For the graded comparisons, apply the ungraded natural isomorphism degree-wise.

Same argument applies dually to S^{op} and β. □

Remark 1.58. The natural isomorphisms in Theorem 1.57 depend on the chosen symmetrising form. For example, every choice $c_1, \dots, c_n \in \Bbbk \setminus \{0\}$ gives a form $\tau(a_1, \dots, a_n) = \sum_{i=1}^n c_i a_i$ on $\Bbbk^{\oplus n}$. Under Assumption 1.50 we always use the *standard symmetrising form* $\tau_S(\sum_{i=1}^n a_i e_i) = \sum_{i=1}^n a_i$, which involves no choice; then α_M and β_N are canonical isomorphisms.

Remark 1.59 (General S). For an arbitrary finite-dimensional semisimple S, identifying an S-dual with a $\Bbbk$-linear dual requires choosing a symmetrising form. If τ' is another one, then $\tau'(s) = \tau(zs)$ for a central unit $z \in S$, and the two identifications differ by multiplication by z; see Exercise 1.66(3).

Remark 1.60. Let A be a graded algebra with $A_0 = S$. The space $\mathrm{Hom}_S(M, S)$ has no natural left A-module structure on its own, but acquires one after *picking a symmetrising form τ of S*; under Assumption 1.50 we pick $\tau = \tau_S$. Indeed, the graded comparison α_M of Theorem 1.57 transports the canonical left A-action on $\mathbb{D}(M)$ to the graded S-dual of a locally finite-dimensional module:

$$af = \alpha_M^{-1}(a\alpha_M(f)) \quad \text{for right } M, \qquad ga = \beta_N^{-1}(\beta_N(g)a) \quad \text{for left } N,$$

where $a \in A$, $f \in M^{(*)}$, and $g \in {}^{(*)}N$. Similarly for right action on left dual.

The comparison theorem has a bimodule version:

Theorem 1.61. *With the actions in* (1.17)*, we obtain natural isomorphisms of functors*

$$\begin{aligned} (-)^* \xrightarrow[\alpha]{\sim} D \xleftarrow[\beta]{\sim} {}^*(-) \quad &: (\operatorname{mod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{mod} S^{\mathrm{en}}, \\ (-)^{(*)} \xrightarrow[\alpha]{\sim} \mathbb{D} \xleftarrow[\beta]{\sim} {}^{(*)}(-) \quad &: (\operatorname{grmod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{grmod} S^{\mathrm{en}}. \end{aligned} \tag{1.24}$$

There is also a natural isomorphism $\omega : (-)^{*_{\mathrm{en}}} \xrightarrow{\sim} (-)^*$ given by

$$\omega_M : M^{*_{\mathrm{en}}} \to M^*, \qquad F \mapsto \big(m \mapsto \sum_j a_j \tau(b_j)\big) \quad \text{for} \quad F(m) = \sum_j a_j \otimes b_j^{\mathrm{op}}, \tag{1.25}$$

which also depends on τ, but does not require S^{en} to be semisimple.

Consequently, on finite-dimensional S-bimodules and lfd graded S-bimodules we have natural isomorphisms:

$$\textit{Ungraded:}\quad (-)^{*_{\mathrm{en}}} \xrightarrow[\omega]{\sim} (-)^{*} \xrightarrow[\alpha]{\sim} D \xleftarrow[\beta]{\sim} {}^{*}(-) \ \textit{as functors}\ (\operatorname{mod} S^{\mathrm{en}})^{\mathrm{op}} \to \operatorname{mod} S^{\mathrm{en}},$$

$$\textit{Graded:}\quad (-)^{(*_{\mathrm{en}})} \xrightarrow[\omega]{\sim} (-)^{(*)} \xrightarrow[\alpha]{\sim} \mathbb{D} \xleftarrow[\beta]{\sim} {}^{(*)}(-) \ \textit{as functors}\ (\operatorname{grmod} S^{\mathrm{en}})^{\mathrm{op}} \to \operatorname{grmod} S^{\mathrm{en}}.$$

Proof The form $\tau_{\mathrm{en}}(a \otimes b^{\mathrm{op}}) = \tau(a)\tau(b)$ is symmetrising on S^{en}. Indeed, $\tau_{\mathrm{en}}((a \otimes b^{\mathrm{op}})(c \otimes d^{\mathrm{op}})) = \tau(ac)\tau(db) = \tau(ca)\tau(bd)$, and its pairing is the tensor product of two non-degenerate pairings. The formula (1.25) satisfies

$$\omega_M(F)(ms) = \omega_M(F)(m)s, \qquad \omega_M(sFt) = s\omega_M(F)t, \qquad \alpha_M(\omega_M(F)) = \tau_{\mathrm{en}} \circ F.$$

The argument (1.23), applied to S^{en} and τ_{en}, makes $F \mapsto \tau_{\mathrm{en}} \circ F$ an isomorphism. Thus ω_M is an isomorphism. For a bimodule map $u : M \to N$ and $F \in N^{*_{\mathrm{en}}}$, the formula (1.25) gives $\omega_M(F \circ u) = \omega_N(F) \circ u$, proving naturality. Taking the direct sum of $\omega_{M_{-i}}$ over $i \in \mathbb{Z}$ gives the graded two-sided comparison. Each component is a bimodule isomorphism, and naturality follows in each degree from the ungraded comparison. □

[a] S. Eilenberg and T. Nakayama, *On the dimension of modules and algebras, II. Frobenius algebras and quasi-Frobenius rings*, Nagoya Math. J. **9** (1955), 1–16, §2, Propositions 3–5, pp. 4–5; doi:10.1017/S0027763000023229.

1.13 Exercises

Exercise 1.62 (Matrix calculations). *Let $n \geq 2$ and $A = \Bbbk A_n$. Verify the grading in Example 1.19 using $e_{ij}e_{pq} = \delta_{jp}e_{iq}$. Compute* $\operatorname{rad} A$ *from its definition as the intersection of all maximal right ideals.*

Hint. *The strictly upper triangular ideal J satisfies $J^n = 0$. Show that $LJ = 0$ for every simple right A-module L, and use the diagonal maps $A \to \Bbbk$, $X \mapsto X_{ii}$, for $1 \leq i \leq n$.*

Exercise 1.63 (Shifts and duals). *Let $A = \Bbbk A_3$ and $M = e_{22}A$ with the inherited grading.*

(1) *Find a homogeneous basis of $\mathbb{D}(M)$, its degrees, and all non-zero actions of matrix units on the basis vectors.*

(2) *Show that $\Bbbk e_{13} \subset Ae_{33}$ is a graded left submodule and that*

$$\mathbb{D}(M) \cong (Ae_{33}/\Bbbk e_{13})(1)$$

as graded left A-modules.

(3) *Deduce that $\mathbb{D}(M) \ncong (Ae_{jj})(r)$ for every $j \in \{1,2,3\}$ and $r \in \mathbb{Z}$.*

Hint for (3). *Compare dimensions and the actions of e_{11}, e_{22}, e_{33}.*

Exercise 1.64 (Homogeneous maps). *The finite-generation hypothesis in Lemma 1.31 is needed. Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ with componentwise multiplication, let $N = e_1S$ have degree 0, and put*

$$M = \bigoplus_{i \geq 0} (e_1S)(-i) = \bigoplus_{i \geq 0} \Bbbk m_i, \qquad \deg m_i = i, \quad m_ie_1 = m_i, \quad m_ie_2 = 0.$$

For the S-linear map $f : M \to N$ with $f(m_i) = e_1$ for every $i \geq 0$, determine all homogeneous components and prove that f is not a finite sum of homogeneous maps.

Exercise 1.65 (Dependence on the form). *Let $S = (\Bbbk A_2)_0 = \Bbbk e_{11} \oplus \Bbbk e_{22}$, $V = (\Bbbk A_2)_1 = \Bbbk e_{12}$, and $c_1, c_2 \in \Bbbk \setminus \{0\}$. Set $\tau(ae_{11} + be_{22}) = c_1 a + c_2 b$ and let $\varepsilon : V \to \Bbbk$ satisfy $\varepsilon(e_{12}) = 1$. Find the right and left S-linear maps corresponding to ε under* (1.20)*, and determine how they depend on c_1, c_2.*

Exercise 1.66 (Dependence of the action on the form). *The transported A-action can depend on the symmetrising form, but its graded isomorphism class does not. Let $A = \Bbbk A_2$, $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$, and $M = e_{11}A$, with the inherited gradings. For $c = (c_1, c_2) \in (\Bbbk \setminus \{0\})^2$, define*

$$\tau_c : S \longrightarrow \Bbbk, \qquad ae_{11} + be_{22} \longmapsto c_1 a + c_2 b \quad \text{for all } a, b \in \Bbbk. \tag{1.26}$$

Write $\alpha_c : M^{(*)} \to \mathbb{D}(M)$, $f \mapsto \tau_c \circ f$, and let $a \cdot_c f = \alpha_c^{-1}(a\alpha_c(f))$ be the transported action of Remark 1.60. Consider the right S-linear maps

$$\begin{aligned} &\varphi_0 : M \longrightarrow S, \quad ae_{11} + be_{12} \longmapsto ae_{11}, \\ &\varphi_1 : M \longrightarrow S, \quad ae_{11} + be_{12} \longmapsto be_{22} \quad \text{for all } a, b \in \Bbbk. \end{aligned} \tag{1.27}$$

(1) *Verify that τ_c is symmetrising and that φ_0, φ_1 form a homogeneous basis of $M^{(*)}$. Determine their degrees.*

(2) *Calculate the actions of e_{11}, e_{22}, e_{12} on φ_0, φ_1. For $c, c' \in (\Bbbk \setminus \{0\})^2$, prove that $\cdot_c = \cdot_{c'}$ if and only if $c' = \rho c$ for some $\rho \in \Bbbk \setminus \{0\}$.*

(3) *More generally, let A be a locally finite-dimensional graded algebra with semisimple $S = A_0$, and let M be a locally finite-dimensional graded right A-module. Let τ, τ' be symmetrising forms on S, with comparison maps $\alpha_\tau, \alpha_{\tau'} : M^{(*)} \to \mathbb{D}(M)$ given by composition with τ, τ', respectively, and transported actions $\cdot_\tau, \cdot_{\tau'}$. Prove that there is an invertible $z \in S$ satisfying $\tau'(s) = \tau(zs)$ for every $s \in S$, and that*

$$\gamma : (M^{(*)}, \cdot_{\tau'}) \longrightarrow (M^{(*)}, \cdot_\tau), \qquad (\gamma f)(m) = zf(m)$$

is an isomorphism of graded left A-modules. Deduce that the modules in (2) are isomorphic for every c, c', and give the isomorphism on φ_0, φ_1.

Hint for (3). *Compare $\alpha_{\tau'}$ with α_τ.*

2 Tensor and quadratic algebras

2.1 Tensor products over an algebra

We retain our assumption on graded algebras meaning $\mathbb{Z}$-graded and concentrated in non-negative degrees. Unless otherwise specified, in this section, we do not require the assumption that $A_0 = S$ to be semisimple.

Definition 2.1. *Let A be any $\Bbbk$-algebra, M a right A-module, and N a left A-module. The tensor product of M and N over A is the vector space*

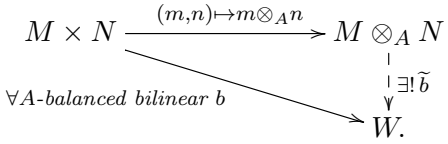
$$M \otimes_A N := M \otimes_\Bbbk N / \operatorname{span}_\Bbbk\{ma \otimes n - m \otimes an \mid m \in M,\ n \in N,\ a \in A\}.$$

Write $m \otimes_A n$, or simply $m \otimes n$, for the image of $m \otimes n$ in this quotient. Thus $ma \otimes_A n = m \otimes_A an$ for all $m \in M$, $n \in N$, and $a \in A$.

More generally, a bilinear map $b : U \times V \to W$ satisfying $b(u \cdot a, v) = b(u, a \cdot v)$ for all $a \in A$, $u \in U, v \in V$ is said to be A-balanced; we can think of tensoring over A as an A-balanced $\Bbbk$-linear tensor product - as the following universal property justifies.

Lemma 2.2 (Basic properties of tensor products). *Let A be any $\Bbbk$-algebra, M a right A-module, and N a left A-module.*

(1) *(Universal property) For a vector space W and a bilinear map $b : M \times N \to W$ satisfying $b(ma, n) = b(m, an)$ for all $m \in M$, $a \in A$, and $n \in N$, there is a unique linear factorisation (commutative diagram)*



(2) *If B, C are $\Bbbk$-algebras and ${}_BM_A$, ${}_AN_C$ are bimodules, then $M \otimes_A N$ is a bimodule ${}_B(M \otimes_A N)_C$ with*

$$b(m \otimes_A n)c = bm \otimes_A nc \qquad (b \in B,\ c \in C,\ m \in M,\ n \in N).$$

For right A-module maps $u : M \to M'$ and left A-module maps $v : N \to N'$, the assignment $(m, n) \mapsto u(m) \otimes v(n)$ induces

$$u \otimes_A v : M \otimes_A N \longrightarrow M' \otimes_A N', \qquad m \otimes n \longmapsto u(m) \otimes v(n).$$

These define functors

$$(-) \otimes_A N : \operatorname{Mod} A \longrightarrow \operatorname{Mod} C, \qquad M \otimes_A (-) : \operatorname{Mod}(A^{\mathrm{op}}) \longrightarrow \operatorname{Mod}(B^{\mathrm{op}}).$$

On bimodule maps, they retain the outer actions:

$$(-) \otimes_A N : B\operatorname{Mod} A \longrightarrow B\operatorname{Mod} C, \qquad M \otimes_A (-) : A\operatorname{Mod} C \longrightarrow B\operatorname{Mod} C.$$

(3) *For $\Bbbk$-algebras A, B, a right A-module U, a bimodule ${}_AV_B$, and a left B-module W, there is a natural linear isomorphism*

$$(U \otimes_A V) \otimes_B W \xrightarrow{\sim} U \otimes_A (V \otimes_B W), \qquad (u \otimes v) \otimes w \longmapsto u \otimes (v \otimes w).$$

There are also natural module isomorphisms

$$M \otimes_A A \xrightarrow{\sim} M, \quad m \otimes a \longmapsto ma, \qquad A \otimes_A N \xrightarrow{\sim} N, \quad a \otimes n \longmapsto an.$$

Moreover, these maps are linear with respect to any additional bimodule action on U, W, M, N.

(4) *Tensor products distribute over direct sums in either variable. For arbitrary families $(M_i)_{i\in I}$ of right A-modules and $(N_j)_{j\in J}$ of left A-modules, indexed by sets I, J,*

$$\Big(\bigoplus_{i\in I} M_i\Big)\otimes_A N \xrightarrow{\sim} \bigoplus_{i\in I}(M_i\otimes_A N), \quad (m_i)_{i\in I}\otimes n \longmapsto (m_i\otimes n)_{i\in I},$$
$$M\otimes_A \Big(\bigoplus_{j\in J} N_j\Big) \xrightarrow{\sim} \bigoplus_{j\in J}(M\otimes_A N_j), \quad m\otimes (n_j)_{j\in J} \longmapsto (m\otimes n_j)_{j\in J}.$$

Proof For (1), Proposition 1.37 gives a unique linear map $M\otimes_{\Bbbk} N\to W$, $m\otimes n\mapsto b(m,n)$. The map vanishes on the defining subspace precisely when $b(ma,n)=b(m,an)$.

For (2), the outer actions preserve that subspace: for $b\in B$, $c\in C$, and $a\in A$,

$$b(ma\otimes n - m\otimes an)c = (bm)a\otimes nc - bm\otimes a(nc).$$

The module axioms follow from those on M and N. For module maps u, v,

$$u(ma)\otimes v(n) = u(m)a\otimes v(n) = u(m)\otimes av(n) = u(m)\otimes v(an),$$

so the displayed formula induces $u\otimes_A v$. The functor identities follow from $(u'\otimes v')(u(m)\otimes v(n)) = u'u(m)\otimes v'v(n)$ and $(\mathrm{id}_M\otimes \mathrm{id}_N)(m\otimes n) = m\otimes n$. If u, v preserve the outer actions, then $u\otimes_A v$ does as well.

For (3), both parenthesisations impose

$$(ua)\otimes v\otimes w = u\otimes (av)\otimes w, \qquad u\otimes (vb)\otimes w = u\otimes v\otimes (bw) \quad (a\in A,\ b\in B).$$

Thus the displayed associativity map and the reverse assignment are well-defined and inverse. The unit inverses are $M\to M\otimes_A A$, $m\mapsto m\otimes 1$, and $N\to A\otimes_A N$, $n\mapsto 1\otimes n$.

For (4), the bilinear map $((m_i)_{i\in I}, n)\mapsto (m_i\otimes n)_{i\in I}$ satisfies the identity in (1), so it induces the displayed linear map. For each $i\in I$, let $\iota_i : M_i\to \bigoplus_{j\in I} M_j$ be the inclusion. The inverse sends $m\otimes n$ in the ith summand to $\iota_i(m)\otimes n$ and extends over the direct sum. Both composites fix every element $m\otimes n$ from a summand. The other variable is handled in the same way. □

Lemma 2.3 (Hom and finite direct sums). *For finite sets I, J and right A-modules M_j, N_i, there is a natural isomorphism*

$$\begin{aligned}\mathrm{Hom}_A\Big(\bigoplus_{j\in J} M_j, \bigoplus_{i\in I} N_i\Big) &\xrightarrow{\sim} \bigoplus_{\substack{i\in I\\ j\in J}} \mathrm{Hom}_A(M_j, N_i),\\ f &\longmapsto (\pi_i f \iota_j)_{i\in I,\ j\in J},\end{aligned} \tag{2.1}$$

where $\iota_j : M_j\to \bigoplus_{j'\in J} M_{j'}$ and $\pi_i : \bigoplus_{i'\in I} N_{i'}\to N_i$ are the inclusions and projections.

Proof Write $\pi'_j : \bigoplus_{j'\in J} M_{j'}\to M_j$ and $\iota'_i : N_i\to \bigoplus_{i'\in I} N_{i'}$ for the remaining projections and inclusions. The inverse sends $(f_{ij})_{i\in I,\ j\in J}$ to $\sum_{i\in I}\sum_{j\in J}\iota'_i f_{ij}\pi'_j$. Both composites are identities because $\sum_{j\in J}\iota_j\pi'_j$ and $\sum_{i\in I}\iota'_i\pi_i$ are identity maps, while $\pi'_j\iota_{j'} = \delta_{jj'}\,\mathrm{id}_{M_j}$ and $\pi_i\iota'_{i'} = \delta_{ii'}\,\mathrm{id}_{N_i}$. The same formulas commute with pre- and postcomposition, proving naturality. □

Lemma 2.4 (Hom, direct sums, and direct products). *Let M, N be right A-modules, and let $(M_i)_{i\in I}$ and $(N_i)_{i\in I}$ be arbitrary families of right A-modules, indexed by a set I. Write $\iota_i : M_i \to \bigoplus_{j\in I} M_j$ for the inclusions and $\pi_i : \prod_{j\in I} N_j \to N_i$ for the projections. There are natural linear isomorphisms*

$$\operatorname{Hom}_A\Big(\bigoplus_{i\in I} M_i, N\Big) \xrightarrow{\sim} \prod_{i\in I} \operatorname{Hom}_A(M_i, N), \quad f \longmapsto (f\circ\iota_i)_{i\in I},$$
$$\operatorname{Hom}_A\Big(M, \prod_{i\in I} N_i\Big) \xrightarrow{\sim} \prod_{i\in I} \operatorname{Hom}_A(M, N_i), \quad f \longmapsto (\pi_i\circ f)_{i\in I}.$$

There is also a natural injection

$$\bigoplus_{i\in I} \operatorname{Hom}_A(M, N_i) \longrightarrow \operatorname{Hom}_A\Big(M, \bigoplus_{i\in I} N_i\Big), \qquad (f_i)_{i\in I} \longmapsto \big(m \mapsto (f_i(m))_{i\in I}\big),$$

which is an isomorphism when M is finitely generated.

Proof The inverse of the first map sends $(f_i)_{i\in I}$ to $(m_i)_{i\in I} \mapsto \sum_{i\in I} f_i(m_i)$; the sum is finite because $(m_i)_{i\in I}$ has finite support. The inverse of the second sends $(f_i)_{i\in I}$ to $m \mapsto (f_i(m))_{i\in I}$; a direct product imposes no support condition. For the third map, coordinate projections recover each f_i, proving injectivity. If $m_1, \dots, m_r$ generate M, the images of these generators under $f : M \to \bigoplus_{i\in I} N_i$ are supported on a common finite subset $J \subset I$. Then $f(M) \subset \bigoplus_{i\in J} N_i$, so f comes from a finite family of maps. All three constructions commute with precomposition and postcomposition. □

Proposition 2.5 (Tensor products and direct products). *Let A be a $\Bbbk$-algebra, M, M_i right A-modules, and N, N_i left A-modules, with $i \in I$. The coordinate projections define linear maps*

$$\begin{aligned} \kappa_N &: \Big(\prod_i M_i\Big) \otimes_A N \longrightarrow \prod_i (M_i \otimes_A N), \quad (m_i)_i \otimes n \longmapsto (m_i \otimes n)_i, \\ \lambda_M &: M \otimes_A \Big(\prod_i N_i\Big) \longrightarrow \prod_i (M \otimes_A N_i), \quad m \otimes (n_i)_i \longmapsto (m \otimes n_i)_i. \end{aligned} \tag{2.2}$$

(1) *If I is finite, both maps are isomorphisms.*

(2) *The map κ_N is an isomorphism if N is a direct summand of $A^{\oplus r}$ as a left module for some finite r. The corresponding condition on the right module M makes λ_M an isomorphism.*

(3) *If A is finite-dimensional semisimple, both maps are injective. Moreover, κ_N is an isomorphism when N is finite-dimensional, and λ_M is an isomorphism when M is finite-dimensional.*

Proof The formulas in (2.2) are bilinear. For $s \in A$, the component identities $(m_i s) \otimes n = m_i \otimes sn$ and $(ms) \otimes n_i = m \otimes sn_i$ show that both formulas give well-defined maps from the indicated tensor products. For finite I, products equal sums, and Lemma 2.2(4) proves (1).

For $N = A$, the map κ_A is the identity on $\prod_i M_i$ under $m \otimes s \mapsto ms$. For $N = A^{\oplus r}$, both sides are $(\prod_i M_i)^{\oplus r}$, since a finite direct sum commutes with a direct product. If $N \oplus N' \cong A^{\oplus r}$, the comparison for $N \oplus N'$ is $\kappa_N \oplus \kappa_{N'}$. Its bijectivity implies the bijectivity of κ_N. This proves the assertion about κ_N in (2).

Now let A be finite-dimensional semisimple. For finite-dimensional N, Corollary 1.13 gives (2). For arbitrary N, take $z = \sum_{\ell=1}^{r} m^{(\ell)} \otimes n_\ell$ in $\operatorname{Ker}\kappa_N$ and set $N_0 = \sum_{\ell=1}^{r} An_\ell$. The submodule

N_0 is finite-dimensional and has a complement in N by Theorem 1.12. Thus the inclusions

$$\Big(\prod_i M_i\Big) \otimes_A N_0 \longrightarrow \Big(\prod_i M_i\Big) \otimes_A N, \qquad M_i \otimes_A N_0 \longrightarrow M_i \otimes_A N$$

are injective, by the direct sum formula in Lemma 2.2(4). Regard z as an element with second factors in N_0. The equality $\kappa_N(z) = 0$ then gives $\kappa_{N_0}(z) = 0$. Aince κ_{N_0} is an isomorphism, $z = 0$.

For λ_M, apply the same arguments over A^{op}, using $N \otimes_{A^{\mathrm{op}}} M \cong M \otimes_A N$, $n \otimes m \mapsto m \otimes n$. □

The finite-dimensional hypothesis on the fixed factor cannot be dropped from the isomorphism assertion: Example 1.39 gives a non-surjective comparison even for vector spaces. Thus tensor products always distribute over direct sums, but not over arbitrary direct products.

Remark 2.6. For infinite families, Hom need not preserve direct sums in its second variable. For $M = \bigoplus_{i\geq 1} \Bbbk v_i$, the identity $M \to M$ does not come from $\bigoplus_i \mathrm{Hom}_{\Bbbk}(M, \Bbbk v_i)$: its projection onto every $\Bbbk v_i$ is non-zero. Thus the injection in Lemma 2.4 need not be surjective.

A product in the source of Hom behaves differently from a product in the target. For example, a linear map $\prod_{i\geq 1} \Bbbk \to \Bbbk$ need not be determined by its restrictions to the coordinate copies of $\Bbbk$. Indeed, $(1, 1, \ldots) \notin \bigoplus_{i\geq 1} \Bbbk$. Define a functional on $\bigoplus_{i\geq 1} \Bbbk + \Bbbk(1, 1, \ldots)$ to vanish on the direct sum and take value 1 at $(1, 1, \ldots)$, and extend it to the product by extending a basis. The resulting non-zero functional vanishes on every coordinate copy.

For a bimodule ${}_AV_A$, define

$$V^{\otimes_A 0} := A, \qquad V^{\otimes_A i} := \underbrace{V \otimes_A \cdots \otimes_A V}_{i \text{ factors}} \quad (i \geq 1).$$

Note that there is no parenthesis in the definition of $V^{\otimes_A i}$ since Lemma 2.2(3) says that tenso product is an associatitive operation.

Let A, B, C be $\Bbbk$-algebras. For bimodules ${}_AM_B$, ${}_CX_B$, we have $\mathrm{Hom}_B(M, X) \in C\,\mathrm{Mod}\,A$, with

$$(cfa)(m) = cf(am) \quad (f \in \mathrm{Hom}_B(M, X),\ c \in C,\ a \in A,\ m \in M). \tag{2.3}$$

If $X \in \mathrm{Mod}\,B$ has no specified left action, the same formula defines the right A-action $(fa)(m) = f(am)$. The left-module version follows by passing to opposite algebras.

Proposition 2.7 (Tensor and Hom). *Consider bimodules ${}_AM_B$, ${}_BN_C$, a right C-module X, and a left A-module Y. There are maps, natural in M, N,*

$$\begin{aligned} \theta_r : \mathrm{Hom}_C(N, X) \otimes_B \mathrm{Hom}_B(M, B) &\longrightarrow \mathrm{Hom}_C(M \otimes_B N, X), \\ \theta_r(g \otimes f)(m \otimes n) &= g(f(m)n), \\ \theta_\ell : \mathrm{Hom}_{B^{\mathrm{op}}}(N, B) \otimes_B \mathrm{Hom}_{A^{\mathrm{op}}}(M, Y) &\longrightarrow \mathrm{Hom}_{A^{\mathrm{op}}}(M \otimes_B N, Y), \\ \theta_\ell(g \otimes f)(m \otimes n) &= f(mg(n)). \end{aligned} \tag{2.4}$$

The maps θ_r and θ_ℓ are right A-linear and left C-linear, respectively.

(1) *The map θ_r is an isomorphism if $M \in \mathrm{add}\,B$ as a right B-module.*

(2) *The map θ_ℓ is an isomorphism if ${}_BN \in \mathrm{add}_{\mathrm{mod}\,B^{\mathrm{op}}}(B)$.*

Here $\mathrm{add}\,B = \mathrm{add}_{\mathrm{mod}\,B}(B)$, as in Example 1.5. If ${}_CX_C$ and ${}_AY_A$ are bimodules, the respective maps also preserve the additional actions and are morphisms in $C\,\mathrm{Mod}\,A$.

Proof *Well-definedness of θ_r.* For $f \in \mathrm{Hom}_B(M,B)$, $g \in \mathrm{Hom}_C(N,X)$, and $b \in B$,

$$g(f(mb)n) = g(f(m)bn), \qquad (gb)(f(m)n) = g(bf(m)n) = g((bf)(m)n).$$

These identities respect the defining identifications in both tensor products. Right C-linearity follows from $g(f(m)nc) = g(f(m)n)c$ for $c \in C$. For $a \in A$,

$$\theta_r(g \otimes fa)(m \otimes n) = g(f(am)n) = (\theta_r(g \otimes f)a)(m \otimes n).$$

If X also has a commuting left C-action, replacing g by cg multiplies the value by c, so θ_r preserves that action as well.

Bijectivity of θ_r. Regard M only as a right B-module. For $M = B$, evaluation at 1 identifies $\mathrm{Hom}_B(B,B) \cong B$, while multiplication identifies $B \otimes_B N \cong N$. Under these isomorphisms, θ_r becomes

$$\mathrm{Hom}_C(N,X) \otimes_B B \longrightarrow \mathrm{Hom}_C(N,X), \qquad g \otimes b \longmapsto gb,$$

which is an isomorphism by Lemma 2.2(3). Finite direct sums commute with both constructions by Lemma 2.2(4) and (2.1). Thus θ_r is bijective for $M = B^{\oplus r}$, where $r \geq 0$ is finite. If $M \in \mathrm{add}\, B$, choose M' with $M \oplus M' \cong B^{\oplus r}$. The comparison for $M \oplus M'$ is the direct sum of the comparisons for M and M', so both summand maps are bijective.

The left comparison. For $f \in \mathrm{Hom}_{A^{\mathrm{op}}}(M,Y)$ and $g \in \mathrm{Hom}_{B^{\mathrm{op}}}(N,B)$, well-definedness follows from

$$f(mb\, g(n)) = f(mg(bn)), \qquad f(m(gb)(n)) = f(mg(n)b) = (bf)(mg(n)) \quad (b \in B).$$

Left A-linearity of f makes the resulting functional left A-linear. For $c \in C$,

$$\theta_\ell(cg \otimes f)(m \otimes n) = f(mg(nc)) = (c\theta_\ell(g \otimes f))(m \otimes n).$$

If Y also has a commuting right A-action, replacing f by fa multiplies the value by a, so θ_ℓ preserves that action as well. For $N = B$ as a left module, evaluation at 1 identifies $\mathrm{Hom}_{B^{\mathrm{op}}}(B,B) \cong B$ and turns θ_ℓ into

$$B \otimes_B \mathrm{Hom}_{A^{\mathrm{op}}}(M,Y) \longrightarrow \mathrm{Hom}_{A^{\mathrm{op}}}(M,Y), \qquad b \otimes f \longmapsto bf.$$

The same finite-sum and direct-summand argument proves (2).

Naturality. Let $p : M \to M'$ and $q : N \to N'$ be bimodule maps. For $f \in \mathrm{Hom}_B(M',B)$ and $g \in \mathrm{Hom}_C(N',X)$, applying θ_r before or after precomposition gives $g(f(p(m))q(n))$ at $m \otimes n$. For $f \in \mathrm{Hom}_{A^{\mathrm{op}}}(M',Y)$ and $g \in \mathrm{Hom}_{B^{\mathrm{op}}}(N',B)$, the two orders give $f(p(m)g(q(n)))$ for θ_ℓ. Thus both comparisons are natural. □

For an arbitrary finite-dimensional semisimple S, the one-sided S-duals satisfy an analogue of Proposition 3.3 without any choice. Consider bimodules ${}_SV_S$, ${}_SW_S$ finitely generated on the right and left, respectively. Exercise 2.39 shows, using only Corollary 1.13, that there are natural bimodule isomorphisms

$$\theta_r : W^* \otimes_S V^* \xrightarrow{\sim} (V \otimes_S W)^*, \qquad \theta_\ell : {}^*W \otimes_S {}^*V \xrightarrow{\sim} {}^*(V \otimes_S W), \tag{2.5}$$

$$\theta_r(g \otimes f)(v \otimes w) = g(f(v)w), \qquad \theta_\ell(g \otimes f)(v \otimes w) = f(vg(w)). \tag{2.6}$$

Example 2.8. *Finite generation alone does not suffice in Proposition 2.7 over a general algebra. Take $A = C = \Bbbk$, $B = \Bbbk[x]$, and $X = Y = \Bbbk$. Consider bimodules ${}_\Bbbk M_B$, ${}_B N_\Bbbk$ with $M = N =$*

$B/(x) \cong \Bbbk$ *and the outer* $\Bbbk$*-actions given by scalar multiplication. Every* $f \in \mathrm{Hom}_B(M,B)$ *satisfies* $f(\overline{1})x = f(\overline{1}x) = 0$*, hence* $f = 0$*. Consequently*

$$\mathrm{Hom}_{\Bbbk}(N,\Bbbk) \otimes_B \mathrm{Hom}_B(M,B) = 0, \qquad \mathrm{Hom}_{\Bbbk}(M \otimes_B N, \Bbbk) \cong \Bbbk.$$

Both M_B *and* ${}_BN$ *are generated by* $\overline{1}$*, so neither direct-summand hypothesis can be replaced by finite generation alone. For the left comparison,* $\mathrm{Hom}_{B^{\mathrm{op}}}(N,B) = 0$*, whereas* $\mathrm{Hom}_{\Bbbk}(M \otimes_B N, \Bbbk) \cong \Bbbk$*.*

For S, V, W as in (2.5), take $A = B = S$, $C = \Bbbk$, $M = V$, $N = W$, and $X = \Bbbk$ in Proposition 2.7(1). Since $\mathrm{Hom}_{\Bbbk}(W,\Bbbk) = D(W)$, we obtain the natural bimodule isomorphism [GI20, §2.1]

$$\begin{aligned} &\theta_D : D(W) \otimes_S V^* \xrightarrow{\sim} D(V \otimes_S W), \\ &\theta_D(\lambda \otimes f)(v \otimes w) = \lambda(f(v)w). \end{aligned}$$

The actions on $D(W)$ are those in (1.17). For $s \in S$, naturality with respect to $W \to W$, $w \mapsto ws$, gives left S-linearity of θ_D.

Suppose now that S is equipped with a symmetrising form $\tau : S \to \Bbbk$, as in Definition 1.55. The two-sided dual of Definition 1.54 satisfies

$$\begin{aligned} &\theta_{\mathrm{en}} : W^* \otimes_S V^{*\mathrm{en}} \xrightarrow{\sim} (V \otimes_S W)^{*\mathrm{en}}, \\ &\theta_{\mathrm{en}}(g \otimes F)(v \otimes w) = \sum_j g(a_j w) \otimes b_j^{\mathrm{op}}, \qquad F(v) = \sum_j a_j \otimes b_j^{\mathrm{op}}. \end{aligned} \tag{2.7}$$

We check that the formula defines a map on the two indicated tensor products and preserves both outer actions. For example, $F(vs) = F(v)(s \otimes 1)$ replaces a_j by $a_j s$, while $(Fs)(v) = (1 \otimes s^{\mathrm{op}})F(v)$ replaces b_j by $b_j s$. These give, respectively, compatibility with $vs \otimes w = v \otimes sw$ and with the right outer action. Compatibility with $(gs) \otimes F = g \otimes (sF)$ follows from $(gs)(a_j w) = g(sa_j w)$. The other actions follow from right S-linearity of g and $F(tv) = F(v)(1 \otimes t^{\mathrm{op}})$.

The comparison maps ω_M in (1.25) give

$$\omega_{V \otimes_S W} \circ \theta_{\mathrm{en}} = \theta_r \circ (\mathrm{id}_{W^*} \otimes \omega_V),$$

since both sides send $g \otimes F$, evaluated at $v \otimes w$, to $\sum_j g(a_j w)\tau(b_j)$. Thus θ_{en} is an isomorphism by Theorem 1.57. Precomposition shows that θ_D and θ_{en} are natural. No assumption that S^{en} is semisimple is used; compare Remark 1.43.

For the maps $\alpha_M(f) = \tau \circ f$ in (1.20),

$$\alpha_{V \otimes_S W} \circ \theta_r = \theta_D \circ (\alpha_W \otimes \mathrm{id}_{V^*}),$$

because both sides send $g \otimes f$ to the functional $v \otimes w \mapsto \tau(g(f(v)w))$.

For an arbitrary finite-dimensional semisimple S, the one-sided S-duals also describe Hom spaces and give evaluation and coevaluation maps [BGS96, §2.7].

Lemma 2.9. *Let* S *be finite-dimensional semisimple and* ${}_SV_S$ *a finite-dimensional bimodule. There are bimodule isomorphisms*

$$\begin{aligned} &V \otimes_S V^* \xrightarrow{\sim} \mathrm{End}_S(V), \qquad v \otimes f \longmapsto \big(u \mapsto vf(u)\big), \\ &{}^*V \otimes_S V \xrightarrow{\sim} \mathrm{End}_{S^{\mathrm{op}}}(V), \quad g \otimes v \longmapsto \big(u \mapsto g(u)v\big). \end{aligned} \tag{2.8}$$

The source actions are those of Lemma 2.2(2). The target actions are defined, for $s,t \in S$ and $u \in V$, by

$$\begin{aligned}(sFt)(u) &= sF(tu) \quad \text{for } F \in \mathrm{End}_S(V),\\ (sGt)(u) &= G(us)t \quad \text{for } G \in \mathrm{End}_{S^{\mathrm{op}}}(V).\end{aligned} \tag{2.9}$$

Proof For the first comparison, $(vs)f(u) = v(sf)(u)$ and $vf(us) = vf(u)s$ give a well-defined map to $\mathrm{End}_S(V)$. By Corollary 1.13, identify V_S with a direct summand of $S^{\oplus m}$ for some integer $m \geq 0$, and let $p : S^{\oplus m} \to V$ be the right S-linear projection. For $1 \leq j \leq m$, let $\varepsilon_j \in S^{\oplus m}$ be the vector with 1 in coordinate j and zero elsewhere, and let $q_j : S^{\oplus m} \to S$ be the coordinate projection. Put $v_j = p(\varepsilon_j)$ and $f_j = q_j|_V$. Then $u = \sum_{j=1}^m v_j f_j(u)$ for every $u \in V$. The inverse is

$$\mathrm{End}_S(V) \longrightarrow V \otimes_S V^*, \qquad F \longmapsto \sum_{j=1}^m F(v_j) \otimes f_j.$$

Indeed, one composite sends F to $u \mapsto \sum_{j=1}^m F(v_j)f_j(u) = F(u)$, and the other sends

$$v \otimes f \longmapsto \sum_{j=1}^m vf(v_j) \otimes f_j = v \otimes \sum_{j=1}^m f(v_j)f_j = v \otimes f.$$

Applying this construction over S^{op} proves the second isomorphism, using $V \otimes_{S^{\mathrm{op}}} {}^*V \cong {}^*V \otimes_S V$, $v \otimes g \mapsto g \otimes v$.

The commuting left and right actions on V make (2.9) bimodule actions. For example, $(sFt)(ur) = sF(tur) = sF(tu)r$ for $r \in S$, so sFt is right S-linear. The images of $sv \otimes ft$ and $sg \otimes vt$, evaluated at u, are $svf(tu)$ and $g(us)vt$, respectively. These are exactly the actions in (2.9), proving bimodule linearity of both comparisons. □

Evaluation applies a functional to an element. Coevaluation expresses the identity endomorphism as an element of a tensor product.

Proposition 2.10. *For a bimodule ${}_SV_S$, where S is finite-dimensional semisimple, evaluation gives bimodule maps*

$$\begin{aligned}V^* \otimes_S V &\longrightarrow S, \quad f \otimes v \longmapsto f(v),\\ V \otimes_S {}^*V &\longrightarrow S, \quad v \otimes g \longmapsto g(v).\end{aligned} \tag{2.10}$$

If V is finite-dimensional, the coevaluation maps are

$$\begin{aligned}S &\longrightarrow V \otimes_S V^* \xrightarrow{\sim} \mathrm{End}_S(V), \quad s \longmapsto \big(v \mapsto sv\big),\\ S &\longrightarrow {}^*V \otimes_S V \xrightarrow{\sim} \mathrm{End}_{S^{\mathrm{op}}}(V), \quad s \longmapsto \big(v \mapsto vs\big).\end{aligned} \tag{2.11}$$

The isomorphisms are those in (2.8)*. Both coevaluation maps are bimodule maps, and each image of* 1 *corresponds to* id_V*.*

Proof The identities $(fs)(v) = f(sv)$ and $(sg)(v) = g(vs)$ from (1.17) make evaluation well-defined on the two tensor products; the other two actions give its bimodule linearity. For coevaluation, left multiplication is right S-linear and right multiplication is left S-linear because the two actions on V commute. Under the endomorphism actions in (2.9), both multiplication assignments preserve the two S-actions. The isomorphisms in (2.8) therefore give the stated bimodule maps. □

For a finite-dimensional graded bimodule ${}_SV_S$, use $V^{(*)}$ and ${}^{(*)}V$ in Proposition 2.10. With the total grading of Definition 2.35, evaluation and coevaluation preserve degree: evaluation pairs degrees i and $-i$, and id_V has degree 0.

The evaluation and coevaluation maps in Proposition 2.10 are the constructions in the last two paragraphs of [BGS96, §2.7, pp. 484–485], translated to our notation.

For a locally finite-dimensional graded bimodule ${}_SV_S$, (2.11) applies to each V_d; a coevaluation for the whole module need not exist. For example, take $S \neq 0$ and $V = \bigoplus_{d\geq 0} S(-d)$. Every element $\sum_{i=1}^{n} v_i \otimes f_i$ of $V \otimes_S V^{(*)}$ defines an endomorphism with image contained in the finite-dimensional submodule $\sum_{i=1}^{n} v_i S$, so it cannot represent id_V.

2.2 Generated algebras

Definition 2.11. *Let A be graded with $S = A_0$. Multiplication makes A_1 an S-bimodule and gives*

$$\mu_i : A_1^{\otimes_S i} \longrightarrow A_i, \qquad a_1 \otimes \cdots \otimes a_i \longmapsto a_1 \cdots a_i \quad (i \geq 1), \qquad \mu_0 = \mathrm{id}_S\,. \tag{2.12}$$

The algebra A is generated in degree 1 (over S) if every μ_i in (2.12) is surjective.

Equivalently, S and A_1 generate A as an algebra.

Example 2.12. *The algebra $\Bbbk A_n$ of Example 1.19 is generated in degree 1 over $S = \bigoplus_i \Bbbk e_{ii}$.*

Example 2.13. *The polynomial algebra $\Bbbk[x_1, \ldots, x_n]$, with each $\deg x_i = 1$ for all i, is generated in degree 1 over $\Bbbk$. Indeed, every monomial of total degree d is a product of d variables, and these monomials span the degree d component.*

Example 2.14. *Let*

$$A = \begin{pmatrix} \Bbbk[x] & \Bbbk[x] \\ 0 & \Bbbk[x] \end{pmatrix}, \qquad \deg(x^m e_{ij}) = m + j - i \quad (m \geq 0,\ 1 \leq i \leq j \leq 2).$$

Then

$$A_0 = \Bbbk e_{11} \oplus \Bbbk e_{22}, \qquad A_d = \Bbbk x^d e_{11} \oplus \Bbbk x^d e_{22} \oplus \Bbbk x^{d-1} e_{12} \quad (d \geq 1).$$

It is generated in degree 1 over A_0, since

$$x^d e_{11} = (x e_{11})^d, \qquad x^d e_{22} = (x e_{22})^d, \qquad x^{d-1} e_{12} = (x e_{11})^{d-1} e_{12} \quad (d \geq 1).$$

Here A_0 has two simple summands, $A_d \neq 0$ for every $d \geq 0$, and A is non-commutative: $e_{11}e_{12} = e_{12}$ whereas $e_{12}e_{11} = 0$.

2.3 Tensor algebras and the universal property

Definition 2.15. *Let S be a $\Bbbk$-algebra and consider a bimodule ${}_SV_S$. The tensor algebra of V over S is the graded $\Bbbk$-algebra*

$$T_S(V) := \bigoplus_{i\in\mathbb{Z}} T_S(V)_i, \quad \text{where} \quad T_S(V)_i = \begin{cases} V^{\otimes_S i}, & i \geq 1, \\ S, & i = 0, \\ 0, & i < 0, \end{cases}.$$

For $v_1, \ldots, v_{p+q} \in V$, multiplication is determined $\Bbbk$-bilinearly by

$$(v_1 \otimes_S \cdots \otimes_S v_p)(v_{p+1} \otimes_S \cdots \otimes_S v_{p+q}) = v_1 \otimes_S \cdots \otimes_S v_{p+q} \qquad \text{for all } p, q \geq 1,$$

and by the left and right S-actions when a factor has degree 0.

The displayed decomposition is the *tensor grading*: $S = T_S(V)_0$ and $V = T_S(V)_1$. Multiplication satisfies $T_S(V)_p T_S(V)_q \subset T_S(V)_{p+q}$ for all $p, q \in \mathbb{Z}$. We also call $T_S(V)$ the tensor algebra generated by V over S. Every element has finitely many non-zero homogeneous components. The unit is 1_S, and associativity follows from Lemma 2.2(3). Write $v_1 \cdots v_d$ for $v_1 \otimes_S \cdots \otimes_S v_d$ in $T_S(V)$. In a quotient algebra (including $T_S(V)$ itself) of $T_S(V)$, the notation $v_1 \cdots v_d$ denotes the image of $v_1 \otimes_S \cdots \otimes_S v_d$.

Example 2.16 (Polynomial and exterior algebras). *Let V, W be finite-dimensional vector spaces with bases $\{x_i \mid 1 \le i \le n\}$ and $\{y_j \mid 1 \le j \le m\}$, respectively. The symmetric algebra of V and the exterior algebra of W are*

$$\begin{aligned} \mathsf{S}(V) &= T_{\Bbbk}(V)/(x_i x_j - x_j x_i \mid 1 \le i < j \le n) \cong \Bbbk[x_1, \dots, x_n], \\ \bigwedge W &= T_{\Bbbk}(W)/(w^2 \mid w \in W) = T_{\Bbbk}(W)/(y_j^2, y_i y_j + y_j y_i \mid 1 \le i < j \le m). \end{aligned} \tag{2.13}$$

They are generated in degree 1, with

$$\mathsf{S}(V)_0 = \Bbbk = (\textstyle\bigwedge W)_0, \qquad \mathsf{S}(V)_1 = V, \qquad (\textstyle\bigwedge W)_1 = W.$$

Their respective bases are

$$\begin{aligned} &\{x_1^{a_1} \cdots x_n^{a_n} \mid a_1, \dots, a_n \in \mathbb{Z},\ a_1, \dots, a_n \ge 0\}, \\ &\{y_{j_1} \cdots y_{j_d} \mid 0 \le d \le m,\ 1 \le j_1 < \cdots < j_d \le m\}. \end{aligned}$$

It is customary to write $\mathsf{S}^d(V) := \mathsf{S}(V)_d$ and $\bigwedge^d(W) := (\bigwedge(W))_d$.

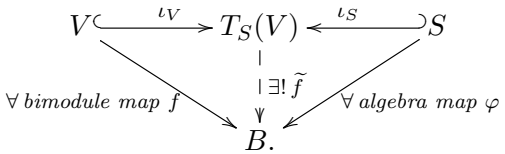
The bases in Example 2.16 can be checked directly. For $\mathsf{S}(V)$, commuting adjacent generators gives the ordered monomials, which are independent under the map to the polynomial ring. For $\bigwedge W$, the relations reduce every product of basis elements to a multiple of a strictly increasing product or to zero. For $1 \le d \le m$, fix $j_1 < \cdots < j_d$ and consider the linear functional $W^{\otimes_{\Bbbk} d} \to \Bbbk$ sending $w_1 \otimes \cdots \otimes w_d$ to the determinant of their coordinates in rows $j_1, \dots, j_d$ relative to the basis $\{y_j \mid 1 \le j \le m\}$. The determinant vanishes whenever two adjacent factors agree, hence on the degree d component of $(w^2 \mid w \in W)$. On increasing products it is 1 on $y_{j_1} \cdots y_{j_d}$ and 0 on the others, proving independence in every characteristic.

Example 2.17. *We apply Lemma 3.1 to realise $\Bbbk A_3$ as a tensor algebra. Let $S = \Bbbk^{\oplus 3}$ with coordinate idempotents e_1, e_2, e_3, and take the bimodule ${}_S V_S$ with $V = \Bbbk a \oplus \Bbbk b$, $e_1 a e_2 = a$, and $e_2 b e_3 = b$; all other idempotent actions are zero. By Lemma 3.1,*

$$V \otimes_S V = \Bbbk(a \otimes b), \qquad V^{\otimes_S 3} = 0, \qquad T_S(V) = S \oplus \Bbbk a \oplus \Bbbk b \oplus \Bbbk ab.$$

Here $ab = a \otimes b$. The linear map $T_S(V) \to \Bbbk A_3$ given by $e_i \mapsto e_{ii}$, $a \mapsto e_{12}$, $b \mapsto e_{23}$, and $ab \mapsto e_{13}$ preserves multiplication and sends a basis to a basis, hence is an algebra isomorphism.

Proposition 2.18 (Universal property of the tensor algebra). *Let S be a $\Bbbk$-algebra, consider a bimodule ${}_S V_S$, and let $\iota_S : S \to T_S(V)$ and $\iota_V : V \to T_S(V)$ be the canonical inclusions. Given a unital $\Bbbk$-algebra homomorphism $\varphi : S \to B$ and a bimodule map $f : {}_S V_S \to {}_S B_S$, where $sbt = \varphi(s) b \varphi(t)$ on B, there is a unique $\Bbbk$-algebra homomorphism $\widetilde{f} : T_S(V) \to B$ making the diagram commute:*



(2.14)

Proof Define $\widetilde{f}$ on S and each tensor power by

$$\widetilde{f}(s) = \varphi(s), \qquad \widetilde{f}(v_1 \otimes_S \cdots \otimes_S v_i) = f(v_1)\cdots f(v_i) \quad (s \in S,\ i \geq 1). \tag{2.15}$$

The identity $f(vs)f(w) = f(v)\varphi(s)f(w) = f(v)f(sw)$ shows that replacing adjacent factors $vs \otimes w$ by $v \otimes sw$ does not change the value in (2.15). The formulas therefore induce linear maps on the tensor powers, which extend to their direct sum. Multiplication is preserved by (2.15), including the degree 0 factors because $f(sv) = \varphi(s)f(v)$ and $f(vs) = f(v)\varphi(s)$. Also, $\widetilde{f}(1_S) = 1_B$. Every algebra homomorphism making (2.14) commute must have the values in (2.15), proving uniqueness. □

If B is graded, $\varphi(S) \subset B_0$, and $f(V) \subset B_1$, then (2.15) gives $\widetilde{f}(V^{\otimes_S i}) \subset B_i$ for every $i \geq 0$. Thus $\widetilde{f}$ is a graded algebra homomorphism.

Corollary 2.19. *If A is generated in degree 1 over $S = A_0$, then the multiplication map defines a surjective graded algebra homomorphism*

$$\mu : T_S(A_1) \twoheadrightarrow A, \qquad \mu|_S = \mathrm{id}_S, \qquad a_1 \otimes_S \cdots \otimes_S a_i \longmapsto a_1 \cdots a_i \quad \forall i \geq 1.$$

Its degree i component is μ_i from (2.12)*. The kernel $I := \operatorname{Ker}\mu$ is homogeneous, $I_0 = I_1 = 0$, and μ induces a graded algebra isomorphism*

$$T_S(A_1)/I \xrightarrow{\sim} A, \qquad z + I \longmapsto \mu(z).$$

Proof In Proposition 2.18, take $V = A_1$, $B = A$, and the inclusions

$$\varphi : S \hookrightarrow A, \quad s \longmapsto s, \qquad f : A_1 \hookrightarrow A, \quad a \longmapsto a.$$

Then (2.15) gives $\widetilde{f} = \mu$. The map μ preserves degrees, and each component μ_i is surjective by the assumption that A is generated in degree 1. Thus $\operatorname{Ker}\mu$ is homogeneous, and $\mu_0 = \mathrm{id}_S$, $\mu_1 = \mathrm{id}_{A_1}$ give $I_0 = I_1 = 0$. The map $z + I \mapsto \mu(z)$ is therefore a degree-preserving bijective algebra homomorphism. □

Definition 2.20. *For the map μ in Corollary 2.19, the homogeneous two-sided ideal $I := \operatorname{Ker}\mu$ is the ideal of relations.*

The bimodule A_1 specifies the generators and their left and right S-actions; I specifies the relations between their products. Homogeneity is essential for the quotient grading: the relation $x - x^2$ in Example 1.34 does not satisfy this condition.

2.4 Quadratic algebras

We now consider algebras whose ideal of relations is generated in degree 2. Here, having $A_0 = S$ semisimple is crucial.[3]

Definition 2.21. *A graded algebra A is quadratic if there exist a finite-dimensional semisimple $\Bbbk$-algebra S, a finite-dimensional S-bimodule V, and an S-sub-bimodule $R \subset V \otimes_S V$ such that $A \cong T_S(V)/(R)$ as graded algebras, where (R) is the two-sided ideal of $T_S(V)$ generated by R.*

An isomorphism $A \cong T_S(V)/(R)$ as in Definition 2.21 is called a *quadratic presentation* of A.

[3] You can still do a lot of similar things with more general S; [PP05] has some exposition on this.

Proposition 2.22. *A quadratic algebra $A = T_S(V)/(R)$ has $A_0 = S$, $A_1 = V$, and $A_2 = (V \otimes_S V)/R$, is locally finite-dimensional, and is generated in degree 1 over S.*

Proof The ideal (R) has no components in degrees 0 and 1. Its degree 2 component is exactly R. Each A_i is a quotient of the finite-dimensional space $V^{\otimes_S i}$, and $\mu_i : V^{\otimes_S i} \twoheadrightarrow A_i$. □

Thus a quadratic presentation recovers $S \cong A_0$, $V \cong A_1$, and $R \cong \mathrm{Ker}(A_1 \otimes_{A_0} A_1 \to A_2)$. In particular, the kernel of the canonical map $T_{A_0}(A_1) \twoheadrightarrow A$ is generated by its degree 2 component. Quadraticity depends on the grading: for $n \geq 2$, the grading $\deg e_{ij} = 2(j-i)$ on $\Bbbk A_n$ is not quadratic, since it has no degree 1 generators.

Example 2.23. *Let $S = \bigoplus_{i=1}^{n} \Bbbk e_{ii}$ and $V = \bigoplus_{i=1}^{n-1} \Bbbk e_{i,i+1}$ inside $\Bbbk A_n$. Then we have $T_S(V) \cong \Bbbk A_n$ given by*

$$e_{i,i+1} \otimes_S e_{i+1,i+2} \otimes_S \cdots \otimes_S e_{j-1,j} \longmapsto e_{ij} \quad (1 \leq i < j \leq n).$$

Example 2.24. *For S, V, a, b from Example 2.17, the quotient*

$$T_S(V)/(ab) \cong \Bbbk A_3/(e_{13})$$

is quadratic with $R = \Bbbk(a \otimes b)$. Its basis is e_1, e_2, e_3, a, b, in degrees $0, 0, 0, 1, 1$, as in Example 1.33.

Example 2.25. *Let $B = \Bbbk A_4$ and $A = B/(e_{14})$, with the grading of Example 1.19. The ideal $I = (e_{14}) = \Bbbk e_{14}$ has $I_2 = 0$ but $I_3 \neq 0$. Since $e_{14} = e_{12}e_{23}e_{34}$, the algebra A is generated in degree 1 with a relation of degree 3 and no relations of degree 2. By Proposition 2.22, any quadratic presentation would have the same generators and zero quadratic relation space, hence would give $\Bbbk A_4$ itself. Thus A is not quadratic.*

Example 2.26. *The symmetric and exterior algebras of Example 2.16 are quadratic: the defining relations in* (2.13) *belong to $V \otimes_{\Bbbk} V$ and $W \otimes_{\Bbbk} W$, respectively.*

2.5 Hilbert series

Let A be a locally finite-dimensional graded algebra with $S = A_0$ finite-dimensional semisimple. For a graded right A-module M, regard M_d as an A-module via $A \to A/A_+ = S$. The simple direct summands of the semisimple module M_d are its *composition factors*.

Notation. *Choose a complete set of pairwise orthogonal primitive idempotents $e_1, \ldots, e_n$. Then $e_1S, \ldots, e_nS$ represent each isomorphism class of simple right S-modules exactly once, and $L_j := e_jS \cong e_jA/e_jA_+$ is a graded simple A-module concentrated in degree 0 for every $1 \leq j \leq n$.*

Remark 2.27. Under Assumption 1.50, the coordinate idempotents $e_1, \ldots, e_n$ of is the full set of primitive idempotents we need.

Definition 2.28. *For a locally finite-dimensional graded right A-module M, write $[M_d : L_j]$ for the multiplicity of the composition factor L_j in M_d. Its degree d dimension vector is the row vector*[4]

$$\mathbf{dim}\, M_d := \big([M_d : L_j]\big)_{1 \leq j \leq n}.$$

For a graded module M, its Hilbert series[5] *is*

$$h_M(t) := \sum_{d \in \mathbb{Z}} (\mathbf{dim}\, M_d) t^d = \sum_{d \in \mathbb{Z}} \big([M_d : L_j]\big)_{1 \leq j \leq n} t^d.$$

[4] In most representation theory literature, it is common to regard dimension vector as a column vector. But in the context of Hilbert series, we follow Beilinson–Ginzburg–Soergel convention to use row vector here.

[5] sometimes it may also be referred as quantum dimension vector; the variable t is the quantum parameter and specialising $t = 1$ yields the usual dimension vector

The Hilbert series of the algebra A collects the right-module series $h_{e_iA}(t)$ as rows:

$$h_A(t)_{ij} := \big(h_{e_iA}(t)\big)_j = \sum_{d\geq 0}[e_iA_d : L_j]t^d \quad \textit{for all } 1 \leq i, j \leq n.$$

Row i of $h_A(t)$ records the composition factors of e_iA degree by degree. The entries of the Hilbert series are formal Laurent series; for algebras they are formal power series. Finite support gives Laurent polynomials or polynomials, respectively. Since $e_iA_0 = L_i$, we have $h_A(0) = I_n$, the $n \times n$ identity array. For a finite-dimensional right S-module N, use the same multiplicity notation and write $\mathbf{dim}\, N = ([N : L_j])_{1\leq j\leq n}$.

The multiplicities can be computed using Hom spaces:

$$[M_d : L_j] = \dim_{\mathrm{End}_A(L_j)} \mathrm{Hom}_A(e_jA, M_d) \quad \text{for all } d \in \mathbb{Z},\ 1 \leq j \leq n. \tag{2.16}$$

Indeed, $M_dA_+ = 0$, so every map $e_jA \to M_d$ factors uniquely through $e_jA/e_jA_+ \cong L_j$. This identifies $\mathrm{Hom}_A(e_jA, M_d)$ with $\mathrm{Hom}_A(L_j, M_d)$; precomposition gives its right vector-space structure over the division algebra $\mathrm{End}_A(L_j)$. By Schur's lemma, a non-zero map between simple modules is an isomorphism, so each copy of L_j contributes one copy of $\mathrm{End}_A(L_j)$, and the other simples contribute zero. This also shows that the multiplicities are independent of the chosen decomposition.

Example 2.29. *If $S = \bigoplus_{j=1}^n \Bbbk e_j$ with coordinate idempotents, then $\mathrm{End}_A(L_j) = \Bbbk$. The Yoneda lemma 1.17 says that*

$$[M_d : L_j] = \dim_{\Bbbk}(M_de_j), \qquad \mathbf{dim}\, M_d = (\dim_{\Bbbk} M_de_j)_{1\leq j\leq n}.$$

Lemma 2.30. *For locally finite-dimensional graded right A-modules L, M, N, we have the following.*

(1) *A graded isomorphism $M \cong N$ implies $h_M(t) = h_N(t)$.*

(2) *A short exact sequence $0 \to L \to M \to N \to 0$ gives $h_M(t) = h_L(t) + h_N(t)$.*

(3) *For every $a \in \mathbb{Z}$, $h_{M(a)}(t) = t^{-a}h_M(t)$.*

Proof A graded isomorphism preserves each degree and its simple-module multiplicities. In each degree, a graded short exact sequence gives $0 \to L_d \to M_d \to N_d \to 0$, so $[M_d : L_j] = [L_d : L_j] + [N_d : L_j]$ for all $1 \leq j \leq n$. Finally, $M(a)_d = M_{d+a}$ gives

$$h_{M(a)}(t) = \sum_d (\mathbf{dim}\, M_{d+a})t^d = t^{-a}\sum_j (\mathbf{dim}\, M_j)t^j. \qquad \square$$

To multiply two arrays $C = (c_{ij})$ and $D = (d_{ij})$, or a row vector $m = (m_i)$ and an array C, use

$$(CD)_{i\ell} = \sum_{j=1}^n c_{ij}d_{j\ell}, \qquad (mC)_j = \sum_{i=1}^n m_ic_{ij}.$$

Thus multiplication sums over the matching middle index, rather than multiplying individual entries independently. For series, use the same products on coefficients and collect terms with the same total power of t.

Lemma 2.31. *Let N be a finite-dimensional right S-module and ${}_SU_S$ a finite-dimensional bimodule. Then*

$$[N \otimes_S U : L_j] = \sum_{i=1}^n [N : L_i][e_iU : L_j] \quad \textit{for all } 1 \leq j \leq n.$$

Proof Decompose $N \cong \bigoplus_{i=1}^{n} (e_i S)^{\oplus [N:L_i]}$. Multiplication $e_i S \otimes_S U \to e_i U$, $e_i s \otimes u \mapsto e_i s u$, is an isomorphism with inverse $u \mapsto e_i \otimes u$. Distribution over direct sums therefore gives

$$N \otimes_S U \cong \bigoplus_{i=1}^{n} (e_i U)^{\oplus [N:L_i]},$$

which implies the claim. □

Example 2.32 (Hilbert series of a tensor algebra). *Consider the tensor algebra $A = T_S(V)$ and put*

$$C := ([e_i A_1 : L_j])_{1 \le i,j \le n} = ([e_i V : L_j])_{1 \le i,j \le n}.$$

Since $e_i A_0 = L_i$ and $e_i A_{d+1} \cong (e_i A_d) \otimes_S A_1$, Lemma 2.31 gives, by induction,

$$[e_i A_d : L_j] = (C^d)_{ij}, \qquad C^0 = I_n \quad \text{for all } d \ge 0,\ 1 \le i, j \le n.$$

Consequently,

$$h_{T_S(V)}(t) = \sum_{d \ge 0} t^d C^d = \frac{1}{I_n - tC}. \tag{2.17}$$

Indeed, multiplying the series on either side by $I_n - tC$ gives constant coefficient I_n; the coefficient of t^d for $d \ge 1$ is $C^d - C^d = 0$. The inverse is therefore an identity of formal power series with the coefficient multiplication defined above.

Example 2.33 (Upper triangular algebras and a quotient). *For $A = \Bbbk A_n$, order the coordinate idempotents as $e_{11}, \dots, e_{nn}$. Let C have $C_{i,i+1} = 1$ for $1 \le i < n$ and all other entries zero. The tensor presentation in Example 2.23 and (2.17) give*

$$h_A(t) = I_n + tC + \cdots + t^{n-1} C^{n-1}, \qquad h_A(t)_{ij} = \begin{cases} t^{j-i}, & i \le j, \\ 0, & i > j. \end{cases}$$

For $n = 3$, killing e_{13} removes the degree 2 contribution in entry $(1,3)$:

$$h_{\Bbbk A_3}(t) = \begin{pmatrix} 1 & t & t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}, \qquad h_{\Bbbk A_3/(e_{13})}(t) = \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

The quotient removes $\Bbbk e_{13}$ from degree 2, and hence removes t^2 from entry $(1,3)$.

Example 2.34. *Let $n = \dim_{\Bbbk} V$ and $m = \dim_{\Bbbk} W$ in Example 2.16. Choosing an exponent independently for each x_i, and choosing whether each y_j occurs, gives*

$$h_{\mathsf{S}(V)}(t) = \sum_{a_1, a_2, \cdots a_n \ge 0} t^{a_1 + a_2 + \cdots a_n} = \prod_{i=1}^{n} \sum_{a_i \ge 0} t^{a_i} = \frac{1}{(1-t)^n} = \sum_{d \ge 0} \binom{d+n-1}{n-1} t^d,$$
$$h_{\bigwedge W}(t) = (1+t)^m.$$

Definition 2.35. *For graded right and left S-modules M, N, their tensor product has total grading*

$$(M \otimes_S N)_d = \bigoplus_{p+q=d} M_p \otimes_S N_q.$$

The same definition applies to bimodules, and to vector spaces by taking $S = \Bbbk$.

The decomposition follows by distributing tensor products over direct sums, as in Lemma 2.2(4). When M, N are locally finite-dimensional and bounded below, only finitely many pairs (p, q) contribute in a fixed degree, so the tensor product is locally finite-dimensional and bounded below.

Lemma 2.36. *Let A, B be locally finite-dimensional graded algebras with $A_0 = \bigoplus_{i=1}^n \Bbbk e_i$ and $B_0 = \bigoplus_{j=1}^m \Bbbk f_j$ where e_i, f_j's are the coordinate idempotents. Give $A \otimes_\Bbbk B$ the total grading of Definition 2.35, so that $(A \otimes_\Bbbk B)_d = \bigoplus_{i+j=d} A_i \otimes_\Bbbk B_j$ and multiplication $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$. Then*

$$h_{A\otimes_\Bbbk B}(t)_{(i,p),(j,q)} = h_A(t)_{ij} h_B(t)_{pq} \quad \text{for all } 1 \le i, j \le n,\ 1 \le p, q \le m.$$

Proof For vector spaces, take dimensions in each summand of the total grading. For algebras,

$$(e_i \otimes f_p)(A_a \otimes_\Bbbk B_b)(e_j \otimes f_q) = (e_i A_a e_j) \otimes_\Bbbk (f_p B_b f_q).$$

Take vector space dimensions and sum over $a + b = d$ to obtain the coefficient of t^d. □

2.6 The degree 2 relations of a quotient

Let S be finite-dimensional semisimple, V a finite-dimensional S-bimodule, and $I \subset T_S(V)$ a homogeneous two-sided ideal with $I_0 = I_1 = 0$. Set $A = T_S(V)/I$. In degree 2,

$$A_2 \cong (V \otimes_S V)/I_2, \quad \text{where} \quad I_2 = I \cap (V \otimes_S V).$$

Definition 2.37. *The quadratic part of A is $qA := T_S(V)/(I_2)$.*

Since $(I_2) \subset I$, there is a surjective graded algebra homomorphism

$$\pi_A : qA \longrightarrow A, \qquad z + (I_2) \longmapsto z + I, \qquad \operatorname{Ker} \pi_A = I/(I_2).$$

Consequently,

$$\pi_A \text{ is an isomorphism} \iff I = (I_2) \iff A \text{ is quadratic.}$$

For the last equivalence, (I_2) gives a quadratic presentation, while Proposition 2.22 identifies the generators and degree 2 relations of any quadratic presentation with A_1 and I_2.

Example 2.38. *For $A = \Bbbk A_4/(e_{14})$ from Example 2.25, we have $I_2 = 0$, so*

$$qA = \Bbbk A_4, \qquad \pi_A : \Bbbk A_4 \twoheadrightarrow \Bbbk A_4/(e_{14}), \qquad \operatorname{Ker} \pi_A = \Bbbk e_{14}.$$

Thus qA does not impose the cubic relation $e_{12}e_{23}e_{34} = 0$.

2.7 Exercises

Exercise 2.39 (S-dual of tensor product). *Let S be a finite-dimensional semisimple algebra, V a finite-dimensional right S-module, and ${}_S W_S$ a finite-dimensional bimodule. Consider the formula*

$$\theta_r^{V,W} : W^* \otimes_S V^* \longrightarrow (V \otimes_S W)^*, \qquad \theta_r^{V,W}(g \otimes f)(v \otimes w) = g(f(v)w).$$

Write $\theta_r = \theta_r^{V,W}$ when V, W are fixed.

(1) *Check that the formula is bilinear in (g, f) and in (v, w), is unchanged by*

$$vs \otimes w \longleftrightarrow v \otimes sw, \qquad gs \otimes f \longleftrightarrow g \otimes sf \quad \text{for all } s \in S,$$

and gives a right S-linear map $V \otimes_S W \to S$ for each $g \otimes f$. Prove that θ_r is left S-linear.

If V is also an S-bimodule, verify $\theta_r(sg \otimes ft) = s\theta_r(g \otimes f)t$ for all $s, t \in S$.

(2) *Prove that θ_r is bijective.*

(3) *Let V' be a finite-dimensional right S-module and ${}_SW'_S$ a finite-dimensional bimodule. For a right S-linear map $p : V \to V'$ and a bimodule map $q : W \to W'$, prove*

$$(p \otimes q)^* \circ \theta_r^{V',W'} = \theta_r^{V,W} \circ (q^* \otimes p^*).$$

Here $p^* : (V')^* \to V^*$, $q^* : (W')^* \to W^*$, and $(p \otimes q)^* : (V' \otimes_S W')^* \to (V \otimes_S W)^*$ are given by precomposition with p, q, and $p \otimes q$, respectively.

Hint for (2). *First take $V = S$, then use Corollary 1.13.*

Remark. *For a finite-dimensional bimodule ${}_SV_S$ and a finite-dimensional left S-module W, the corresponding natural isomorphism of right S-modules is*

$$\theta_\ell : {}^*W \otimes_S {}^*V \longrightarrow {}^*(V \otimes_S W), \qquad \theta_\ell(g \otimes f)(v \otimes w) = f(vg(w)).$$

Exercise 2.40. *For $n \geq 3$ and $Z \subset \{1, \ldots, n-2\}$, let*

$$A_Z = \Bbbk A_n/(e_{k,k+2} \mid k \in Z)$$

with the quotient grading, and write $\bar{e}_{ij}$ for the image of e_{ij} in A_Z.

(1) *Show that a homogeneous basis consists of $\{\bar{e}_{ii} \mid 1 \leq i \leq n\}$ together with all $\bar{e}_{ij}$ for which $1 \leq i < j \leq n$ and no $k \in Z$ satisfies $i \leq k \leq j-2$.*

(2) *Compute $h_{A_Z}(t)$, and give an S-preserving isomorphism of graded algebras $A_Z \cong T_S(V)/(R)$ with explicit S, V, and $R \subset V \otimes_S V$.*

Hint. *For $i \leq k \leq j-2$, use $e_{ij} = e_{ik}e_{k,k+2}e_{k+2,j}$ to determine which matrix units belong to the ideal.*

Exercise 2.41. *Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ with coordinate idempotents, and let the bimodule $V = \Bbbk a \oplus \Bbbk b$ in degree 1 satisfy $e_1 a e_2 = a$, $e_2 b e_1 = b$, with all other idempotent actions zero.*

(1) *Find a basis of $T_S(V)_d$ for every $d \geq 0$ and show that*

$$h_{T_S(V)}(t) = \frac{1}{1-t^2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

(2) *Find a homogeneous basis and the Hilbert series of $A = T_S(V)/(ab)$, and prove that A is finite-dimensional.*

Hint for (1). *Count alternating words separately in each degree, including degree 0.*

3 The quadratic dual

We use Assumption 1.50: $A_0 = S = \bigoplus_{i=1}^n \Bbbk e_i$. Such form of S guarantees that S^{en} is semisimple as well. This assumption on S means that $M \in \operatorname{mod} S^{\mathrm{en}}$ can be decomposed as $M \cong \bigoplus_{i,j=1}^n e_i M e_j$.

The case of arbitrary finite-dimensional semisimple S is treated in the advanced material.

3.1 Technical preliminaries

Lemma 3.1. *For bimodules ${}_SV_S$, ${}_SW_S$, there is a bimodule isomorphism*

$$V \otimes_S W \cong \bigoplus_{i,j,k=1}^{n} e_i V e_j \otimes_{\Bbbk} e_j W e_k. \tag{3.1}$$

Consequently, after choosing a basis of each $e_i V e_j$, a basis of $V^{\otimes_S d}$ for $d \geq 1$ consists of products of d chosen basis elements whose adjacent idempotents match.

Proof For $v \in e_i V e_j$ and $w \in e_l W e_k$ with $j \neq l$, the element $v \otimes w \in V \otimes_{\Bbbk} W$ becomes 0 in $V \otimes_S W$, since $v \otimes w = v e_j \otimes w = v \otimes e_j w = v \otimes e_j e_l w = 0$; for $j = l$ the relations $vs \otimes w = v \otimes sw$ are trivial, as both sides equal $s_j v \otimes w$ for $s = \sum s_m e_m$. Hence

$$V \otimes_S W \cong \bigoplus_{i,j,k} e_i V e_j \otimes_{\Bbbk} e_j W e_k,$$

as claimed. □

Remark 3.2 (General S). By Theorem 1.12, $S = \prod_a \operatorname{Mat}_{n_a}(\Delta_a)$ with division algebras Δ_a. Choose one primitive idempotent e_a in each factor, so $e_a S e_a = \Delta_a$. Then in this setting the only changes needed for this lemma is: change (3.1) to

$$V \otimes_S W \cong \bigoplus_{a} V e_a \otimes_{\Delta_a} e_a W,$$

that is, the idempotents e_i are replaced by the e_a and $\otimes_{\Bbbk}$ by $\otimes_{\Delta_a}$. Indeed, multiplication gives a bimodule isomorphism $S \cong \bigoplus_a S e_a \otimes_{\Delta_a} e_a S$ (columns times rows in $\operatorname{Mat}_{n_a}(\Delta_a)$), and $V \otimes_S W \cong V \otimes_S S \otimes_S W$ with $V \otimes_S S e_a = V e_a$ and $e_a S \otimes_S W = e_a W$.

Proposition 3.3 (Duals of tensor products). *For finite-dimensional bimodules ${}_SV_S$ and ${}_SW_S$, there is a bimodule isomorphism*

$$\theta_{V,W} : D(W) \otimes_S D(V) \xrightarrow{\sim} D(V \otimes_S W), \quad \lambda \otimes \mu \mapsto \Big(v \otimes w \mapsto \sum_{i=1}^{n} \mu(ve_i)\, \lambda(e_i w) \Big). \tag{3.2}$$

It is natural contravariantly in both V and W: for bimodule maps $f : V \to V'$ and $g : W \to W'$ between finite-dimensional bimodules, the following diagram commutes

$$\begin{array}{ccc} D(W') \otimes_S D(V') & \xrightarrow{\theta_{V',W'}} & D(V' \otimes_S W') \\ \downarrow{\scriptstyle D(f)\otimes_S D(g)} & & \downarrow{\scriptstyle D(f\otimes_S g)} \\ D(W) \otimes_S D(V) & \xrightarrow[\theta_{V,W}]{} & D(V \otimes_S W). \end{array} \tag{3.3}$$

Proof Take $\lambda \in D(W)$ and $\mu \in D(V)$ throughout. By (1.13), the bimodule S-action on the dual spaces are

$$(\lambda t)(w) = \lambda(tw), \quad (s\lambda)(w) = \lambda(ws), \quad (\mu t)(v) = \mu(tv), \quad (s\mu)(v) = \mu(vs),$$

for $s, t \in S$. Write $s = \sum_i s_i e_i$ with $s_i \in \Bbbk$, so that $se_i = s_i e_i = e_i s$.

θ is well defined. Put $\Theta(\lambda, \mu)(v, w) := \sum_i \mu(ve_i)\lambda(e_i w)$. Then $\Theta(\lambda, \mu) : V \times W \to \Bbbk$ is $\Bbbk$-bilinear and S-balanced: for $s \in S$,

$$\Theta(\lambda, \mu)(vs, w) = \sum_i s_i\, \mu(ve_i)\lambda(e_i w) = \Theta(\lambda, \mu)(v, sw).$$

So $\Theta(\lambda, \mu)$ defines an element of $D(V \otimes_S W)$ by the universal property of tensor product over S.

The assignment $(\lambda, \mu) \mapsto \Theta(\lambda, \mu)$ is also $\Bbbk$-bilinear and S-balanced, since

$$\Theta(\lambda s, \mu)(v, w) = \sum_i \mu(ve_i)\lambda(se_i w) = \sum_i s_i\, \mu(ve_i)\lambda(e_i w) = \sum_i \mu(ve_i s)\lambda(e_i w) = \Theta(\lambda, s\mu)(v, w).$$

Hence θ is a well-defined $\Bbbk$-linear map on $D(W) \otimes_S D(V)$.

θ is a bimodule map. For $s \in S$ and $x = v \otimes w \in V \otimes_S W$, the left action on $D(V \otimes_S W)$ is $(sf)(x) = f(xs) = f(v \otimes ws)$. Thus

$$\theta(s\lambda \otimes \mu)(v \otimes w) = \sum_i \mu(ve_i)\lambda(e_i ws) = \theta(\lambda \otimes \mu)(v \otimes ws),$$

so $\theta(s\lambda \otimes \mu) = s\,\theta(\lambda \otimes \mu)$. Similarly, $(fs)(x) = f(sx) = f(sv \otimes w)$ and

$$\theta(\lambda \otimes \mu s)(v \otimes w) = \sum_i \mu(sve_i)\lambda(e_i w) = \theta(\lambda \otimes \mu)(sv \otimes w).$$

θ is bijective. By Lemma 3.1 and (1.14),

$$D(W) \otimes_S D(V) \cong \bigoplus_{a,b,c=1}^{n} e_a D(W) e_b \otimes_\Bbbk e_b D(V) e_c \cong \bigoplus_{a,b,c=1}^{n} D(e_b W e_a) \otimes_\Bbbk D(e_c V e_b),$$

and, since $V \otimes_S W \cong \bigoplus_{a,b,c} e_c V e_b \otimes_\Bbbk e_b W e_a$ is finite-dimensional,

$$D(V \otimes_S W) \cong \bigoplus_{a,b,c=1}^{n} D(e_c V e_b \otimes_\Bbbk e_b W e_a).$$

Take $\lambda \in e_a D(W) e_b$ and $\mu \in e_b D(V) e_c$, regarded as functionals supported on $e_b W e_a$ and $e_c V e_b$ by (1.14). In $\sum_i \mu(ve_i)\lambda(e_i w)$, the first factor can be non-zero only for $i = b$ and $v \in e_c V$, and the second only for $i = b$ and $w \in W e_a$. Hence $\theta(\lambda \otimes \mu)$ vanishes on all summands of $V \otimes_S W$ other than $e_c V e_b \otimes_\Bbbk e_b W e_a$, and on it equals $v \otimes w \mapsto \mu(v)\lambda(w)$. So θ preserves the decompositions, and on the summand (a, b, c) it is

$$D(e_b W e_a) \otimes_\Bbbk D(e_c V e_b) \longrightarrow D(e_c V e_b \otimes_\Bbbk e_b W e_a), \qquad \lambda \otimes \mu \longmapsto \big(v \otimes w \mapsto \mu(v)\lambda(w)\big).$$

This is an isomorphism of finite-dimensional vector spaces: it sends the tensor products of dual basis vectors to the dual basis of the tensor product basis.

Naturality. For $\lambda \in D(W')$, $\mu \in D(V')$, $v \in V$, $w \in W$, and using that f, g commute with the idempotents,

$$\begin{aligned}\big(D(f \otimes_S g)\,\theta_{V',W'}(\lambda \otimes \mu)\big)(v \otimes w) &= \sum_i \mu(f(v)e_i)\lambda(e_i g(w)) \\ &= \sum_i \mu(f(ve_i))\lambda(g(e_i w)) \\ &= \sum_i \big(D(f)\mu\big)(ve_i)\big(D(g)\lambda\big)(e_i w) \\ &= \theta_{V,W}\big(D(g)\lambda \otimes D(f)\mu\big)(v \otimes w).\end{aligned}$$

This is (3.3). □

Remark 3.4. The formula $\mu(v)\lambda(w)$ need not define a functional on $V \otimes_S W$. For example, take $S = \Bbbk e_1 \oplus \Bbbk e_2$, $V = W = S$, and let μ, λ be the dual of e_1, e_2 respectively. Then $e_1 \otimes_S e_2 = e_1 \otimes_S e_1 e_2 = 0$, whereas $\mu(e_1)\lambda(e_2) = 1$.

Remark 3.5 (General S). Fix a symmetrising form τ and dual bases $\{x_k\}, \{y_k\}$ of S, that is, $\tau(x_k y_l) = \delta_{kl}$. Then (3.2) and (3.3) hold with

$$\theta_{V,W}(\lambda \otimes \mu)(v \otimes w) = \sum_k \mu(vx_k)\lambda(y_k w).$$

Proof. The proof above goes through with $se_i = e_i s$ replaced by the Casimir identity $\sum_k sx_k \otimes y_k = \sum_k x_k \otimes y_k s$ for $s \in S$. This identity follows from $z = \sum_k \tau(x_k z) y_k$ for $z \in S$ and $\tau(y_l s x_k) = \tau(s x_k y_l)$: expanding $sx_k = \sum_l \tau(sx_k y_l)x_l$ gives $\sum_k sx_k \otimes y_k = \sum_l x_l \otimes \sum_k \tau(y_l s x_k) y_k = \sum_l x_l \otimes y_l s$. For bijectivity, let $f \in V^*$ and $\mu = \tau \circ f = \alpha_V(f)$ (Theorem 1.57). Then $f(v) = \sum_k \tau(f(v)x_k)y_k = \sum_k \mu(vx_k)y_k$, so $\theta(\lambda \otimes \mu)(v \otimes w) = \lambda(f(v)w)$. Thus $\theta = \theta_D \circ (\mathrm{id} \otimes \alpha_V^{-1})$ for θ_D in §2.1, which is bijective as θ_D and the left S-linear map α_V are. For $S = \bigoplus_i \Bbbk e_i$ and τ_S, take $x_k = y_k = e_k$ to recover (3.2).

Lemma 3.6 (Monoidality of evaluation). *For finite-dimensional bimodules ${}_S V_S$ and ${}_S W_S$, the following diagram commutes:*

$$\begin{array}{ccc} V \otimes_S W & \xrightarrow{\mathrm{ev}_V \otimes_S \mathrm{ev}_W} & D(D(V)) \otimes_S D(D(W)) \\ \Big\downarrow{\scriptstyle \mathrm{ev}_{V \otimes_S W}} & & \Big\downarrow{\scriptstyle \theta_{D(W),D(V)}} \\ D(D(V \otimes_S W)). & \xrightarrow[D(\theta_{V,W})]{} & D(D(W) \otimes_S D(V)) \end{array} \tag{3.4}$$

Proof For $v \in V$, $w \in W$, $\lambda \in D(W)$, and $\mu \in D(V)$, the dual actions in (1.13) give $\mathrm{ev}_W(w)(\lambda e_i) = \lambda(e_i w)$ and $\mathrm{ev}_V(v)(e_i \mu) = \mu(ve_i)$. The claim then follows by direct check:

$$\begin{aligned}\theta_{D(W),D(V)}\big(\mathrm{ev}_V(v) \otimes \mathrm{ev}_W(w)\big)(\lambda \otimes \mu) &= \sum_{i=1}^n \mu(ve_i)\lambda(e_i w), \\ \big[D(\theta_{V,W})\big(\mathrm{ev}_{V \otimes_S W}(v \otimes w)\big)\big](\lambda \otimes \mu) &= \theta_{V,W}(\lambda \otimes \mu)(v \otimes w) \\ &= \sum_{i=1}^n \mu(ve_i)\lambda(e_i w).\end{aligned}$$

□

Remark 3.7 (General S). With θ as in Remark 3.5, the diagram commutes by the same computation: $\mathrm{ev}_W(w)(\lambda x_k) = \lambda(x_k w)$ and $\mathrm{ev}_V(v)(y_k\mu) = \mu(vy_k)$, and $\sum_k x_k \otimes y_k = \sum_k y_k \otimes x_k$ (the dual bases are interchangeable, as τ is symmetric), so both paths send $v \otimes w$ to $\lambda \otimes \mu \mapsto \sum_k \mu(vx_k)\lambda(y_k w)$.

Definition 3.8. *For a sub-bimodule U of a finite-dimensional bimodule ${}_SM_S$, its orthogonal complement is*

$$U^\perp := \{\lambda \in D(M) \mid \lambda(u) = 0 \text{ for all } u \in U\}.$$

The space $U^\perp$ is a sub-bimodule of $D(M)$: for $\lambda \in U^\perp$, $s, t \in S$, and $u \in U$, we have $(s\lambda t)(u) = \lambda(tus) = 0$ since $tus \in U$. Let $\iota : U \hookrightarrow M$ be the inclusion. Since $D(\iota)(\lambda) = \lambda|_U$, exactness of D gives the short exact sequence of S-bimodules

$$0 \longrightarrow U^\perp \longrightarrow D(M) \xrightarrow{D(\iota)} D(U) \longrightarrow 0. \tag{3.5}$$

In particular, we have $U^\perp = \operatorname{Ker} D(\iota) = \operatorname{Im} D(q) = D(M/U)$, where $0 \to U \xrightarrow{\iota} M \xrightarrow{q} M/U \to 0$ is the canonical short exact sequence.

Lemma 3.9. *Let ${}_SM_S$ and ${}_SV_S$ be finite-dimensional S-bimodules.*

(1) *For a sub-bimodule ${}_SU_S \subset M$, we have $(U^\perp)^\perp = DD(U) \cong U$.*

(2) *For a sub-bimodule ${}_SR_S \subset V \otimes_S V$, we have*

$$\theta^{-1}_{D(V),D(V)}\Big(\big(\theta^{-1}_{V,V}(R^\perp)\big)^\perp\Big) = (\mathrm{ev}_V \otimes_S \mathrm{ev}_V)(R) \quad \text{in } D(D(V)) \otimes_S D(D(V)).$$

Proof (1) Let $\iota : U \hookrightarrow M$ be the inclusion. Dualising (3.5) gives

$$(U^\perp)^\perp = \operatorname{Im} D(D(\iota)) = \operatorname{Im}(\mathrm{ev}_M \circ \iota \circ \mathrm{ev}_U^{-1}) = \operatorname{Im}(\mathrm{ev}_U \circ \mathrm{ev}_U^{-1}) = DD(U) \cong U,$$

where the second equality follows from the naturality of ev:

$$\begin{array}{ccc} U & \xrightarrow{\ \iota\ } & M \\ {\scriptstyle \mathrm{ev}_U}\downarrow & \circlearrowleft & \downarrow{\scriptstyle \mathrm{ev}_M} \\ DDU & \xrightarrow[DD(\iota)]{} & DD(M). \end{array}$$

We can also prove this by direct calculation: If $u \in U$ and $\lambda \in U^\perp$, then $\mathrm{ev}_M(u)(\lambda) = \lambda(u) = 0$; hence $\mathrm{ev}_U(U) = \mathrm{ev}_M(U) \subset (U^\perp)^\perp$. If $m \notin U$, extend a basis of U by m to a basis of M; the coordinate functional of m lies in $U^\perp$ and does not vanish at m, so $\mathrm{ev}_M(m) \notin (U^\perp)^\perp$.

(2) Write $\theta_V = \theta_{V,V}$ and $\theta' = \theta_{D(V),D(V)}$, and put $U := \theta_V^{-1}(R^\perp) \subset D(V) \otimes_S D(V)$. Finite-dimensionality ensure that the map

$$D(\theta_V) : D(D(V \otimes_S V)) \to D(D(V) \otimes_S D(V)), \quad \psi \mapsto \psi \circ \theta,$$

is bijective. For $\lambda \in D(V) \otimes_S D(V)$, we have by definition $(\psi \circ \theta_V)(\lambda) = \psi(\theta_V(\lambda))$, and so

$$\begin{aligned} D(\theta_V)(\psi) = \psi \circ \theta_V \in U^\perp &\Leftrightarrow\ 0 = (\psi \circ \theta_V)(U) = \psi\big(\theta_V(U)\big) \\ &\Leftrightarrow\ \psi \in \big(\theta_V(U)\big)^\perp = (R^\perp)^\perp. \end{aligned}$$

Hence, we have

$$U^\perp = D(\theta_V)\big((R^\perp)^\perp\big) \overset{(1)}{=} D(\theta_V)\big(\mathrm{ev}_{V\otimes_S V}(R)\big) \overset{\text{Lem 3.6}}{=} \theta' \circ (\mathrm{ev}_V \otimes_S \mathrm{ev}_V)(R),$$

which means that $\theta'^{-1}\big((\theta_V R^\perp)^\perp\big) = (\mathrm{ev}_V \otimes_S \mathrm{ev}_V)(R)$ as claimed. □

Writing down θ^{-1} in bases. Choose bases $\mathcal{B}_{ij}$ of e_iVe_j and $\mathcal{C}_{jk}$ of e_jWe_k. By Lemma 3.1, the products $b\otimes c$ with $b\in\mathcal{B}_{ij}$ and $c\in\mathcal{C}_{jk}$ (over all i,j,k) form a basis of $V\otimes_S W$. Let $b^*\in D(V)$ and $c^*\in D(W)$ be the dual basis functionals, so $b^*\in e_jD(V)e_i$ and $c^*\in e_kD(W)e_j$ by (1.14), and the $c^*\otimes b^*$ with $b\otimes c$ as above form a basis of $D(W)\otimes_S D(V)$. For $b'\in\mathcal{B}_{i'j'}$ and $c'\in\mathcal{C}_{j'k'}$, the definition of θ gives

$$\theta(c^*\otimes b^*)(b'\otimes c') = \sum_m b^*(b'e_m)\,c^*(e_mc') = b^*(b')\,c^*(c') = \delta_{bb'}\delta_{cc'},$$

since only $m=j'$ contributes. Thus $\theta(c^*\otimes b^*) = (b\otimes c)^*$, the dual basis functional of $b\otimes c$, and for $\Phi\in D(V\otimes_S W)$,

$$\theta^{-1}(\Phi) = \sum_{i,j,k}\ \sum_{b\in\mathcal{B}_{ij},\,c\in\mathcal{C}_{jk}} \Phi(b\otimes c)\; c^*\otimes b^*.$$

In particular, for $R\subset V\otimes_S V$, the space $\theta^{-1}(R^\perp)$ consists of the tensors $\sum\gamma_{bc}\,c^*\otimes b^*$ with $\sum\gamma_{bc}\,r_{bc}=0$ for every $\sum r_{bc}\,b\otimes c\in R$; note that the order of the two factors is reversed.

For general S, the one-sided S-duals give the following choice-free orthogonal complements.

Definition 3.10. *For an S-sub-bimodule U of a finite-dimensional S-bimodule M, its right orthogonal complement and left orthogonal complement are*

$$U^\perp := \{f\in M^* \mid f(u)=0 \text{ for all } u\in U\},$$
$$^\perp U := \{g\in {}^*M \mid g(u)=0 \text{ for all } u\in U\}.$$

Both orthogonal complements are S-sub-bimodules. Indeed, for $s,t\in S$, $u\in U$, $f\in U^\perp$, and $g\in {}^\perp U$, the actions in (1.17) give

$$(sft)(u) = sf(tu) = 0, \qquad (sgt)(u) = g(us)t = 0,$$

since $tu, us\in U$. The orthogonal complements are defined without choosing a symmetrising form, but it is possible to realise, through the natural isomorphism associated to a chosen symmetrising form, the orthogonal complements as a sub-bimodule of $D(M)$.

If S is equipped with a symmetrising form $\tau: S\to\Bbbk$, the comparison maps α_M,β_M of (1.20) satisfy

$$\alpha_M(U^\perp) = \{\lambda\in D(M) \mid \lambda(u)=0 \text{ for all } u\in U\} = \beta_M({}^\perp U). \tag{3.6}$$

The middle space is the orthogonal complement of Definition 3.8; it contains $\alpha_M(U^\perp)$ and $\beta_M({}^\perp U)$: if $f\in U^\perp$ and $g\in {}^\perp U$, then $\tau\circ f$ and $\tau\circ g$ vanish on U. Conversely, $\alpha_M(f)(U)=0$ gives

$$\tau(f(u)s) = \tau(f(us)) = 0 \quad \text{for all } s\in S,\ u\in U.$$

Non-degeneracy of τ forces $f(u)=0$ for every $u\in U$. For the left dual, $\tau(sg(u)) = \tau(g(su)) = 0$ gives $g(u)=0$ instead. Bijectivity of α_M,β_M in Theorem 1.57 proves (3.6).

Example 3.11. *Let $S=\Bbbk e_{11}\oplus\Bbbk e_{22}\oplus\Bbbk e_{33}$ and $V=\Bbbk a\oplus\Bbbk b$ with $a=e_{12}\in e_{11}Ve_{22}$ and $b=e_{23}\in e_{22}Ve_{33}$. By Lemma 3.1, $V\otimes_S V=\Bbbk(a\otimes b)$, since $b\otimes a$, $a\otimes a$ and $b\otimes b$ are not composable. We*

have $a^* \in e_{22}D(V)e_{11}$ *and* $b^* \in e_{33}D(V)e_{22}$*, and*

$$D(V) \otimes_S D(V) = \Bbbk(b^* \otimes a^*), \qquad \theta(b^* \otimes a^*)(a \otimes b) = a^*(a)\, b^*(b) = 1,$$

so θ *reverses the order of the factors. The path algebra* $\Bbbk A_3 = T_S(V)$ *has* $R = 0$*, and* $\Bbbk A_3/(e_{13}) = T_S(V)/(V \otimes_S V)$ *has* $R = V \otimes_S V$*. Hence*

$$\theta^{-1}(0^\perp) = \Bbbk\, b^* \otimes a^*, \qquad \theta^{-1}\big((V \otimes_S V)^\perp\big) = 0,$$

and therefore

$$(\Bbbk A_3)^! = T_S(D(V))/(b^* \otimes a^*) = S \oplus D(V), \qquad \big(\Bbbk A_3/(e_{13})\big)^! = T_S(D(V)).$$

The second algebra is the path algebra with arrows $a^* : e_{22} \to e_{11}$ *and* $b^* : e_{33} \to e_{22}$*, that is,* $(\Bbbk A_3)^{\mathrm{op}}$ *for the reversed quiver; the first is* $\big(\Bbbk A_3/(e_{13})\big)^{\mathrm{op}}$*.*

3.2 Generators and orthogonal relations

Fix a quadratic presentation $A = T_S(V)/(R)$ as in Definition 2.21.

Definition 3.12. *The quadratic dual of* A *is*

$$A^! := T_S(D(V))/\big(\theta_{V,V}^{-1}(R^\perp)\big), \tag{3.7}$$

where $\theta_{V,V} : D(V) \otimes_S D(V) \xrightarrow{\sim} D(V \otimes_S V)$ *is the map of* (3.2) *and* $R^\perp \subset D(V \otimes_S V)$ *is as in Definition 3.8.*

Note that $D(V)$ is placed in degree 1 in (3.7) – in contrast to the graded dual $\mathbb{D}(V)$, which is placed in degree -1. By (1.14), dualising reverses the idempotents of each generator: $e_j D(V) e_i \cong D(e_i V e_j)$.

The quadratic dual is independent of the choice of quadratic presentation of A, up to A_0-preserving isomorphisms of graded algebras. Indeed, by Proposition 2.22, a quadratic presentation yields

$$S \cong A_0, V \cong A_1, \text{ and } R \cong \mathrm{Ker}(\mu_2 : A_1 \otimes_{A_0} A_1 \to A_2).$$

Through these isomorphisms, functoriality of D and binaturality (3.3) give

$$D(V) \cong D(A_1), \qquad \theta_{V,V}^{-1}(R^\perp) \cong \theta_{A_1,A_1}^{-1}\big((\mathrm{Ker}\,\mu_2)^\perp\big) \subset D(A_1) \otimes_{A_0} D(A_1).$$

By the universal property of tensor algebra (Proposition 2.18), we can extend $D(V) \cong D(A_1)$ to a tensor algebra isomorphism, which descends to the quotients by the two ideals generated by these relation spaces.

Proposition 3.13. *The algebra* $A^!$ *is quadratic and*

$$A^!_0 = S, \qquad A^!_1 = D(V), \qquad A^!_2 \cong D(R).$$

In particular, $\dim_\Bbbk A^!_2 = \dim_\Bbbk R$*.*

Proof Since $D(V)$ is finite-dimensional and $\theta_{V,V}^{-1}(R^\perp) \subset D(V) \otimes_S D(V)$ is a sub-bimodule, the presentation (3.7) is quadratic. Proposition 2.22 gives the components in degrees 0 and 1. Applying $D(-)$ to the bimodule short exact sequence $0 \to R \to V \otimes_S V \to A_2 \to 0$ combined with $\theta_{V,V} : D(V) \otimes_S D(V) \to D(V \otimes_S V)$ gives the following commutative diagram of S-bimodules

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \theta_{V,V}^{-1}(R^\perp) & \longrightarrow & D(V) \otimes_S D(V) & \xrightarrow{f \mapsto \theta_{V,V}(f)|_R} & A^!_2 & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \theta_{V,V}|_R} & & \downarrow{\scriptstyle \theta_{V,V}} & & \downarrow{\scriptstyle \cong} & & \\ 0 & \longrightarrow & R^\perp & \longrightarrow & D(V \otimes_S V) & \longrightarrow & D(R) & \longrightarrow & 0 \end{array} \tag{3.8}$$

where both rows are exact. Hence, $\dim_\Bbbk D(R) = \dim_\Bbbk R$. □

Theorem 3.14 (Duality of quadratic algebra). *Evaluation induces an S-preserving isomorphism of graded algebras*

$$(A^!)^! \cong A.$$

Proof By the universal property of tensor algebra (Proposition 2.18), evaluation $\mathrm{ev}_V : V \xrightarrow{\sim} D(D(V))$ extends to an S-preserving isomorphism $T_S(V) \xrightarrow{\sim} T_S(D(D(V)))$ of graded algebras, whose degree 2 component is $\mathrm{ev}_V \otimes_S \mathrm{ev}_V$.

The relation space of $(A^!)^!$ is, by definition, $\theta^{-1}_{D(V),D(V)}\Big(\big(\theta^{-1}_{V,V}(R^\perp)\big)^\perp\Big)$. By Lemma 3.9(2), this is precisely $\mathrm{ev}_V \otimes_S \mathrm{ev}_V(R)$, which shows that ev_V induces an isomorphism $A \xrightarrow{\sim} (A^!)^!$. □

Convention. *From now on we suppress θ: for a quadratic algebra $A = T_S(V)/(R)$, we write $R^\perp$ also for $\theta^{-1}_{V,V}(R^\perp) \subset D(V) \otimes_S D(V)$, so that $A^! = T_S(D(V))/(R^\perp)$. Accordingly, $(R^\perp)^\perp$ denotes $\theta^{-1}_{D(V),D(V)}\big((\theta^{-1}_{V,V}R^\perp)^\perp\big)$, and Lemma 3.9(2) reads "$(R^\perp)^\perp = R$ under $\mathrm{ev}_V \otimes_S \mathrm{ev}_V$".*

Remark 3.15 (General S). For general S, the map θ, and hence Definition 3.12, needs a choice of symmetrising form. The choice-free version uses the right and left S-duals: the right and left quadratic duals below satisfy ${}^!(A^!) \cong A \cong ({}^!A)^!$ with no choice (Theorem 3.18, advanced).

Quadratic duals via one-sided S-duals. For general S, we use the isomorphisms θ_r, θ_ℓ of (2.5) instead of θ.

Definition 3.16. *The right quadratic dual and left quadratic dual of A are*

$$\begin{aligned} A^! &:= T_S(V^*)/(R^\perp), \quad R^\perp = \{F \in V^* \otimes_S V^* \mid \theta_r(F)(R) = 0\}, \\ {}^!A &:= T_S({}^*V)/({}^\perp R), \quad {}^\perp R = \{F \in {}^*V \otimes_S {}^*V \mid \theta_\ell(F)(R) = 0\}. \end{aligned} \tag{3.9}$$

Under Assumption 1.50, the map $\alpha_V : V^* \to D(V)$, $f \mapsto \tau_S \circ f$, of Theorem 1.57 satisfies $\theta \circ (\alpha_V \otimes \alpha_V) = \alpha_{V \otimes_S V} \circ \theta_r$: both sides send $g \otimes f$ to $v \otimes w \mapsto \tau_S(g(f(v)w))$, because $\sum_i \tau_S(se_i)e_i = s$ for $s \in S$. Hence α_V induces an isomorphism from the right quadratic dual onto the quadratic dual of Definition 3.12; this is why we use the same symbol $A^!$.

Our notation is swapped relative to [BGS96, §2.7 and Definition 2.8.1]: their V^* uses left S-linear maps and their ${}^*V$ uses right S-linear maps. Thus our $A^!$ is their ${}^!A$, and our ${}^!A$ is their $A^!$. The names agree: they also call ${}^!A$ the right dual and $A^!$ the left dual.

Lemma 3.17. *Let $U \subset M$ be an S-sub-bimodule of a finite-dimensional bimodule ${}_SM_S$. Under $M \cong {}^*(M^*)$ and $M \cong ({}^*M)^*$ from the right and left S-dualities in Lemma 1.53,*

$${}^\perp(U^\perp) = U, \qquad ({}^\perp U)^\perp = U.$$

Proof For $u \in U$ and $f \in U^\perp$, evaluation $\mathrm{ev}_M : M \to {}^*(M^*)$ gives $\mathrm{ev}_M(u)(f) = f(u) = 0$, so $U \subset {}^\perp(U^\perp)$ under Lemma 1.53.

Conversely, let $m \notin U$. By Corollary 1.13, $M/U \hookrightarrow S^{\oplus d}$ as a right S-module for some d. Choose a coordinate projection $S^{\oplus d} \twoheadrightarrow S$ for which the composite $f : M \twoheadrightarrow M/U \hookrightarrow S^{\oplus d} \twoheadrightarrow S$ is non-zero at m. Then $f \in \mathrm{Hom}_S(M, S) = M^*$ satisfies $f(U) = 0$ and $f(m) \neq 0$. Thus $f \in U^\perp$ and $m \notin {}^\perp(U^\perp)$. The proof of $({}^\perp U)^\perp = U$ is similar. □

Theorem 3.18. *Left and right quadratic duality are canonically inverse:*

$${}^!(A^!) \cong A \cong ({}^!A)^! \quad \textit{by } S\textit{-preserving isomorphisms of graded algebras.}$$

Proof *Generators.* The right S-duality in Lemma 1.53 gives $\mathrm{ev}_V : V \xrightarrow{\sim} {}^*(V^*)$, $v \mapsto (f \mapsto f(v))$, identifying the generators of A and ${}^!(A^!)$. It preserves both S-actions by (1.17).

Relations. The right S-action on V^* is $(gs)(w) = g(sw)$ for all $g \in V^*$, $s \in S$, and $w \in V$. Applying (2.6) first to $V^* \otimes_S V^*$ and then to $V \otimes_S V$ gives, for all $v, w \in V$ and $f, g \in V^*$,

$$\begin{aligned} \theta_\ell(\mathrm{ev}_V(v) \otimes \mathrm{ev}_V(w))(g \otimes f) &= \mathrm{ev}_V(w)(gf(v)) = g(f(v)w) \\ &= \mathrm{ev}_{V \otimes_S V}(v \otimes w)(\theta_r(g \otimes f)). \end{aligned} \tag{3.10}$$

Thus $\mathrm{ev}_V \otimes \mathrm{ev}_V$ is the bidual identification for $V \otimes_S V$ under anti-distributivity of the S-dual (2.5). Lemma 3.17, applied to $R \subset V \otimes_S V$, identifies the relations ${}^\perp(R^\perp)$ of ${}^!(A^!)$ with R. The presentations (3.9) therefore give an S-preserving isomorphism of graded algebras ${}^!(A^!) \cong A$. Interchanging left and right gives $({}^!A)^! \cong A$. □

Proposition 3.19. *A choice of symmetrising form $\tau : S \to \Bbbk$ gives S-preserving isomorphisms of graded algebras $A^! \cong {}^!A$ and $(A^!)^! \cong A$.*

Proof The maps $\alpha_V(f) = \tau \circ f$ and $\beta_V(g) = \tau \circ g$ in (1.20) give a bimodule isomorphism $\beta_V^{-1}\alpha_V : V^* \to {}^*V$. For $f, g \in V^*$, write $f' = \beta_V^{-1}\alpha_V(f)$ and $g' = \beta_V^{-1}\alpha_V(g)$. Then

$$\begin{aligned} \tau\big(f'(vg'(w))\big) = \tau\big(f(v)g'(w)\big) &= \tau\big(g'(f(v)w)\big) \\ = \tau\big(g(f(v)w)\big) &\qquad \text{for all } v, w \in V. \end{aligned}$$

By (3.6), $(\beta_V^{-1}\alpha_V)^{\otimes_S 2}$ therefore identifies $R^\perp$ with ${}^\perp R$. The bimodule map $\beta_V^{-1} \circ \alpha_V$ extends by Proposition 2.18 to an isomorphism $T_S(V^*) \to T_S({}^*V)$ and hence descends to $A^! \cong {}^!A$. Apply the same comparison to $A^!$ and then Theorem 3.18 to obtain $(A^!)^! \cong {}^!(A^!) \cong A$. □

For general S, the isomorphism $A^! \cong {}^!A$ of Proposition 3.19 depends on τ; changing τ changes it by the central unit of Remark 1.59. Theorem 3.18 involves no choice.

3.3 Example: Symmetric and exterior algebras

Proposition 3.20. *For finite-dimensional $\Bbbk$-vector spaces V, W,*

$$\mathsf{S}(V)^! \cong \bigwedge D(V), \qquad \Big(\bigwedge W\Big)^! \cong \mathsf{S}(D(W)).$$

Proof For $\mathsf{S}(V)$, choose a basis $x_1, \dots, x_n$ of V, with dual basis $x_1^*, \dots, x_n^*$ of $D(V)$. Here $S = \Bbbk$, and (3.2) reads

$$\theta(x_j^* \otimes x_i^*)(x_p \otimes x_q) = x_i^*(x_p)\, x_j^*(x_q) = \delta_{ip}\delta_{jq}.$$

A tensor $\sum_{i,j=1}^n c_{ij} x_i^* \otimes x_j^*$ vanishes on each $x_p \otimes x_q - x_q \otimes x_p$ precisely when $c_{pq} = c_{qp}$ for $1 \le p < q \le n$. Thus $\theta^{-1}(R^\perp)$ for the symmetric relations R is spanned by $x_i^* \otimes x_i^*$ for $1 \le i \le n$ and $x_i^* \otimes x_j^* + x_j^* \otimes x_i^*$ for $1 \le i < j \le n$, which are the exterior relations on $D(V)$.

For $\bigwedge W$, choose a basis $y_1, \dots, y_m$ of W, with dual basis $y_1^*, \dots, y_m^*$ of $D(W)$. A tensor $\sum_{i,j=1}^m c_{ij} y_i^* \otimes y_j^*$ annihilates the exterior relations precisely when $c_{ii} = 0$ for $1 \le i \le m$ and $c_{pq} + c_{qp} = 0$ for $1 \le p < q \le m$. Hence $\theta^{-1}(R^\perp)$ is spanned by $y_i^* \otimes y_j^* - y_j^* \otimes y_i^*$ for $1 \le i < j \le m$, which are the symmetric relations on $D(W)$. The presentations in (2.13) give both isomorphisms. □

Note that the proof works in characteristic 2: the relations $(x_i^*)^2 = 0$ remain part of the exterior presentation even when the signs $+$ and $-$ coincide.

This example also shows that

- does not preserve Hilber series (see Example 2.34);
- does not preserve (in)finite-dimensionality of the algebra ($\mathsf{S}\Bbbk = \Bbbk[x]$ is infinite-dimensional, but $\bigwedge \Bbbk = \Bbbk[x]/(x^2)$ is not).

3.4 Tensor products and signs

Definition 3.21. *For graded $\Bbbk$-algebras A, B, the ordinary tensor product $A \otimes_\Bbbk B$ and the graded tensor product $A \otimes_\Bbbk^{\mathrm{gr}} B$ have underlying vector space $A \otimes_\Bbbk B$ and the total grading of Definition 2.35, with products*

$$\begin{aligned} (a \otimes b)(a' \otimes b') &= aa' \otimes bb' && \text{in } A \otimes_\Bbbk B, \\ (a \otimes b)(a' \otimes b') &= (-1)^{\deg b \deg a'} aa' \otimes bb' && \text{in } A \otimes_\Bbbk^{\mathrm{gr}} B \end{aligned} \tag{3.11}$$

for homogeneous $a, a' \in A$ and $b, b' \in B$, extended bilinearly.

Both products have identity $1_A \otimes 1_B$. For the graded product, associativity follows from $\deg b \deg a' + (\deg b + \deg b') \deg a'' = \deg b' \deg a'' + \deg b(\deg a' + \deg a'')$ for homogeneous $a', a'' \in A$ and $b, b' \in B$. For homogeneous $a \in A$ and $b \in B$, the elements $a \otimes 1$ and $1 \otimes b$ commute in the ordinary tensor product; in the graded tensor product they satisfy

$$(1 \otimes b)(a \otimes 1) = (-1)^{\deg a \deg b}(a \otimes 1)(1 \otimes b).$$

Suppose $A_0 = S$ and $B_0 = T$ are finite-dimensional semisimple and $S \otimes_\Bbbk T$ is semisimple. The sign does not change the underlying graded $(S \otimes_\Bbbk T)$-bimodule, so the two tensor products have the same Hilbert series whenever A, B are locally finite-dimensional. For $S = \Bbbk^{\oplus n}$ and $T = \Bbbk^{\oplus m}$, Lemma 2.36 gives its entries:

$$h_{A \otimes_\Bbbk B}(t)_{(i,p),(j,q)} = h_A(t)_{ij} h_B(t)_{pq}.$$

In particular, when $S = T = \Bbbk$, this is $h_A(t) h_B(t)$.

Proposition 3.22. *Let $A = T_S(V)/(R)$ and $B = T_T(W)/(Q)$ be quadratic algebras, and assume that $S \otimes_\Bbbk T$ is semisimple. Then both tensor products are quadratic over $S \otimes_\Bbbk T$, and*

$$(A \otimes_\Bbbk B)^! \cong A^! \otimes_\Bbbk^{\mathrm{gr}} B^!, \qquad (A \otimes_\Bbbk^{\mathrm{gr}} B)^! \cong A^! \otimes_\Bbbk B^!. \tag{3.12}$$

The mixed commuting relations become anticommuting relations under the reversed tensor pairing.

The assumption on $S \otimes_\Bbbk T$ holds when S, T are products of matrix algebras over $\Bbbk$, since $\mathrm{Mat}_a(\Bbbk) \otimes_\Bbbk \mathrm{Mat}_b(\Bbbk) \cong \mathrm{Mat}_{ab}(\Bbbk)$. It does not follow from semisimplicity alone: take $S = T$ to be the field in Example 1.44.

Proof The degree 1 part of $A \otimes_\Bbbk B$ is $(V \otimes_\Bbbk T) \oplus (S \otimes_\Bbbk W)$. By Lemma 2.2, its tensor square over $S \otimes_\Bbbk T$ decomposes into the four summands

$$(V \otimes_S V) \otimes_\Bbbk T \;\oplus\; \underbrace{V \otimes_\Bbbk W}_{\text{order } VW} \;\oplus\; \underbrace{V \otimes_\Bbbk W}_{\text{order } WV} \;\oplus\; S \otimes_\Bbbk (W \otimes_T W).$$

Its quadratic relations consist of $R \otimes_\Bbbk T$, $S \otimes_\Bbbk Q$, and the mixed relations

$$(v \otimes 1)(1 \otimes w) - (1 \otimes w)(v \otimes 1) \quad (v \in V,\ w \in W). \tag{3.13}$$

Here the products are taken in the tensor algebra on the degree 1 part, and their span is an $(S \otimes_{\Bbbk} T)$-sub-bimodule. Let C be the quotient of that tensor algebra by the three specified relation spaces. The generator maps induce algebra homomorphisms $A \to C$, $a \mapsto \overline{a}$, and $B \to C$, $b \mapsto \overline{b}$. Their images commute: the degree 0 actions already commute across the two factors, and (3.13) makes the degree 1 generators commute. Consequently $A \otimes_{\Bbbk} B \to C$, $a \otimes b \mapsto \overline{a}\,\overline{b}$, is an algebra homomorphism. It is inverse to the canonical homomorphism $C \to A \otimes_{\Bbbk} B$, since both composites fix the generating copies of S, T, V, W. Thus the stated relations give exactly $A \otimes_{\Bbbk} B$. For the graded tensor product, the same construction uses $\overline{b}\,\overline{a} = (-1)^{\deg a \deg b}\overline{a}\,\overline{b}$ for homogeneous a, b.

We use the description of quadratic duals by right S-duals (advanced material in Section 3), which applies to the semisimple algebra $S \otimes_{\Bbbk} T$ and agrees with Definition 3.12 when $S \otimes_{\Bbbk} T$ is a product of copies of $\Bbbk$. The right $(S \otimes_{\Bbbk} T)$-duals of the two generator summands are $V^* \otimes_{\Bbbk} T$ and $S \otimes_{\Bbbk} W^*$, with evaluations

$$(f \otimes t)(v \otimes t') = f(v) \otimes tt', \qquad (s \otimes g)(s' \otimes w) = ss' \otimes g(w).$$

These identifications hold for finite free right modules and hence for their direct summands by Corollary 1.13. The reversed tensor evaluation pairs each pure summand with its corresponding dual summand and exchanges the two mixed summands; all other pairings are zero. The pure relation spaces have orthogonal complements $R^{\perp} \otimes_{\Bbbk} T$ and $S \otimes_{\Bbbk} Q^{\perp}$ in the corresponding pure summands. The mixed relation space is the set of pairs $(z, -z)$ in the two copies of $V \otimes_{\Bbbk} W$. For mixed tensors, the two reversed evaluations of $f \in V^*$ and $g \in W^*$ both give $f(v) \otimes g(w)$ on the corresponding copy. Thus an annihilating pair of functionals must have equal components. As these simple tensors span the full dual, the mixed annihilator is spanned by

$$(f \otimes 1)(1 \otimes g) + (1 \otimes g)(f \otimes 1) \quad (f \in V^*,\ g \in W^*).$$

These are the mixed relations of $A^! \otimes^{\mathrm{gr}}_{\Bbbk} B^!$, proving the first isomorphism in (3.12). For $A \otimes^{\mathrm{gr}}_{\Bbbk} B$, the minus sign in (3.13) is a plus sign. The mixed annihilator then has a minus sign, giving the ordinary tensor product on the dual side. □

Remark 3.23. The two tensor products are genuinely different in general. Let $\operatorname{char} \Bbbk \neq 2$ and $E = \Bbbk[\epsilon]/(\epsilon^2)$ with $\deg \epsilon = 1$. Then $E \otimes_{\Bbbk} E = \Bbbk[\epsilon_1, \epsilon_2]/(\epsilon_1^2, \epsilon_2^2)$ is commutative, whereas $\epsilon_1\epsilon_2 = -\epsilon_2\epsilon_1 \neq 0$ in $E \otimes^{\mathrm{gr}}_{\Bbbk} E$, which is the exterior algebra on two generators and is not commutative. Hence they are not isomorphic as algebras; similarly $\Bbbk[x] \otimes_{\Bbbk} \Bbbk[y] = \Bbbk[x, y]$ and $\Bbbk[x] \otimes^{\mathrm{gr}}_{\Bbbk} \Bbbk[y] = \Bbbk\langle x, y\rangle/(xy + yx)$, with $\deg x = \deg y = 1$. The signs are invisible if $\operatorname{char} \Bbbk = 2$, or if A or B is concentrated in even degrees, in which case both products coincide. This is consistent with Proposition 3.22: $\Bbbk[x, y] = \mathsf{S}(\Bbbk^2)$ has dual $\bigwedge \Bbbk^2 = E \otimes^{\mathrm{gr}}_{\Bbbk} E = \Bbbk[x]^! \otimes^{\mathrm{gr}}_{\Bbbk} \Bbbk[y]^!$.

Example 3.24. *We compute how Proposition 3.22 changes a commutation relation in a matrix example. Take $A = \Bbbk A_2 \otimes_{\Bbbk} \Bbbk A_2$. Its degree 0 algebra has basis $e_{ii} \otimes e_{jj}$ for $i, j \in \{1, 2\}$, and its degree 1 part has basis*

$$a = e_{12} \otimes e_{11}, \quad b = e_{22} \otimes e_{12}, \quad c = e_{11} \otimes e_{12}, \quad d = e_{12} \otimes e_{22}.$$

By Lemma 3.1, the tensor square of the degree 1 part has basis ab, cd, and both map to $e_{12} \otimes e_{12}$. Thus the relation space is $\Bbbk(ab - cd)$. Using the reversed generators $x^* \in e_j D(V) e_i$ of $x \in e_i V e_j$ from (1.14), its orthogonal complement is $\Bbbk(b^*a^* + d^*c^*)$. Hence the dual presentation has an anticommuting relation:

$$ab = cd \text{ in } A, \qquad b^*a^* = -d^*c^* \text{ in } A^!.$$

Order the degree 0 idempotents as $e_{11} \otimes e_{11}, e_{22} \otimes e_{11}, e_{11} \otimes e_{22}, e_{22} \otimes e_{22}$. The homogeneous bases

give

$$h_A(t) = \begin{pmatrix} 1 & t & t & t^2 \\ 0 & 1 & 0 & t \\ 0 & 0 & 1 & t \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad h_{A^!}(t) = h_A(t)^{\mathsf{T}}.$$

The signs agree with Proposition 3.22; in characteristic 2, the same presentations remain valid with $-1 = 1$. Here replacing d^* by $-d^*$ removes the minus sign. In this example the sign distinguishes the specified presentations, not the isomorphism classes of algebras.

Example 3.25. *The symmetric/exterior pair shows that the sign in a graded tensor product cannot always be omitted. Let $V = \Bbbk x_1 \oplus \Bbbk x_2$ and let y_1, y_2 be the dual basis of $W = D(V)$. Proposition 3.22 gives*

$$\mathsf{S}(V)^! \cong \big(\Bbbk[x_1] \otimes_{\Bbbk} \Bbbk[x_2]\big)^! \cong \frac{\Bbbk[y_1]}{(y_1^2)} \otimes_{\Bbbk}^{\mathrm{gr}} \frac{\Bbbk[y_2]}{(y_2^2)} \cong \bigwedge W.$$

If $\operatorname{char} \Bbbk \neq 2$, the ordinary tensor product of the two square-zero algebras is commutative, whereas the graded tensor product is not: $y_1 y_2 = -y_2 y_1 \neq 0$. They are therefore not isomorphic. This sign cannot be removed by a change of generators.

3.5 Exercises

Exercise 3.26. *For $A = \Bbbk A_4/(e_{13})$, write a presentation of $A^!$ using reversed generators and find homogeneous bases of A and $A^!$. Check that $(A^!)^! \cong A$ by computing the relations of $A^!$ directly.* Hint. *Among $e_{12}e_{23}$ and $e_{23}e_{34}$, only the first is a relation. Complement the forbidden pairs and reverse their order.*

Exercise 3.27. *Let $S = \bigoplus_{i=1}^n \Bbbk e_i$ with coordinate idempotents, and let ${}_SV_S$ be a finite-dimensional bimodule in degree 1. Set $C_{ij} = [e_iV : e_jS] = \dim_{\Bbbk}(e_iVe_j)$ for $1 \le i, j \le n$, and put*

$$A = T_S(V)/(V \otimes_S V) \overset{v.sp}{=} S \oplus V.$$

(1) *Prove that*

$$A^! = T_S(D(V)), \qquad h_A(t) = I_n + tC, \qquad h_{A^!}(t) = (I_n - tC^{\mathsf{T}})^{-1}.$$

(2) *Deduce that $h_A(t)h_{A^!}(-t)^{\mathsf{T}} = I_n$.*

Hint for (1). *Compute the multiplicities $[e_iD(V) : e_jS]$.*

Exercise 3.28. *Let $S = \bigoplus_{i=1}^4 \Bbbk e_i$ with coordinate idempotents, and let ${}_SV_S$ have basis $\{a, b, c, d\}$ with*

$$a \in e_1Ve_2, \qquad b \in e_2Ve_4, \qquad c \in e_1Ve_3, \qquad d \in e_3Ve_4,$$

and $e_iVe_j = 0$ for all other pairs. For $\lambda \in \Bbbk$, put $R_\lambda = \Bbbk(ab - \lambda cd) \subset V \otimes_S V$ and $A_\lambda = T_S(V)/(R_\lambda)$.

(1) *Using θ, compute $R_\lambda^\perp$ and a presentation of $A_\lambda^!$ for every λ, including $\lambda = 0$.*

(2) *Compute $(A_\lambda^!)^!$ and recover the relation $ab - \lambda cd$ under the canonical identification $D(D(V)) \cong V$.*

(3) *Prove that $A_\lambda \cong A_\mu$ as graded algebras if and only if λ, μ are both zero or both non-zero.*

Hint for (1). *Evaluate $\alpha b^*a^* + \beta d^*c^*$ on $ab - \lambda cd$ for $\alpha, \beta \in \Bbbk$.*

Hint for (3). *A graded isomorphism permutes the coordinate idempotents of S. The degree 1 components force it to fix e_1, e_4 and either fix or interchange e_2, e_3. Compare which of the multiplication maps*

$$e_1(A_\lambda)_1 e_i \otimes_{\Bbbk} e_i(A_\lambda)_1 e_4 \longrightarrow e_1(A_\lambda)_2 e_4 \quad \text{for } i \in \{2,3\}$$

are zero. For non-zero λ, μ, rescale d.

4 Projective modules and resolutions

4.1 Free and projective modules

Until Subsection 4.3, A is an ordinary $\Bbbk$-algebra and all modules are right A-modules.

Definition 4.1. *A module F is free with basis $\{u_i \mid i \in I\}$ if every $v \in F$ has a unique expression $v = \sum_{i\in I} u_i a_i$, where $a_i \in A$ and only finitely many a_i are non-zero. Equivalently, $F \cong \bigoplus_{i\in I} A$, with u_i corresponding to 1 in the ith summand.*

A map from a free module is determined by arbitrary images of its basis elements. In particular, a generating family $\{m_i \mid i \in I\}$ of M gives a surjection

$$\bigoplus_{i\in I} A \longrightarrow M, \qquad (a_i)_{i\in I} \longmapsto \sum_{i\in I} m_i a_i.$$

Every module is therefore a quotient of a free module. If M is finitely generated, the free module can be chosen to be $A^{\oplus m}$ for some integer $m \geq 0$.

Definition 4.2. *A short exact sequence $0 \to L \xrightarrow{i} X \xrightarrow{q} Y \to 0$ splits if there is an A-linear map $s : Y \to X$ with $qs = \mathrm{id}_Y$. The map s is called a section of q.*

Such a section gives $X = i(L) \oplus s(Y)$: write $x = \big(x - sq(x)\big) + sq(x)$, by $qs = \mathrm{id}_Y$ we have $qi(L) = 0$ and $qs(Y) = Y$, which means that $i(L) \cap s(Y) = 0$. Thus $(l, y) \mapsto i(l) + s(y)$ defines an isomorphism $L \oplus Y \to X$.

Definition 4.3. *A module P is projective if, for every surjective A-linear map $q : X \to Y$ the following diagram commutes:*

$$\text{(exact row:)} \qquad \begin{array}{ccccc} & & & & P \\ & & & \overset{\forall f}{\swarrow} & \\ & & \overset{\exists \widetilde{f}}{\swarrow} & & \\ X & \xrightarrow[q]{} & Y & \longrightarrow & 0 \end{array} \qquad ,$$

that is, every A-linear map $f : P \to Y$ factors through $q\widetilde{f} = f$ for some A-linear $\widetilde{f} : P \to X$. In this case, we say that $\widetilde{f}$ is the lift of f.

Example 4.4. *Free modules have this lifting property: If $P = \bigoplus_{i\in I} A$, then for each basis element u_i, choose $x_i \in X$ with $q(x_i) = f(u_i)$, and extend $u_i \mapsto x_i$.*

The (arguably) best characterisation of projective modules is how they guarantee certain functors are always exact. Let us spell this out in details now.

For any module M, postcomposition defines a functor $\mathrm{Hom}_A(M, -) : \mathrm{Mod}\, A \to \mathrm{Mod}\, \Bbbk$, acting on $u : X \to Y$ by

$$\mathrm{Hom}_A(M, u) : \mathrm{Hom}_A(M, X) \longrightarrow \mathrm{Hom}_A(M, Y), \qquad f \longmapsto u \circ f.$$

Lemma 4.5. *The functor $\mathrm{Hom}_A(P, -)$ is left exact, that is, every short exact sequence $0 \to L \xrightarrow{i} M \xrightarrow{q} N \to 0$ induces an exact sequence*

$$0 \longrightarrow \mathrm{Hom}_A(P, L) \xrightarrow{i\circ -} \mathrm{Hom}_A(P, M) \xrightarrow{q\circ -} \mathrm{Hom}_A(P, N).$$

Proof Since i is injective, $i \circ f = 0$ implies $f = 0$. For $g : P \to M$, we have

$$qg = 0 \;\Leftrightarrow\; \mathrm{Im}(g) \subseteq \mathrm{Ker}(q) = \mathrm{Im}(i),$$

which is then equivalent to saying that there is a unique factorisation $g = i\widetilde{g}$ (namely, $\widetilde{g}(p) := l$ such that $i(l) = g(p)$). □

Surjectivity of q does not in general imply surjectivity of $q \circ -$; projectivity is precisely the extra condition.

Proposition 4.6. *For a right A-module P, the following are equivalent.*

(1) *P is projective.*

(2) *The functor $\mathrm{Hom}_A(P, -)$ is exact.*

(3) *Every surjective A-linear map $q : X \to P$ has an A-linear section.*

(4) *P is isomorphic to a direct summand of a free module.*

In particular, direct summands and arbitrary direct sums of projective modules are projective.

Note that if P is finitely generated, condition (4) can be replaced by $P \in \operatorname{add} A$.

Proof We prove (1) ⇔ (2) and (1) ⇒ (3) ⇒ (4) ⇒ (1).

(1) ⇔ (2). By Lemma 4.5, exactness is equivalent to surjectivity of $q \circ - : \mathrm{Hom}_A(P, X) \to \mathrm{Hom}_A(P, Y)$ for every surjection $q : X \to Y$. This says exactly that every $f : P \to Y$ lifts to X.

(1) ⇒ (3). Apply the lifting property to $f = \mathrm{id}_P$.

(3) ⇒ (4). Choose a free module F and a surjection $q : F \to P$. A section $s : P \to F$ gives $F = \operatorname{Ker} q \oplus s(P)$. If P is finitely generated, choose $F = A^{\oplus m}$.

(4) ⇒ (1). Identify P with a summand of a free module F, with inclusion $j : P \to F$ and projection $r : F \to P$ satisfying $rj = \mathrm{id}_P$. For $q : X \to Y$ surjective and $f : P \to Y$, freeness gives $h : F \to X$ with $qh = fr$. Then $\widetilde{f} = hj$ satisfies $q\widetilde{f} = frj = f$.

Summands and sums. For a direct summand, use the inclusion and projection as in (4) ⇒ (1). A map from $\bigoplus_{i \in I} P_i$ is determined by its restrictions to the P_i; lift each restriction and combine the lifts. □

Example 4.7. *If A is finite-dimensional semisimple, every surjection splits by Theorem 1.12, so every A-module is projective (while not every A-module is free).*

Example 4.8. *For an idempotent $e \in A$, $A = eA \oplus (1 - e)A$ makes eA projective. For $A = \Bbbk A_2$, the module $e_{22}A$ is projective of dimension 1, but not free: non-zero free modules have dimension at least $\dim_{\Bbbk} A = 3$.*

Analogous to $\mathrm{Hom}_A(M, -)$, we also have a contravariant left-exact functor $\mathrm{Hom}_A(-, N)$ for any fixed module N (morphisms are sent by precompositions $u \mapsto (g \mapsto g \circ u)$):

Lemma 4.9. *For any A-module Y, applying $\mathrm{Hom}_A(-, Y)$ to a short exact sequence $0 \to L \xrightarrow{i} M \xrightarrow{q} N \to 0$ yields an exact sequence*

$$0 \longrightarrow \mathrm{Hom}_A(N, Y) \xrightarrow{-\circ q} \mathrm{Hom}_A(M, Y) \xrightarrow{-\circ i} \mathrm{Hom}_A(L, Y).$$

Example 4.10. *Let* $A = \Bbbk A_2$, $L_1 = e_{11}A/\Bbbk e_{12}$ *and* $L_2 = e_{22}A \cong \Bbbk e_{12}$, *and consider the exact sequence* $0 \to \Bbbk e_{12} \xrightarrow{i} e_{11}A \xrightarrow{q} L_1 \to 0$. *Apply* $\mathrm{Hom}_A(-, L_2)$. *For an* A*-linear map* $\varphi : e_{11}A \to L_2$ *we have* $\varphi(e_{11}) = \varphi(e_{11})e_{11} \in L_2 e_{11} = 0$, *so* $\varphi = 0$ *as* e_{11} *generates* $e_{11}A$. *So* $\mathrm{Hom}_A(e_{11}A, L_2) = 0$, *while* $\mathrm{Hom}_A(\Bbbk e_{12}, L_2) \cong \Bbbk$. *Thus* $- \circ i$ *is not surjective.*

A module I is *injective* if $\mathrm{Hom}_A(-, I)$ is exact. We will not use (or at least try to avoid them) this class of modules in this lecture series. For finite-dimensional algebra, $\Bbbk$-linear duality swaps projective right/left module with injective left/right module.

4.2 Complexes and resolutions

Definition 4.11. *A chain complex* $C_\bullet$ *of right* A*-modules is a sequence of* A*-linear differentials* d_h *with*

$$\cdots \longrightarrow C_{h+1} \xrightarrow{d_{h+1}} C_h \xrightarrow{d_h} C_{h-1} \longrightarrow \cdots, \qquad d_h d_{h+1} = 0, \quad h \in \mathbb{Z}.$$

Its homology is $H_h(C_\bullet) := \mathrm{Ker}\, d_h / \mathrm{Im}\, d_{h+1}$, *where* h *is the homological degree.*

Exactness at C_h means $H_h(C_\bullet) = 0$. Writing $C^h := C_{-h}$ and $\delta^h := d_{-h}$ turns subscripts into superscripts and reverses the order:

$$\cdots \longrightarrow C^{h-1} \xrightarrow{\delta^{h-1}} C^h \xrightarrow{\delta^h} C^{h+1} \longrightarrow \cdots, \qquad \delta^h \delta^{h-1} = 0, \qquad H^h(C^\bullet) := \mathrm{Ker}\, \delta^h / \mathrm{Im}\, \delta^{h-1} = H_{-h}(C_\bullet).$$

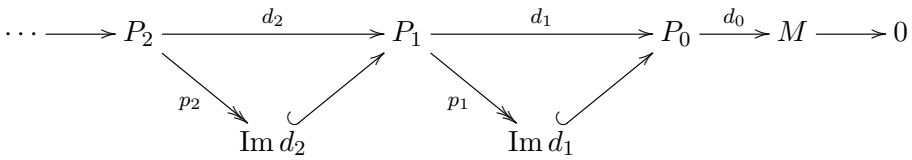
This is a *cochain complex* with *cohomology* H^h; algebraically it is the same notion.

Definition 4.12. *A projective resolution of* M *is an exact sequence*

$$\cdots \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} M \longrightarrow 0 \tag{4.1}$$

with every P_h *projective. Sometimes, we just write* $P_\bullet \twoheadrightarrow M$; *here* $P_\bullet$ *is treated as a complex that ends in* $P_0 \to 0$, *and* M *is not a term of it. A free resolution if is a projective resolution where every* P_h *is free.*

Exactness says that each differential factors through its image: d_h is a surjection $p_h : P_h \twoheadrightarrow \mathrm{Im}\, d_h$ followed by the inclusion $\mathrm{Im}\, d_h \hookrightarrow P_{h-1}$, where $\mathrm{Im}\, d_h = \mathrm{Ker}\, d_{h-1}$:



Thus a projective resolution is a splicing of short exact sequences with projective middle terms,

$$0 \to \mathrm{Im}\, d_{h+1} \to P_h \xrightarrow{p_h} \mathrm{Im}\, d_h \to 0 \quad \forall h \geq 0.$$

Proposition 4.13. *Every right* A*-module has a free resolution, hence a projective resolution.*

Proof For a module N, the free module $A^{\oplus N} := \bigoplus_{n \in N} A$ surjects onto N by $(a_n) \mapsto \sum_{n \in N} n a_n$. Start with $N = M$ and repeat for each successive kernel, composing each surjection with the inclusion of its kernel into the preceding free module; the resulting sequence is exact. □

4.3 Graded modules, tops and Nakayama's lemma

From now on $A = \bigoplus_{i\in\mathbb{Z}} A_i$ is locally finite-dimensional non-negatively graded with $S = A_0$ finite-dimensional semisimple (we do not need Assumption 1.50 in this section), and $A_+ = \bigoplus_{i>0} A_i$. Since $A/A_+ \cong S$, we regard S as a graded right A-module in degree 0. All maps are degree-preserving unless otherwise stated.

Notation. *A complex $C_\bullet$ of graded modules, with degree-preserving differentials, has two gradings: the homological degree h of the term C_h, and the internal degree i (also called the Adams degree) inside each C_h. We write*

$$C_{h,i} := (C_h)_i$$

for the degree i part of the hth term. Restricting the differentials d_h to degree i component is a degree-preserving map $d_{h,i} : C_{h,i} \to C_{h-1,i}$, so $C_\bullet$ is a direct sum over $i \in \mathbb{Z}$ of complexes of vector spaces $C_{\bullet,i}$, and exactness is checked in each internal degree i (Remark 1.29).

Definition 4.14. *A graded free module is a direct sum $\bigoplus_{\lambda\in I} A(-i_\lambda)$ for some set I and $i_\lambda \in \mathbb{Z}$; the generator 1 of $A(-i)$ has degree i. A graded module is graded projective if it has the lifting property of Definition 4.3 for degree-preserving maps.*

For a graded surjection $q : X \to Y$, an element of Y_i has a preimage in X_i, so the lifting proof works for graded free modules; and every graded module is a quotient of a graded free module, by taking homogeneous generators.

Corollary 4.15. *For a graded right A-module P, the following are equivalent.*

(1) *P is graded projective.*

(2) *The functor $\mathrm{Hom}_A^{\mathbb{Z}}(P, -)$ is exact on graded modules.*

(3) *Every degree-preserving surjection onto P has a degree-preserving section.*

(4) *P is a graded direct summand of a graded free module.*

In particular, $L \otimes_S A(-i)$ is graded projective for every finite-dimensional right S-module L and every $i \in \mathbb{Z}$.

Proof The proof of Proposition 4.6 applies verbatim with degree-preserving maps. For the last assertion, Corollary 1.13 makes Q a direct summand of $S^{\oplus m}$ for some m, so $Q \otimes_S A(-i)$ is a graded direct summand of $A(-i)^{\oplus m}$. □

Definition 4.16. *The graded top of a graded right A-module M is the graded right S-module*

$$\mathrm{grtop}\, M := M/MA_+.$$

A degree-preserving $f : M \to N$ induces $\mathrm{grtop}\, f : \mathrm{grtop}\, M \to \mathrm{grtop}\, N$, $m + MA_+ \mapsto f(m) + NA_+$.

A graded module M is bounded below if $M_{<b} = 0$ for some b. Every finitely generated graded module is bounded below: take b to be the least degree among the homogeneous components of a finite generating set.

Lemma 4.17 (Graded Nakayama lemma). *For a bounded below graded right module M,*

$$\mathrm{grtop}\, M = M/MA_+ = 0 \quad \Longrightarrow \quad M = 0.$$

In particular, if $N \subseteq M$ is a graded submodule and $N + MA_+ = M$, then $N = M$.

Proof Suppose on the contrary that $M = MA_+ \neq 0$. Choose the least integer i with $M_i \neq 0$. Then we have

$$(\operatorname{grtop} M)_i = M_i/(MA_+)_i = M_i/(\underbrace{\sum_{j>0} M_{i-j}A_j}_{=0}) = M_i \neq 0,$$

a contradiction. The second assertion follows by applying the first to M/N. □

Ungraded analogue. The radical $\operatorname{rad} A$ is a two-sided ideal: it consists precisely of the elements acting by zero on every simple right A-module. For a simple module L and $0 \neq l \in L$, the map $A \to L$, $a \mapsto la$, is surjective with a maximal right ideal as kernel, so $l \operatorname{rad} A = 0$. Conversely, if a acts by zero on every simple module, apply this to A/U for each maximal right ideal U to get $a \in U$, hence $a \in \operatorname{rad} A$. Finally $l(ba) = (lb)a = 0$ and $l(ab) = (la)b = 0$ for $a \in \operatorname{rad} A$, $b \in A$ show $ab, ba \in \operatorname{rad} A$.

Lemma 4.18 (Nakayama's lemma). *Let M be a finitely generated right A-module. If $M = M \operatorname{rad} A$, then $M = 0$. More generally, if $N \subseteq M$ and $N + M \operatorname{rad} A = M$, then $N = M$.*

Proof For $a \in \operatorname{rad} A$, the right ideal $(1-a)A$ equals A: otherwise it lies in a maximal right ideal U, and $a, 1-a \in U$ gives $1 \in U$. Thus $(1-a)b = 1$ for some $b \in A$.

Suppose $M = M \operatorname{rad} A \neq 0$ and choose generators $m_1, \ldots, m_r$ with r minimal. Write $m_r = \sum_{j=1}^{r} m_j a_j$ with $a_j \in \operatorname{rad} A$ and choose b with $(1-a_r)b = 1$. Then $m_r(1-a_r) = \sum_{j<r} m_j a_j$ and $m_r = \sum_{j<r} m_j a_j b$, contradicting minimality of r. The last assertion follows by applying the first to M/N. □

Remark 4.19. If A is finite-dimensional, then $\operatorname{rad} A$ is *nilpotent*: $(\operatorname{rad} A)^r = 0$ for some $r \geq 1$. Indeed, the powers stabilise at $\operatorname{rad}^r A = \operatorname{rad}^{r+1} A$, and Lemma 4.18 applied to $(\operatorname{rad} A)^r$ gives $(\operatorname{rad} A)^r = 0$. Thus Nakayama's lemma holds for infinitely generated modules: $M = \operatorname{rad} M$ implies $M = M \operatorname{rad}^r A = 0$, and $N + \operatorname{rad} M = M \Rightarrow N = M$ for every submodule $N \subseteq M$.

Remark 4.20. Note that finite generation is crucial in the ungraded setting, but it is not needed in the graded case; in particular, lifts of homogeneous generators of $\operatorname{grtop} M$ generate M.

For finite-dimensional A, every maximal right ideal contains $\operatorname{rad} A$, so $A/\operatorname{rad} A$ is semisimple by Theorem 1.12.

Definition 4.21. *For finite-dimensional A, the top of a right A-module M is* $\operatorname{top} M := M/\operatorname{rad} M$.

$\operatorname{top} M$ is the maximal semisimple quotient of M: it is semisimple, and every map from M to a semisimple T vanishes on $\operatorname{rad} M$ since $\operatorname{rad} T = 0$. A map $f : M \to N$ induces $\operatorname{top} f : \operatorname{top} M \to \operatorname{top} N$, $m + \operatorname{rad} M \mapsto f(m) + \operatorname{rad} N$. By Nakayama, lifts to M of generators of $\operatorname{top} M$ generate M; in particular, $\operatorname{top} f$ surjective implies f surjective when N is finitely generated.

Example 4.22. *For $A = \Bbbk A_2$, $\operatorname{rad} A = \Bbbk e_{12}$, $\operatorname{top}(e_{11}A) = e_{11}A/\Bbbk e_{12} = L_1$ and $\operatorname{top}(e_{22}A) = L_2$.*

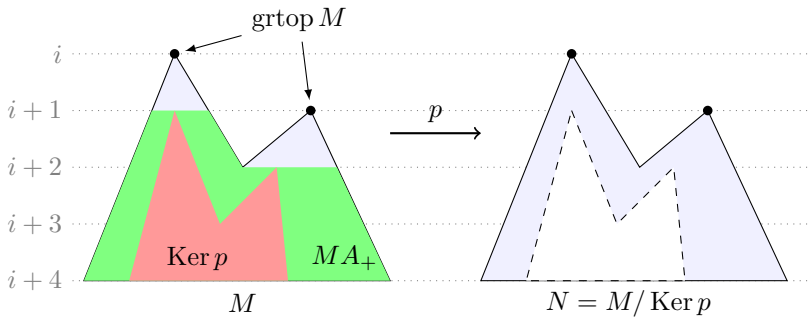
Lemma 4.23. *If A is finite-dimensional, then $A_+ = \operatorname{rad} A$. Consequently, $\operatorname{grtop} M = \operatorname{top} M$ for every graded right A-module M, with $\operatorname{top} M := M/M \operatorname{rad} A$ carrying the quotient grading.*

Proof If A is finite-dimensional, then $A_i = 0$ for all sufficiently large i, so $(A_+)^r = 0$ for some integer $r \geq 1$. For any simple right module L, the submodule LA_+ is either 0 or L. The latter gives $L = L(A_+)^r = 0$, a contradiction. Thus A_+ annihilates every simple module and is contained in $\operatorname{rad} A$. Since $A/A_+ \cong S$ is semisimple, every non-zero element of S acts non-trivially on some simple right S-module. Regarding these as simple A-modules shows that $\operatorname{rad} A \subseteq A_+$. □

Remark 4.24. This is not true for general graded algebra A – for example, $A = \Bbbk[x]$ has $\operatorname{rad} A = 0$.

Lemma 4.25. *Let $p : M \twoheadrightarrow N$ be a surjection of graded modules. Then $\operatorname{Ker} p \subseteq MA_+$ if and only if $\operatorname{grtop} p : \operatorname{grtop} M \to \operatorname{grtop} N$ is an isomorphism.*

Proof Since $p(MA_+) = NA_+$, the map $\operatorname{grtop} p$ is surjective, with kernel $(\operatorname{Ker} p + MA_+)/MA_+$. This vanishes if and only if $\operatorname{Ker} p \subseteq MA_+$. □



If the kernel contained a peak, that generator of M would be killed in N, and $\operatorname{grtop} p$ would fail to be injective.

Lemma 4.26. *Let M be a bounded below graded right A-module, and put*

$$P_M := \bigoplus_{i\in\mathbb{Z}} \big((\operatorname{grtop} M)_i \otimes_S A\big)(-i), \tag{4.2}$$

where $(\operatorname{grtop} M)_i$ is placed in degree 0.

(1) *There is a graded surjection $p : P_M \to M$ with $\operatorname{Ker} p \subseteq P_M A_+$.*

(2) *If M is graded projective, then p is an isomorphism.*

Proof (1) Since S is semisimple, MA_+ has a complement in M as S-modules:

$$M = T \oplus MA_+ \quad \text{with} \quad T_i \xrightarrow{\sim} (\operatorname{grtop} M)_i \ \ \forall i \in \mathbb{Z}.$$

Define p on the direct summand $(\operatorname{grtop} M)_i \otimes_S A$ of P_M by $p(v\otimes a) = va$, for $v \in T_i \cong (\operatorname{grtop} M)_i$; this is well defined since T_i is an S-submodule of M. Note that $\operatorname{grtop}(p) : \operatorname{grtop} P_M \xrightarrow{\sim} \operatorname{grtop} M$ is precisely the isomorphism $T \cong \operatorname{grtop} M$ above under the identification $\operatorname{grtop} P_M = T$.

Hence $M = p(P_M) + MA_+$, so $\overline{M} := M/p(P_M)$ is bounded below with $\overline{M} = \overline{M}A_+$. By the graded Nakayama lemma (Lemma 4.17), we get that $\overline{M} = 0$, thus p is surjective. Since p is surjective and $\operatorname{grtop} p$ is an isomorphism, Lemma 4.25 gives $\operatorname{Ker} p \subseteq P_M A_+$.

(2) Put $K = \operatorname{Ker} p$. Since M is projective, the sequence splits:

$$0 \longrightarrow K \longrightarrow P_M \underset{p}{\overset{s}{\rightleftarrows}} M \longrightarrow 0, \qquad P_M = K \oplus s(M).$$

Let $\pi_K : P_M \to K$ be the projection in this decomposition, so $\pi_K(P_M A_+) = KA_+$. Then by (1)

$$K = \pi_K(K) \subseteq \pi_K(P_M A_+) = KA_+,$$

so $K = KA_+$. As $K \subseteq P_M$ is bounded below, $K = 0$ by graded Nakayama. □

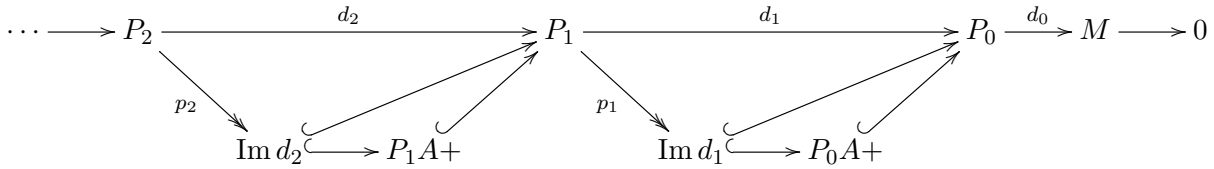
4.4 Minimal resolutions

A graded projective resolution can contain redundant terms: adding a contractible summand $A \xrightarrow{\mathrm{id}} A$ in two consecutive degrees gives another resolution. A resolution is minimal if the differentials land in $P_{h-1}A_+$ (thus, has no aforementioned redundant summand), so that no generator is wasted.

Definition 4.27. *A graded projective resolution is a projective resolution* (4.1) *in which the modules P_h are graded projective and the differentials are degree-preserving. It is minimal if all P_h are lfd bounded below, and*

$$\operatorname{Im} d_h \subseteq P_{h-1}A_+ \quad \text{for all } h \geq 1. \tag{4.3}$$

Equivalently, the surjections $p_h : P_h \twoheadrightarrow \operatorname{Im} d_h$ (and $p_0 : P_0 \twoheadrightarrow M$) have kernel in P_hA_+, since $\operatorname{Ker} p_h = \operatorname{Im} d_{h+1}$*:*



Homological degree h and internal degree i are distinct (Notation 4.3): $d_h : P_{h,i} \to P_{h-1,i}$, and exactness is checked in each internal degree.

Corollary 4.28. *Let $P_\bullet \twoheadrightarrow M$ be a minimal graded projective resolution. Then* $\operatorname{grtop} P_0 \cong \operatorname{grtop} M$ *and* $\operatorname{grtop} P_h \cong \operatorname{grtop} \operatorname{Im} d_h$ *for all $h \geq 1$.*

Proof By minimality, $\operatorname{Ker} p_h = \operatorname{Im} d_{h+1} \subseteq P_hA_+$ for all $h \geq 0$. Apply Lemma 4.25 to each p_h. □

Theorem 4.29 (Existence and uniqueness of minimal graded projective resolution). *Every bounded below lfd graded right A-module M has a minimal graded projective resolution. Moreover, any two such resolutions are isomorphic as complexes of graded modules, i.e. there are isomorphisms f_h making the following diagram commutative*

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & P_1 & \xrightarrow{d_1} & P_0 & \xrightarrow{d_0} & M & \longrightarrow & 0 \\
 & & \wr\downarrow{\scriptstyle f_1} & & \wr\downarrow{\scriptstyle f_0} & & \| & & \\
\cdots & \longrightarrow & P'_1 & \xrightarrow{d'_1} & P'_0 & \xrightarrow{d'_0} & M & \longrightarrow & 0
\end{array}$$

The proof applies Lemma 4.26(1) to M and then to each successive kernel; it is given in Subsection 4.5. By (4.2), $P_h \cong \bigoplus_{i\in\mathbb{Z}} \big((\operatorname{grtop} P_h)_i \otimes_S A\big)(-i)$, so $\operatorname{grtop} P_h$ records the types and degrees of the generators needed in homological degree h.

Example 4.30. *Let $A = \Bbbk A_2$ with $A_0 = S = \Bbbk e_{11} \oplus \Bbbk e_{22}$ and $A_1 = \Bbbk e_{12}$, and place L_1, L_2 in degree 0. Then*

$$0 \longrightarrow e_{22}A(-1) \xrightarrow{d_1} e_{11}A \xrightarrow{d_0} L_1 \longrightarrow 0, \qquad d_1(e_{22}a) = e_{12}a$$

is minimal: $\operatorname{Ker} d_0 = \Bbbk e_{12} = (e_{11}A)A_+$ *and d_1 is an isomorphism onto it, the generator e_{22} and its image e_{12} both having degree 1. Its projective terms have tops L_1 and $L_2(-1)$. The other simple module $L_2 = e_{22}A$ is already projective.*

Example 4.31. *Let $A = \Bbbk[x]$ with $\deg x = 1$. The quotient $\Bbbk = A/(x)$ has the minimal graded free resolution*

$$0 \longrightarrow A(-1) \xrightarrow{d_1} A \xrightarrow{d_0} \Bbbk \longrightarrow 0, \qquad d_1(a) = xa, \quad d_0(f) = f(0).$$

Multiplication by x is injective with image $xA = \operatorname{Ker} d_0 \subseteq AA_+$, and the shift makes it degree-preserving. The two projective terms have one generator each, in degrees 0 and 1.

Example 4.32. *Give $E = \Bbbk[\epsilon]/(\epsilon^2)$ the grading $\deg \epsilon = 1$. Then*

$$\cdots \longrightarrow E(-3) \xrightarrow{d_3} E(-2) \xrightarrow{d_2} E(-1) \xrightarrow{d_1} E \xrightarrow{\pi} \Bbbk \longrightarrow 0, \qquad d_h(a) = \epsilon a,$$

is a minimal graded free resolution with $\operatorname{Im} d_h \cong \Bbbk(-h)$; in particular, the resolution never terminates.

Remark 4.33. The multiplicities in the graded tops are sometimes called graded Betti numbers. For representatives $L_1, \dots, L_n$ of the simple right S-modules, one can write

$$(\operatorname{grtop} P_h)_i \cong \bigoplus_{y=1}^{n} L_y^{\oplus \beta_{h,i,y}(M)}, \qquad \beta_{h,i}(M) := (\beta_{h,i,y}(M))_{1 \le y \le n}.$$

Here $h \ge 0$ is the homological degree and $i \in \mathbb{Z}$ is the internal degree or Adams degree. These multiplicities are independent of the chosen minimal resolution by Theorem 4.29. We will use the modules $\operatorname{grtop} P_h$ directly, without this additional indexing.

Definition 4.34. *A is right noetherian if every submodule of every finitely generated right A-module is finitely generated; left noetherian is defined symmetrically, and noetherian means both.*

Remark 4.35. A is right noetherian if and only if every ascending chain $I_1 \subseteq I_2 \subseteq \cdots$ of right ideals stabilises (the ascending chain condition, ACC); similarly on the left. Finite-dimensional algebras and $\Bbbk[x_1, \dots, x_m]$ are noetherian, whereas the free algebra $\Bbbk\langle x, y\rangle$ is not. Thus the hypothesis of Corollary 4.36 is not automatic.

Proof. If submodules of finitely generated modules are finitely generated, the union of a chain $I_1 \subseteq I_2 \subseteq \cdots$ of right ideals is finitely generated, with generators in some I_k; so the chain stabilises at I_k. Conversely, assume the ACC. Every right ideal I is finitely generated: otherwise choose inductively $u_{k+1} \in I \setminus (u_1A + \cdots + u_kA)$, giving a strictly ascending chain. We show by induction on m that every submodule $U \subseteq A^{\oplus m}$ is finitely generated; the case $m = 1$ was just proved. For $m > 1$, let $\pi : A^{\oplus m} \to A$ be the projection onto the last coordinate. Then $U \cap \operatorname{Ker} \pi \subseteq A^{\oplus(m-1)}$ and $\pi(U)$ are finitely generated, and generators of $U \cap \operatorname{Ker} \pi$ together with lifts to U of generators of $\pi(U)$ generate U. Finally, a finitely generated module is a quotient $A^{\oplus m}/K$, and each submodule of it has the form U/K with $K \subseteq U \subseteq A^{\oplus m}$, hence is finitely generated.

Corollary 4.36. *Assume that A is right noetherian. Every finitely generated graded right A-module has a minimal graded projective resolution whose terms and successive kernels are finitely generated. In particular, this applies to $S = A/A_+$.*

Proof A finitely generated graded module M is bounded below and lfd, being a quotient of a finite direct sum of shifts of A. So we have a minimal resolution $P_\bullet \to M$ by Theorem 4.29. By Corollary 4.28, $\operatorname{grtop} P_0 \cong \operatorname{grtop} M$ is finite-dimensional, so P_0 is finitely generated by (4.2), and $\operatorname{Im} d_1 = \operatorname{Ker} p_0 \subseteq P_0$ is finitely generated by right noetherianity. Then $\operatorname{grtop} P_1 \cong \operatorname{grtop} \operatorname{Im} d_1$ is finite-dimensional, so P_1 is finitely generated, and so is $\operatorname{Im} d_2 = \operatorname{Ker} d_1 \subseteq P_1$. Continuing, every P_h and every $\operatorname{Im} d_h$ is finitely generated. □

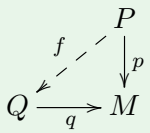
Without noetherianity, boundedness below and local finite-dimensionality still give existence of a minimal graded resolution, but its terms need not be finitely generated.

4.5 Proof of existence and uniqueness of minimal graded projective resolutions

Lemma 4.26(1) constructs a surjection from a graded projective module with kernel in A_+ times the source. The next lemma shows that such a surjection is unique up to isomorphism, which is the input for uniqueness of minimal resolutions.

Lemma 4.37. *Let M be a graded module, and let $p : P \to M$ and $q : Q \to M$ be graded surjections with P, Q graded projective and bounded below, such that* $\operatorname{Ker} p \subseteq PA_+$ *and* $\operatorname{Ker} q \subseteq QA_+$. *Then there is a graded isomorphism $f : P \to Q$ with $qf = p$.*

Proof Projectivity of P gives f in



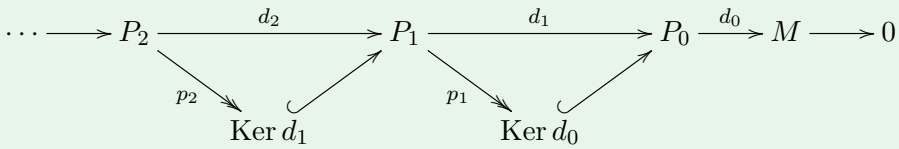
$qf = p$.

f is surjective. Since $q(f(P)) = p(P) = M$, we have $f(P) + \operatorname{Ker} q = Q$, hence $f(P) + QA_+ = Q$. So $Q/f(P)$ equals its own A_+-multiple, and vanishes by graded Nakayama.

f is injective. Since Q is projective, f has a section $s : Q \to P$, $fs = \mathrm{id}_Q$. Then $p(s(Q)) = q(f(s(Q))) = q(Q) = M$, so $s(Q) + \operatorname{Ker} p = P$, hence $s(Q) + PA_+ = P$ and $s(Q) = P$ by graded Nakayama. Thus s is surjective, and injective since $fs = \mathrm{id}$, so $f = s^{-1}$ is an isomorphism. □

Proof of Theorem 4.29

Existence. Lemma 4.26(1) gives $d_0 : P_0 \to M$ with $P_0 = \bigoplus_i ((\operatorname{grtop} M)_i \otimes_S A)(-i)$ and $\operatorname{Ker} d_0 \subseteq P_0 A_+$. P_0 is graded projective by Corollary 4.15. If $M_i = 0$ for $i < b$, then $P_{0,j} = \bigoplus_{b \le i \le j} (\operatorname{grtop} M)_i \otimes_S A_{j-i}$ is finite-dimensional, so P_0 is bounded below and lfd. Hence $\operatorname{Ker} d_0 \subseteq P_0$ is again bounded below and lfd, and we repeat with $\operatorname{Ker} d_0$ in place of M: this gives a surjection $p_1 : P_1 \twoheadrightarrow \operatorname{Ker} p_0$ with $\operatorname{Ker} p_1 \subseteq P_1 A_+$, and we let d_1 be p_1 followed by the inclusion $\operatorname{Ker} p_0 \subseteq P_0$. Inductively, $p_h : P_h \twoheadrightarrow \operatorname{Ker} d_{h-1}$ with $\operatorname{Ker} p_h \subseteq P_h A_+$, and d_h is p_h followed by the inclusion $\operatorname{Ker} d_{h-1} \subseteq P_{h-1}$:



Then $\operatorname{Im} d_h = \operatorname{Ker} d_{h-1}$, so the sequence is exact, and $\operatorname{Im} d_h \subseteq P_{h-1} A_+$ for $h \ge 1$ because $\operatorname{Im} d_h = \operatorname{Ker} p_{h-1}$: this is minimality.

Uniqueness. Let $P_\bullet$ and $P'_\bullet$ be minimal resolutions of M; write $p_h : P_h \twoheadrightarrow \operatorname{Im} d_h$ as above, so $\operatorname{Ker} p_h = \operatorname{Im} d_{h+1} \subseteq P_h A_+$ by minimality. Suppose we have an isomorphism $\varphi_h : \operatorname{Im} d_h \to \operatorname{Im} d'_h$, starting with $\varphi_0 = \mathrm{id}_M$. Both $\varphi_h p_h$ and p'_h are surjections onto $\operatorname{Im} d'_h$ with kernel in $P_h A_+$,

resp. P'_hA_+, so Lemma 4.37 gives an isomorphism f_h in

$$\begin{array}{ccc} P_h & \xrightarrow{p_h} & \operatorname{Im} d_h \\ {\scriptstyle f_h}\downarrow\wr & & \wr\downarrow{\scriptstyle \varphi_h} \\ P'_h & \xrightarrow[p'_h]{} & \operatorname{Im} d'_h. \end{array}$$

It restricts to an isomorphism $\varphi_{h+1} : \operatorname{Im} d_{h+1} = \operatorname{Ker} p_h \to \operatorname{Ker} p'_h = \operatorname{Im} d'_{h+1}$, and the induction continues. The squares give $d'_h f_h = f_{h-1} d_h$ (where $f_{-1} := \mathrm{id}_M$). □

Ungraded analogue.

Definition 4.38. *A projective cover of M is a surjective map $p : P \to M$ with P projective such that*

$$U \subseteq P \text{ a submodule and } p(U) = M \quad \Longrightarrow \quad U = P.$$

Proposition 4.39. *Let A be finite-dimensional, let P be a projective right A-module, and let $p : P \to M$ be surjective. The following are equivalent.*

(1) *p is a projective cover.*

(2) $\operatorname{Ker} p \subseteq \operatorname{rad} P$.

(3) $\operatorname{top} p : \operatorname{top} P \to \operatorname{top} M$ *is an isomorphism.*

Proof (1) ⇒ (2). If $\operatorname{Ker} p \not\subseteq \operatorname{rad} P$, there is a maximal submodule $U \subset P$ not containing $\operatorname{Ker} p$. Then $U + \operatorname{Ker} p = P$, so $p(U) = M$, contradicting (1).

(2) ⇒ (1). If $p(U) = M$, then $U + \operatorname{Ker} p = P$, so $U + \operatorname{rad} P = P$ by (2). Remark 4.19 gives $P/U = 0$.

(2) ⇔ (3). The equality $p(\operatorname{rad} P) = \operatorname{rad} M$ makes $\operatorname{top} p$ surjective, with kernel $(\operatorname{Ker} p + \operatorname{rad} P)/\operatorname{rad} P$. □

Theorem 4.40. *Let A be a finite-dimensional algebra. Every right A-module M has a projective cover; if M is finite-dimensional, so is the cover. Over any algebra, projective covers $p : P \to M$ and $q : Q \to M$ are isomorphic over M.*

Proof Uniqueness. Projectivity of P gives $f : P \to Q$ with $qf = p$. Since $q(f(P)) = M$ and q is a projective cover, $f(P) = Q$. Projectivity of Q gives $s : Q \to P$ with $fs = \mathrm{id}_Q$, and $p(s(Q)) = q(Q) = M$ forces $s(Q) = P$. So f is an isomorphism.

Existence for finite-dimensional M. Choose a surjection $p : P \to M$ from a finite-dimensional projective module with $\dim_{\Bbbk} P$ minimal (a quotient of some $A^{\oplus m}$). Suppose $U \subsetneq P$ and $p(U) = M$. Projectivity gives $u : P \to P$ with $pu = p$ and $u(P) \subseteq U$. For $r \geq 1$ large, kernels and images of powers of u have stabilised, and $P = \operatorname{Ker} u^r \oplus \operatorname{Im} u^r$. Indeed, if $u^r(x) \in \operatorname{Ker} u^r$, then $x \in \operatorname{Ker} u^{2r} = \operatorname{Ker} u^r$, so $u^r(x) = 0$; equality follows by the dimension formula. Thus $\operatorname{Im} u^r$ is projective, and $pu^r = p$ gives $p(\operatorname{Im} u^r) = M$. But $\operatorname{Im} u^r \subseteq U \subsetneq P$ contradicts minimality.

Existence for arbitrary M. Write $\operatorname{top} M \cong \bigoplus_{\lambda \in I} L_\lambda$ with simple L_λ. Take finite-dimensional covers $q_\lambda : Q_\lambda \to L_\lambda$ and lift each composite $Q_\lambda \to L_\lambda \hookrightarrow \operatorname{top} M$ through $M \to \operatorname{top} M$ to $f_\lambda : Q_\lambda \to M$. Set $P := \bigoplus_\lambda Q_\lambda$ and $p := \sum_\lambda f_\lambda$. P is projective by Proposition 4.6, and $\operatorname{top} p$ is

identified with $\bigoplus_\lambda L_\lambda \xrightarrow{\sim} \operatorname{top} M$. Hence $M = p(P) + \operatorname{rad} M$, so p is surjective by Remark 4.19, and a projective cover by Proposition 4.39. □

Example 4.41. *For $A = \Bbbk A_2$, the maps $e_{11}A \to L_1$ and $\mathrm{id} : e_{22}A \to L_2$ are projective covers: their kernels are $\operatorname{rad}(e_{11}A)$ and 0. The surjection $A = e_{11}A \oplus e_{22}A \to L_1$ killing $e_{22}A$ is not a cover, since its restriction to $e_{11}A$ is already surjective.*

Ungraded analogue. For a finite-dimensional algebra A and a right A-module M, replace the criterion (4.3) in Definition 4.27 by projective covers.

Definition 4.42. *A projective resolution* (4.1) *is minimal if $p_0 : P_0 \to M$ and the surjections $p_h : P_h \twoheadrightarrow \operatorname{Im} d_h$ for $h \geq 1$ are projective covers.*

In the diagram following Definition 4.12, this says that every p_h is a projective cover. By Proposition 4.39, minimality is equivalent to

$$\operatorname{Im} d_h \subseteq \operatorname{rad} P_{h-1} \quad \text{for all } h \geq 1. \tag{4.4}$$

Theorem 4.43. *Every right module over a finite-dimensional algebra has a minimal projective resolution; its terms are finite-dimensional if M is. Over any algebra, two minimal resolutions $P_\bullet \to M$ and $P'_\bullet \to M$ are isomorphic: there are isomorphisms $f_h : P_h \to P'_h$ with $p'_0 f_0 = p_0$ and $d'_h f_h = f_{h-1} d_h$.*

Proof As in Theorem 4.29, using Theorem 4.40; if M is finite-dimensional, choose finite-dimensional covers, whose kernels are again finite-dimensional. □

More generally, maps f_h with these identities (the vertical maps in the ladder of Theorem 4.29) form a *map of resolutions*. The kernels $\operatorname{Ker} p_0$ and $\operatorname{Ker} d_h$ are the successive *syzygies* of M; their isomorphism classes do not depend on the minimal resolution. Over an arbitrary algebra, Proposition 4.13 still gives a free resolution, but projective covers of the kernels need not exist; the obstruction is to existence of minimal resolutions, not to uniqueness.

4.6 Exercises

Exercise 4.44. *Let $A = \Bbbk[x]$ with $\deg x = 1$, and let $M = \Bbbk[x, x^{-1}]$ be the graded A-module with $M_i = \Bbbk x^i$ for all $i \in \mathbb{Z}$ and the action by multiplication. Show that $M = MA_+ \neq 0$, so $\operatorname{grtop} M = 0$. Which hypothesis of Lemma 4.17 fails?*

Exercise 4.45. *Let $A = \Bbbk[u, v]/(uv)$ with $\deg u = \deg v = 1$ and $S = A_0 = \Bbbk$. Show that the minimal graded projective resolution of $\Bbbk = A/A_+$ has $P_0 = A$ and $P_h \cong A(-h)^{\oplus 2}$ for all $h \geq 1$.*

Hint. *Show that $ua + vb = 0$ in A if and only if $a \in vA$ and $b \in uA$.*

Exercise 4.46. *Let $A = \Bbbk A_3$, graded by path length. Find the minimal graded projective resolutions of L_1, L_2 and L_3 and the tops of their terms.*

Exercise 4.47. *Let $A = \Bbbk[x]$ with $\deg x = 1$.*

(1) *Prove that $\operatorname{rad} A = 0$ whereas $A_+ = (x)$; in particular, Lemma 4.23 fails for infinite-dimensional A.*

(2) *Prove that the ungraded module $\Bbbk = A/(x)$ has no projective cover.*

5 Ext-groups

The contravariant functor $\mathrm{Hom}_A(-,N)$ is left exact (Lemma 4.9) but in general not exact (Example 4.10). The ungraded Ext-group $\mathrm{Ext}^h_A(M,N)$ measures this failure at homological degree h; it is computed by taking homology on the complex $\mathrm{Hom}_A(P_\bullet,N)$ for a projective resolution $P_\bullet$ of M.

5.1 Ungraded Ext-groups via projective resolutions

All modules in this subsection are right A-modules over an arbitrary algebra A, except where a left module is specified.

Let $P_\bullet \to M$ be a projective resolution of M, and let N be a right A-module. Postcomposition with the differentials gives the cochain complex $\mathrm{Hom}_A(P_\bullet,N)$, concentrated in degrees $h \geq 0$:

$$\mathrm{Hom}_A(P_0,N) \xrightarrow{\delta^0} \mathrm{Hom}_A(P_1,N) \xrightarrow{\delta^1} \mathrm{Hom}_A(P_2,N) \longrightarrow \cdots, \qquad \delta^h(\varphi) = \varphi \circ d_{h+1}. \tag{5.1}$$

Indeed $\delta^{h+1}\delta^h(\varphi) = \varphi\, d_{h+1}d_{h+2} = 0$.

Definition 5.1. *For right A-modules M,N and a projective resolution $P_\bullet \to M$, put*

$$\mathrm{Ext}^h_A(M,N) := H^h\big(\mathrm{Hom}_A(P_\bullet,N)\big) \qquad \textit{for all } h \geq 0.$$

The groups $\mathrm{Ext}^h_A(M,N)$ do not depend on the chosen projective resolution $P_\bullet$, up to canonical isomorphism. This is not trivial, and we should delay some details until Section 7.

> For the interested: Being a chain map (in particular, having an isomorphism of chain complexes) is overly strict if one is only interested in the (co)homology. Instead, we consider a weaker equivalence relations between chain maps called *homotopy*. Homotopic chain maps, homotopic chain maps induce the same map on (co)homology. In particular, chain maps that are homotopic to identity means we have isomorphism on the level of (co)homology, even though there may not be isomorphism on the level of compleces.
>
> The well-definedness of Ext-group is essentially about existence of homotopic inverse between any choice of resolutions of M. The precise statements and proofs are in Section 7 (Lemma 7.1 and Proposition 7.2), where they are needed for the Yoneda product; until then we take this independence for granted.

Proposition 5.2. *For all modules M,N:*

(1) $\mathrm{Ext}^0_A(M,N) \cong \mathrm{Hom}_A(M,N)$.

(2) *The following are equivalent: (a) M is projective; (b) $\mathrm{Ext}^1_A(M,N) = 0$ for all N; (c) $\mathrm{Ext}^h_A(M,N) = 0$ for all $h \geq 1$ and all N.*

Proof (1) Apply Lemma 4.9 to the exact sequence $P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} M \to 0$. It shows that $-\circ d_0 : \mathrm{Hom}_A(M,N) \to \mathrm{Hom}_A(P_0,N)$ is injective with image $\mathrm{Ker}\,\delta^0$, the cocycles in degree 0.

(2) (a)⇒(c): use the resolution with $P_0 = M$ and $P_h = 0$ for $h \geq 1$. (c)⇒(b) is clear. (b)⇒(a): choose a surjection $d_0 : P_0 \to M$ from a projective module and put $K = \mathrm{Ker}\, d_0$ with inclusion $\iota : K \hookrightarrow P_0$. Choose a surjection $\pi : P_1 \to K$ from a projective module, and extend $P_1 \xrightarrow{\iota\pi} P_0 \xrightarrow{d_0} M$ to a projective resolution $P_\bullet \to M$. Since $\mathrm{Ker}(\iota\pi) = \mathrm{Ker}\,\pi = \mathrm{Im}\, d_2$, we have $\pi d_2 = 0$, so π is a cocycle in $\mathrm{Hom}_A(P_1,K)$. As $\mathrm{Ext}^1_A(M,K) = 0$, there is $\sigma \in \mathrm{Hom}_A(P_0,K)$ with $\pi = \sigma d_1 = \sigma\iota\pi$. Since π is surjective, $\sigma\iota = \mathrm{id}_K$. So K is a direct summand of P_0 with complement $\cong P_0/K \cong M$, and M is projective. □

Thus the ungraded Ext extends Hom to higher degrees, and its vanishing in positive degrees characterises projective modules, in agreement with Proposition 4.6.

5.2 Graded Ext

From now on A is a graded algebra as in Subsection 4.3: locally finite-dimensional non-negatively graded with $S = A_0$ finite-dimensional semisimple. Unless a statement says otherwise, we make no further assumption on S.

Let $L_1, \dots, L_n$ be representatives of the isomorphism classes of simple right S-modules, regarded as graded A-modules in degree 0 via $A \twoheadrightarrow A_0 = S$.

Definition 5.3. *Let M be a bounded below locally finite-dimensional graded right module and N a graded right module. Let $P_\bullet \to M$ be a graded projective resolution, for instance a minimal one by Theorem 4.29. For $h \geq 0$, put*

$$\operatorname{ext}^h_A(M,N) := H^h\big(\operatorname{Hom}^{\mathbb{Z}}_A(P_\bullet, N)\big),$$

where $\operatorname{Hom}^{\mathbb{Z}}_A$ denotes the degree-preserving A-linear maps.

The graded Ext-group $\operatorname{ext}^h_A(M,N)$ is independent of $P_\bullet$ up to canonical isomorphism, because the proofs of Lemma 7.1 and Proposition 7.2 apply verbatim with degree-preserving maps and graded modules.

Applying this to the shifted module $N(j)$ gives $\operatorname{ext}^h_A(M, N(j))$, which records the A-linear maps raising the Adams degree by j; here h is the homological degree and j is the shift in the Adams degree, as in Notation 4.3.

Remark 5.4. A graded projective module is projective as an ungraded module, because it is a graded direct summand of a graded free module (Corollary 4.15). So $P_\bullet \to M$ is also an ordinary projective resolution. If P_h is finitely generated, then by Lemma 1.31 we have

$$\operatorname{Hom}_A(P_h, N) = \bigoplus_{j\in\mathbb{Z}} \operatorname{Hom}^{\mathbb{Z}}_A\big(P_h, N(j)\big).$$

Hence, if P_h is finitely generated for all h (for instance if A is right noetherian and M is finitely generated, by Corollary 4.36), then the ordinary $\operatorname{Ext}^h_A(M,N) = \bigoplus_{j\in\mathbb{Z}} \operatorname{ext}^h_A(M,N(j))$. It is also possible to write $\operatorname{Ext}^h_A(M,N)_j$ (or more often in the literature, $\operatorname{Ext}^{h,j}_A(M,N)$) for $\operatorname{ext}^h_A(M,N(j))$, but we will try to avoid this notation. To distinguish setting, we call $\operatorname{Ext}^h_A(M,N)$ the ungraded Ext-groups.

By the same reason in the ungraded setting, graded Ext-groups are always well-defined independent on the choice of graded projective resolution of M.

Remark 5.5 (Balancing of Ext). So far Ext was computed from a projective resolution of the first variable. It can equally be computed from an injective coresolution of the second variable: If $P_\bullet \to M$ is a (graded or ungraded) projective resolution and $N \to I^0 \to I^1 \to \cdots$ is an (graded or ungraded) injective coresolution in $\mathcal{C}$ for $\mathcal{C} = \operatorname{Mod} A$ or $\operatorname{GrMod} A$, then

$$H^h\big(\operatorname{Hom}_{\mathcal{C}}(P_\bullet, N)\big) \cong H^h\big(\operatorname{Hom}_{\mathcal{C}}(M, I^\bullet)\big) \qquad \text{for all } h \geq 0.$$

Both sides are identified with the cohomology of the total complex of the double complex $\operatorname{Hom}_{\mathcal{C}}(P_p, I^q)$, see for instance [Wei94, Theorem 2.7.6].

Proposition 5.6. *Let $P_\bullet \to M$ be a minimal graded projective resolution and N a graded right A-module with $NA_+ = 0$. Then every differential of $\operatorname{Hom}^{\mathbb{Z}}_A(P_\bullet, N)$ is zero, and for all $h \geq 0$,*

$$\operatorname{ext}^h_A(M,N) \cong \operatorname{Hom}^{\mathbb{Z}}_A(P_h, N) \cong \operatorname{Hom}_S\big(\operatorname{grtop} P_h, N\big).$$

In particular, if N is a right S-module placed in degree 0, then $\operatorname{ext}^h_A(M, N(j)) \cong \operatorname{Hom}_S\big((\operatorname{grtop} P_h)_{-j}, N\big)$ for all $j \in \mathbb{Z}$.

Proof Let $\varphi : P_h \to N$ be A-linear and degree-preserving. Then $\varphi(P_h A_+) = \varphi(P_h)A_+ \subseteq NA_+ = 0$, so φ factors uniquely through $P_h \to P_h/P_hA_+ = \operatorname{grtop} P_h$, and every degree-preserving S-linear map $\operatorname{grtop} P_h \to N$ arises this way. This gives the second isomorphism. By minimality of $P_\bullet$ (4.3), $\operatorname{Im} d_{h+1} \subseteq P_hA_+$, so $\varphi d_{h+1} = 0$. Hence all δ^h vanish and the cohomology equals the cochain groups. If N lies in degree 0, then $N(j)$ lies in degree $-j$ and $N(j)A_+ = 0$, so by the above a degree-preserving map $\operatorname{grtop} P_h \to N(j)$ only involves $(\operatorname{grtop} P_h)_{-j}$. □

Thus $\operatorname{grtop} P_h$ can be computed from graded Ext-groups, and when specialising N to a graded simple module, then we get its multiplicity in $\operatorname{grtop} P_h$ as follows:

Corollary 5.7. *Let $P_\bullet \to M$ be a minimal graded projective resolution. Then for $h \geq 0$ and $j \in \mathbb{Z}$,*

$$\dim_{D_y} \operatorname{ext}^h_A(M, L_y(j)) = \big[(\operatorname{grtop} P_h)_{-j} : L_y\big],$$

where $D_y := \operatorname{End}_A(L_y)$.

Proof By Proposition 5.6 with $N = L_y$, we get that $\operatorname{ext}^h_A(M, L_y(j)) \cong \operatorname{Hom}_S\big((\operatorname{grtop} P_h)_{-j}, L_y\big)$, which is a natural D_y-vector space (action given by postcomposition). Then just apply Schur's Lemma since S is semisimple. □

> The multiplicities $[(\operatorname{grtop} P_h)_i : L_y]$ are the graded Betti numbers $\beta_{h,i,y}(M)$ of Remark 4.33.

Example 5.8. *Let $A = \Bbbk A_2$. By Example 4.30, the minimal resolution of L_1 is $0 \to e_{22}A(-1) \to e_{11}A \to L_1 \to 0$. Here $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$, so Corollary 5.7 applies, and the only non-zero groups $\operatorname{ext}^h_A(L_1, L_y(j))$ are*

$$\operatorname{ext}^0_A(L_1, L_1) \cong \Bbbk, \qquad \operatorname{ext}^1_A(L_1, L_2(-1)) \cong \Bbbk,$$

where the basis corresponds to the projections $e_{11}A \twoheadrightarrow L_1$ and $e_{22}A(-1) \twoheadrightarrow L_2$ respectively.

Example 5.9. *For $A = \Bbbk[x]$ with $\deg x = 1$ and minimal graded free resolution of $A_0 = \Bbbk$ as in Example 4.31. Then*

$$\operatorname{ext}^0_A(\Bbbk, \Bbbk) \cong \Bbbk, \qquad \operatorname{ext}^1_A(\Bbbk, \Bbbk(-1)) \cong \Bbbk,$$

and all other groups vanish. Similarly, for $E = \Bbbk[\epsilon]/(\epsilon^2)$ with $\deg \epsilon = 1$, Example 4.32 implies that $\operatorname{ext}^h_E(\Bbbk, \Bbbk(j)) \cong \Bbbk$ if $j = -h$ and 0 otherwise.

In all three examples the non-zero groups $\operatorname{ext}^h_A(S, S(j))$ lie on the line $j = -h$, because the generators of P_h have degree h.

In general, we know that graded Ext-groups always lie below this diagonal $j = -h$.

Lemma 5.10. *For any minimal graded projective resolution $P_\bullet \to S$, we have $P_{h,<h} = 0$; equivalently, $\operatorname{ext}^h_A(S, S(j)) = 0$ for all $j > -h$.*

Proof For $h = 0$, $P_0 = A$. If P_h lives in degrees $\geq h$, then $\operatorname{Im} d_{h+1} \subseteq P_hA_+$ lives in degrees $\geq h+1$. So does $\operatorname{grtop} P_{h+1} \cong \operatorname{grtop} \operatorname{Im} d_{h+1}$ (Corollary 4.28), hence so does P_{h+1} by (4.2). □

5.3 Tor-groups

Definition 5.11. *Let M be a right A-module with a projective resolution $P_\bullet \to M$, and let N be a left A-module. The maps $d_h \otimes \mathrm{id}_N$ make $P_\bullet \otimes_A N$ a chain complex of vector spaces*

(Definition 4.11), and

$$\operatorname{Tor}_h^A(M,N) := H_h(P_\bullet \otimes_A N) \qquad \text{for all } h \geq 0.$$

Replacing $\operatorname{Hom}_A(-,N)$ by $-\otimes_A N$ in the definition of ungraded Ext gives the ungraded Tor-groups. As for Ext, this is independent of $P_\bullet$ up to canonical isomorphism and functorial in M and N, with $\operatorname{Tor}_0^A(M,N) = M \otimes_A N$.

The proofs of Lemma 7.1 and Proposition 7.2 apply with $\operatorname{Hom}_A(-,N)$ replaced by $-\otimes_A N$, which sends lifts and homotopies to lifts and homotopies. Right exactness of $-\otimes_A N$ gives $\operatorname{Tor}_0^A(M,N) = M \otimes_A N$.

As for Ext, there is a graded version, denoted by lower-case letters.

Definition 5.12. *Let M be a graded right A-module and N a graded left A-module, and let $P_\bullet \to M$ be a graded projective resolution. Then $P_\bullet \otimes_A N$ is a chain complex of graded vector spaces for the total grading of Definition 2.35, and for $h \geq 0$ we put*

$$\operatorname{tor}_h^A(M,N) := H_h\big((P_\bullet \otimes_A N)_0\big),$$

the homology of the Adams degree 0 part.

Since $(P_\bullet \otimes_A N(i))_0 = (P_\bullet \otimes_A N)_i$, the group $\operatorname{tor}_h^A(M,N(i))$ is the homology in homological degree h of the Adams degree i part of $P_\bullet \otimes_A N$. The groups $\operatorname{tor}_h^A(M,N)$ are independent of $P_\bullet$ up to canonical isomorphism, by the same proofs with degree-preserving maps.

Remark 5.13. As in Remark 5.4, $P_\bullet \to M$ is also an ordinary projective resolution. Since $P_h \otimes_A N = \bigoplus_i (P_h \otimes_A N)_i$ with degree-preserving differentials, and homology commutes with direct sums,

$$\operatorname{Tor}_h^A(M,N) = \bigoplus_{i\in\mathbb{Z}} \operatorname{tor}_h^A(M,N(i)),$$

with no finiteness assumption on P_h (unlike for ungraded Ext). We call $\operatorname{tor}_h^A(M,N)$ the *graded Tor-groups* and $\operatorname{Tor}_h^A(M,N)$ the *ungraded Tor-groups*.

We only need $N = S = A/A_+$, a graded left A-module in degree 0.

Proposition 5.14. *Let $P_\bullet \to M$ be a minimal graded projective resolution. Then $\operatorname{tor}_h^A(M,S(i)) \cong (\operatorname{grtop} P_h)_i$ as right S-modules, for all $h \geq 0$ and $i \in \mathbb{Z}$.*

Proof We have $P_h \otimes_A S = P_h/P_hA_+ = \operatorname{grtop} P_h$, so $(P_h \otimes_A S(i))_0 = (\operatorname{grtop} P_h)_i$, and the differential of $P_\bullet \otimes_A S$ is induced by d_h. It is zero because $\operatorname{Im} d_h \subseteq P_{h-1}A_+$. □

For a graded left A-module N, the graded linear dual $\mathbb{D}(N)$ is a right A-module via $(fa)(n) = f(an)$, and $\mathbb{D}(N)_i = D(N_{-i})$.

Proposition 5.15. *Let M be a bounded below lfd graded right A-module and N a graded left A-module. Then for all $h \geq 0$,*

$$\operatorname{ext}_A^h\big(M, \mathbb{D}(N)\big) \cong D\big(\operatorname{tor}_h^A(M,N)\big).$$

Proof A degree-preserving A-linear map $\varphi : P \to \mathbb{D}(N)$ is the same as a bilinear form $b(p,n) = \varphi(p)(n)$ on $P \times N$ with $b(pa,n) = b(p,an)$ that vanishes on $P_i \times N_k$ unless $i+k=0$.

This is a linear form on $(P \otimes_A N)_0$, so $\mathrm{Hom}_A^{\mathbb{Z}}(P, \mathbb{D}(N)) \cong D((P \otimes_A N)_0)$, naturally in P. Applying this to $P_\bullet$ identifies the cochain complex $\mathrm{Hom}_A^{\mathbb{Z}}(P_\bullet, \mathbb{D}(N))$ with $D((P_\bullet \otimes_A N)_0)$. Since D is exact, it commutes with taking (co)homology. □

Corollary 5.16. *Assume $S = \bigoplus_x \Bbbk e_x$ (Assumption 1.50), and let M be a bounded below lfd graded right A-module. Then $\mathrm{ext}_A^h(M, S(j)) \cong D\big((\mathrm{grtop}\, P_h)_{-j}\big)$ for all $h \geq 0$ and $j \in \mathbb{Z}$.*

Proof The dual basis $(e_x^*)_x$ of $D(S)$ satisfies $e_x^* e_y = \delta_{xy} e_x^*$, and both $\mathbb{D}(S)$ and S are killed by A_+, so $\mathbb{D}(S) \cong S$ as graded right A-modules. Now apply Propositions 5.15 and 5.14 with $N = S(-j)$, noting $\mathbb{D}(S(-j)) \cong S(j)$; the result agrees with Proposition 5.6. □

So the graded Ext-groups $\mathrm{ext}_A^h(M, S(j))$ and the graded Tor-groups $\mathrm{tor}_h^A(M, S(-j))$ carry the same information, and in the main text we only use graded Ext.

Remark 5.17 (General S). The proof of Corollary 5.16 uses the dual basis of $S = \bigoplus_x \Bbbk e_x$ to identify $\mathbb{D}(S) \cong S$. For arbitrary semisimple S the modules $\mathbb{D}(S)$ and S are still isomorphic as graded right A-modules: both are killed by A_+, hence are right S-modules, and after choosing a symmetrising form on S, Theorem 1.57 applied to the left module S gives $\mathbb{D}(S) \cong {}^*S = \mathrm{Hom}_{S^{\mathrm{op}}}(S, S) \cong S$ (see also Remark 1.59). So Corollary 5.16 holds for arbitrary S as an isomorphism of vector spaces, which depends on the choice of symmetrising form.

5.4 Detecting quadraticity via graded Ext-groups

We determine the first terms of the minimal resolution of S for algebras generated in degree 1 and for quadratic algebras.

Proposition 5.18. *Let A be as in Subsection 5.2 and let $P_\bullet \to S$ be the minimal graded projective resolution of $S = A/A_+$.*

(1) *$P_0 = A$, with $p_0 : A \to S$ the projection.*

(2) *$\mathrm{grtop}\, P_1 \cong A_+/A_+^2$. If A is generated in degree 1 over S, then $\mathrm{grtop}\, P_1 \cong A_1$ and is concentrated in degree 1, $P_1 \cong (A_1 \otimes_S A)(-1)$, and $\mathrm{ext}_A^1(S, S(j)) = 0$ for all $j \neq -1$. Conversely, if $\mathrm{ext}_A^1(S, S(j)) = 0$ for all $j \neq -1$, then A is generated in degree 1 over S.*

(3) *If A is generated in degree 1 by $V := A_1$ with relational ideal $I := \mathrm{Ker}(T_S(V) \twoheadrightarrow A)$. Then*

$$(\mathrm{grtop}\, P_2)_i \cong I_i / \big(V \otimes_S I_{i-1} + I_{i-1} \otimes_S V\big) \quad \text{for all } i.$$

In particular, if $A = T_S(V)/(R)$ is quadratic with $V = A_1$ and $R \subseteq V \otimes_S V$, then $\mathrm{grtop}\, P_2 \cong R$, concentrated in degree 2, $P_2 \cong (R \otimes_S A)(-2)$, and $\mathrm{ext}_A^2(S, S(j)) = 0$ for all $j \neq -2$. Conversely, if A is generated in degree 1 and $\mathrm{ext}_A^2(S, S(j)) = 0$ for all $j \neq -2$, then A is quadratic, with $A = T_S(V)/(I_2)$.

Proof By Proposition 5.6, $\mathrm{ext}_A^h(S, S(j)) \cong \mathrm{Hom}_S\big((\mathrm{grtop}\, P_h)_{-j}, S\big)$, which is non-zero if and only if $(\mathrm{grtop}\, P_h)_{-j} \neq 0$, since every simple S-module is a direct summand of S. So $\mathrm{ext}_A^h(S, S(j)) = 0$ for all $j \neq -h$ if and only if $\mathrm{grtop}\, P_h$ is concentrated in degree h; this is how the statements about graded Ext below follow from those about $\mathrm{grtop}\, P_h$.

(1) The module A is projective and $\mathrm{Ker}\, p_0 = A_+ = AA_+$, so $\mathrm{Ker}\, p_0 \subseteq P_0 A_+$ and p_0 is the first step of a minimal resolution.

(2) Put $K_1 = \mathrm{Ker}\, p_0 = A_+$. By Corollary 4.28, $\mathrm{grtop}\, P_1 \cong \mathrm{grtop}\, K_1 = A_+/A_+^2$. If A is generated in degree 1, then $A_+^2 = \bigoplus_{i \geq 2} A_i$, so $\mathrm{grtop}\, K_1 = A_1$ in degree 1. By (4.2), $P_1 \cong (A_1 \otimes_S A)(-1)$, and the surjection $P_1 \to K_1$ is $v \otimes a \mapsto va$. Conversely, if $\mathrm{grtop}\, P_1 = A_+/A_+^2$ is concentrated in degree 1, then

$A_i = (A_+^2)_i = \sum_{0<p<i} A_p A_{i-p}$ for all $i \geq 2$. By induction on i this gives $A_i = A_1^i$, so A is generated in degree 1.

(3) Put $T = T_S(V)$, so $A_i = T_i/I_i$ and $I_1 = 0$. Since every S-module is projective, $V \otimes_S -$ is exact, so

$$P_{1,i} = V \otimes_S A_{i-1} = T_i/(V \otimes_S I_{i-1}),$$

and $d_1 : P_{1,i} \to A_i = T_i/I_i$ is induced by the identity on T_i. Hence $K := \operatorname{Ker} d_1$ has $K_i = I_i/(V \otimes_S I_{i-1})$. Since $A_j = A_1^j$, we have $(KA_+)_i = K_{i-1}A_1$, which is the image of $I_{i-1} \otimes_S V$. Therefore $(K/KA_+)_i = I_i/(V \otimes_S I_{i-1} + I_{i-1} \otimes_S V)$, and $\operatorname{grtop} P_2 \cong \operatorname{grtop} K$ by Corollary 4.28.

If $I = (R)$, then $I_i = \sum_{p+q=i-2} T_p R T_q$. For $i \geq 3$ each summand with $p \geq 1$ lies in $V \otimes_S I_{i-1}$, and each with $p = 0$ lies in $I_{i-1} \otimes_S V$. So $I_i = V \otimes_S I_{i-1} + I_{i-1} \otimes_S V$ for $i \geq 3$, while $I_1 = 0$ and $I_2 = R$. Thus $\operatorname{grtop} P_2 = R$ in degree 2, and (4.2) gives $P_2 \cong (R \otimes_S A)(-2)$.

Conversely, suppose A is generated in degree 1 and $\operatorname{grtop} P_2$ is concentrated in degree 2. By the formula for $(\operatorname{grtop} P_2)_i$, we have $I_i = V \otimes_S I_{i-1} + I_{i-1} \otimes_S V$ for all $i \geq 3$. By induction on i this gives $I_i \subseteq (I_2)$, so $I = (I_2)$ and $A = T_S(V)/(I_2)$ is quadratic. □

5.5 Exercises

Exercise 5.19. *Let $P_\bullet \to M$ be a projective resolution and $K_1 = \operatorname{Ker} d_0$. Prove that*

$$\operatorname{Ext}_A^1(M, N) \cong \operatorname{coker}\big(\operatorname{Hom}_A(P_0, N) \to \operatorname{Hom}_A(K_1, N)\big),$$

where the map is restriction, and that $\operatorname{Ext}_A^h(M, N) \cong \operatorname{Ext}_A^{h-1}(K_1, N)$ for $h \geq 2$.

Exercise 5.20. *Let $A = \Bbbk A_3/(e_{13})$ with the grading by path length. Find the minimal graded projective resolution of L_1, and compute $\dim_\Bbbk \operatorname{ext}_A^h(L_1, L_y(j))$ for all h, y, j.*

Exercise 5.21. *Let $A = \Bbbk[x]$ with $\deg x = 1$ and $M = A/(x^m)$ for some $m \geq 1$. Find a minimal graded resolution of M, and compute $\operatorname{grtop} P_h$ and $\operatorname{ext}_A^h(M, \Bbbk(j))$ for all h, j.*

Exercise 5.22. *Let A be as in Subsection 5.2.*

(1) *Prove that $\operatorname{ext}_A^1(S, S(j)) \cong \operatorname{Hom}_S\big((A_+/A_+^2)_{-j}, S\big)$ for all $j \in \mathbb{Z}$.*

(2) *Prove that A is generated in degree 1 over S if and only if $\operatorname{ext}_A^1(S, S(j)) = 0$ for all $j \neq -1$.*

6 Koszul algebras and the Koszul complex

Let A be a graded algebra (i.e. a lfd with $S = A_0$ finite-dimensional semisimple and $A_{<0} = 0$). By Theorem 4.29 the module $S = A/A_+$ has a minimal graded projective resolution $P_\bullet \to S$, unique up to isomorphism, and by Proposition 5.6 it records

$$\operatorname{ext}^h_A(S, S(j)) \cong \operatorname{Hom}_S\big((\operatorname{grtop} P_h)_{-j}, S\big).$$

A Koszul algebra is one for which these graded Ext-groups vanish outside the diagonal $j = -h$.

6.1 Koszul algebras

Definition 6.1. *A graded module M is generated in degre i if $M = M_i A$.*

A graded projective resolution $P_\bullet \to M$ with every P_h bounded below and lfd is linear if P_h is generated in degree h, that is, $P_h = P_{h,h}A$, for every $h \geq 0$.

A linear resolution is automatically minimal: $P_{h-1,h} = P_{h-1,h-1}A_1 \subseteq P_{h-1}A_+$, and d_h maps $P_h = P_{h,h}A$ into $P_{h-1,h}A \subseteq P_{h-1}A_+$, so $P_\bullet$ is minimal (4.3).

Definition 6.2. *The algebra A is Koszul if the graded right module $S = A/A_+$ has a linear graded projective resolution.*

The generators in homological degree h must then all have Adams degree h.

Corollary 6.3. *Every Koszul algebra is quadratic.*

Proof A linear resolution $P_\bullet \to S$ is minimal, hence is the minimal graded projective resolution of S. Since $P_h = P_{h,h}A$, we have $P_{h,i} = 0$ for $i < h$, and $P_{h,i} = P_{h,h}A_{i-h} \subseteq P_hA_+$ for $i > h$. So $\operatorname{grtop} P_h$ is concentrated in degree h. For $h = 1$, Proposition 5.18(2) shows that A is generated in degree 1; for $h = 2$, Proposition 5.18(3) shows that A is quadratic. □

The converse fails: not every quadratic algebra is Koszul (Exercise 6.11 introduces an example; its non-Koszulity will be proved later using Hilbert series), so Koszulness is a genuine condition on the graded Ext-groups $\operatorname{ext}^h_A(S, S(j))$ in homological degrees ≥ 3.

Example 6.4. *For $A = \Bbbk A_2$, the resolution in Example 4.30 is linear for L_1, and $L_2 = e_{22}A$ is projective in degree 0. Taking their direct sum gives a linear resolution of $S = L_1 \oplus L_2$; hence $\Bbbk A_2$ is Koszul.*

Example 6.5. *The exterior algebra $E = \Bbbk[\epsilon]/(\epsilon^2)$ is Koszul: Example 4.32 gives a resolution of $\Bbbk$ with $P_h = E(-h)$, which is linear although infinite.*

Recall from Lemma 5.10 that for an arbitrary graded algebra A we always have $\operatorname{ext}^h_A(S, S(j)) = 0$ for all $j > -h$. Koszulity refines this to the diagonal $j = -h$:

Proposition 6.6 (Characterisation by diagonality of graded Ext). *The following are equivalent.*

(1) *A is Koszul.*

(2) *$\operatorname{grtop} P_h = P_{h,h}$ for all $h \geq 0$, where $P_\bullet \to S$ is a minimal graded projective resolution of S.*

(3) *The graded Ext-groups are concentrated in the diagonal, i.e. $\operatorname{ext}^h_A(S, S(j)) = 0$ for all $j \neq -h$.*

(4) *Let M and N be lfd modules M, N concentrated in degrees m and n respectively. Then we have $\operatorname{ext}^h_A(M, N(j)) = 0$ unless $j = n - m - h$.*

Proof (1)⇒(2): A linear graded projective resolution is minimal, hence isomorphic to $P_\bullet$ by Theorem 4.29, and $\operatorname{grtop} P_h = P_{h,h}$.

(2)⇒(1): By (4.2), $P_h \cong ((\operatorname{grtop} P_h)_h \otimes_S A)(-h)$ is generated in degree h.

(2)⇔(3): By Proposition 5.6, we have $\operatorname{ext}^h_A(S, S(j)) \cong \operatorname{Hom}_S((\operatorname{grtop} P_h)_{-j}, S)$, so it is non-zero if and only if $(\operatorname{grtop} P_h)_{-j} \neq 0$, since every simple S-module is a direct summand of S. So (2) says that $\operatorname{grtop} P_h$ is concentrated in degree h.

(2)⇒(4): On one hand, the condition on M we have $M \in \operatorname{add}_{\operatorname{grmod} A}(S(-m))$. On the other hand, the condition on N implies that $NA_+ = 0$. By Proposition 5.6, we have

$$\operatorname{ext}^h_A(S(-m), N(j)) = \operatorname{ext}^h_A(S, N(j+m)) = \operatorname{Hom}_S(\operatorname{grtop} P_h, N(j+m)) = \operatorname{Hom}_S(P_{h,h}, N(j+m)).$$

Since $N = N_n$, $N(j+m)$ is concentrated in $n - j - m$. Hence, comparing the degrees, the last Hom-space vanishes unless $n - j - m = h$.

(4)⇒(3): This is the case $M = N = S$. □

6.2 Koszulity of the opposite ring

Strictly speaking, we have only defined *right* Koszul algebra, since the definition uses the right A-module S. In this subsection, we show that being right Koszul and being left Koszul are equivalent conditions. The trick is nothing but to utilises the fact that linear duality allows us to switch side.

In the following, for clarity, we write S^{op} for the left A-module $S = A/A_+$, regarded as a right A^{op}-module. Recall from (1.8) that the graded linear dual $\mathbb{D}(P)$ of a graded right A-module P is a graded left A-module by $(af)(p) = f(pa)$, that is, a graded right A^{op}-module. Linear duality is exact (1.9), and by Lemma 1.47(2) it is a duality

$$\mathbb{D} : (\operatorname{grmod} A)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod}(A^{\mathrm{op}})$$

between locally finite-dimensional graded right A-modules and locally finite-dimensional graded left A-modules.

Lemma 6.7. *Let M and N be finite-dimensional graded right A-modules. Then $\operatorname{ext}^h_{A^{\mathrm{op}}}(\mathbb{D}N, \mathbb{D}M) \cong \operatorname{ext}^h_A(M, N)$ for all $h \geq 0$.*

Proof First note that $\mathbb{D}$ takes graded projectives to graded injectives: the characterisation Proposition 4.6 is completely dual (reversing arrows and compositions) for injectives.

Let $P_\bullet \to M$ be the minimal graded projective resolution, so every P_q is lfd. The remark in the previous paragraph says that dualising yields an injective coresolution $0 \to \mathbb{D}(M) \to \mathbb{D}(P_0) \to \mathbb{D}(P_1) \to \cdots$ of lfd graded left modules $\mathbb{D}(M) \in \operatorname{grmod}(A^{\mathrm{op}})$. By the balancing of Ext (Remark 5.5) over A^{op}, we have

$$\operatorname{ext}^h_{A^{\mathrm{op}}}(\mathbb{D}N, \mathbb{D}M) \cong H^h\big(\operatorname{Hom}^{\mathbb{Z}}_{A^{\mathrm{op}}}(\mathbb{D}N, \mathbb{D}(P_\bullet))\big) \cong H^h\big(\operatorname{Hom}^{\mathbb{Z}}_A(P_\bullet, N)\big) = \operatorname{ext}^h_A(M, N),$$

where the second isomorphism is induced by the natural isomorphism $\operatorname{Hom}^{\mathbb{Z}}_{A^{\mathrm{op}}}(\mathbb{D}N, \mathbb{D}(-)) \cong \operatorname{Hom}^{\mathbb{Z}}_A(-, N)$ using the fact that $\mathbb{D}$ is a duality. □

Proposition 6.8. *A is Koszul if and only if the opposite algebra A^{op} is Koszul.*

Proof First, we show that $S^{\mathrm{op}} \cong \mathbb{D}(S)$ in $\operatorname{grmod} A^{\mathrm{op}}$. Indeed, if S is of the form $\bigoplus_x \Bbbk e_x$ as in Assumption 1.50, then the isomorphism is given by sending e_x to its dual $\Bbbk$-basis.

For more general finite-dimensional semisimple S, choose a symmetrising form τ. Then the desired isomorphism is given by $s \mapsto \tau(a \cdot -)$ – this was already explained in the first step of the proof of Theorem 1.57.

Hence, by combining Lemma 6.7, we have

$$\begin{aligned}\operatorname{ext}^h_{A^{\mathrm{op}}}\big(S^{\mathrm{op}}, S^{\mathrm{op}}(j)\big) &\cong \operatorname{ext}^h_{A^{\mathrm{op}}}\big(\mathbb{D}S, \mathbb{D}(S)(j)\big) \cong \operatorname{ext}^h_{A^{\mathrm{op}}}\big(\mathbb{D}S, \mathbb{D}(S(-j))\big)\\ &\cong \operatorname{ext}^h_A(S(-j), S) \cong \operatorname{ext}^h_A(S, S(j))\end{aligned}$$

holds for all $h \geq 0$ and $j \in \mathbb{Z}$. The claim now follows from Proposition 6.6(3). □

Corollary 6.9. *The following are equivalent for a graded algebra A.*

- *A is Koszul.*
- *$S \in \operatorname{grmod} A$ has a linear graded (right) projective resolution.*
- *$S \in \operatorname{grmod} A^{\mathrm{op}}$ has a linear graded (left) projective resolution.*
- *$S \in \operatorname{grmod} A$ has a linear graded (right) injective coresolution.*
- *$S \in \operatorname{grmod} A^{\mathrm{op}}$ has a linear graded (left) injective coresolution.*

6.3 Exercises

Exercise 6.10. *Let $A = \Bbbk[x]/(x^3)$ with $\deg x = 1$. Find the minimal graded projective resolution of $\Bbbk$, and show that P_h is generated in degree $\lfloor 3h/2 \rfloor$ for all $h \geq 0$. Deduce that A is not Koszul. Is A quadratic?*

Exercise 6.11. *Let $A = \Bbbk\langle x, y\rangle/(yx,\ x^2 + y^2)$ with $\deg x = \deg y = 1$, a quadratic algebra with $S = \Bbbk$. Show that $x^3 = 0$ and $y^3 = 0$, and deduce that $A_n = 0$ for $n \geq 3$ and $h_A(t) = 1 + 2t + 2t^2$. (This algebra is not Koszul; we will prove this in a later lecture.)*

Hint. *$x^3 = x \cdot x^2 = -xy^2$ and $x^3 = x^2 \cdot x = -y^2x = -y(yx)$, and similarly for the other words of length 3.*

6.4 The Koszul complex

Let $A = T_S(V)/(R)$ be a quadratic algebra with $V = A_1$ and $R \subseteq V \otimes_S V$. For a Koszul algebra, which is quadratic (by Corollary 6.3), Proposition 5.18 shows that the minimal graded projective resolution of S begins

$$\cdots \to R \otimes_S A(-2) \to V \otimes_S A(-1) \to A \to S \to 0,$$

and all its generators lie in the diagonal degrees; so one expects the whole resolution to be built from V and R alone. The Koszul complex continues this pattern: it is a complex of projective modules, defined from V and R for every quadratic algebra, whose terms in homological degrees $0, 1, 2$ are the ones above. It turns out to be the part of the minimal graded projective resolution of S that is generated in the "diagonal" degrees, and it is a resolution exactly when A is Koszul.

Define S-bimodules $\mathcal{K}_0 = S$, $\mathcal{K}_1 = V$, and

$$\mathcal{K}_h := \bigcap_{\nu=0}^{h-2} V^{\otimes_S \nu} \otimes_S R \otimes_S V^{\otimes_S (h-2-\nu)} \qquad \forall h \geq 2.$$

Note that $\mathcal{K}_2 = R$. These are all S-sub-bimodules of $V^{\otimes_S h}$. Since S being semisimple implies that $\operatorname{mod} S = \operatorname{add} S = \operatorname{proj} S$, the functors $- \otimes_S Y$ and $Y \otimes_S -$ are exact (see this using commutation of $\oplus$ and $\otimes$ on decomposition of free modules), so they preserve inclusions and finite intersections of submodules. In particular, if $X \subseteq V^{\otimes_S a}$ and $X' \subseteq V^{\otimes_S b}$ are sub-bimodules, then we have canonical injective map from $X \otimes_S V^{\otimes_S b}$, $V^{\otimes_S a} \otimes_S X'$ and $X \otimes_S X'$ to $V^{\otimes_S (a+b)}$; hence we can regard them as sub-bimodules of $V^{\otimes_S (a+b)}$ canonically.

Lemma 6.12. *For $h \geq 2$, the following are equalities of sub-bimodules of $V^{\otimes_S h}$, where $\mathcal{K}_{h-1} \otimes_S V$ and $\mathcal{K}_{h-2} \otimes_S R$ are regarded as submodules as explained above:*

$$\mathcal{K}_h = \big(\mathcal{K}_{h-1} \otimes_S V\big) \cap \big(V^{\otimes_S (h-2)} \otimes_S R\big) = \big(\mathcal{K}_{h-1} \otimes_S V\big) \cap \big(\mathcal{K}_{h-2} \otimes_S R\big).$$

In particular $\mathcal{K}_h \subseteq \mathcal{K}_{h-1} \otimes_S V$, and more generally $\mathcal{K}_{h+k} \subseteq \mathcal{K}_h \otimes_S \mathcal{K}_k$.

Proof Since $- \otimes_S V$ preserves intersections, $\mathcal{K}_{h-1} \otimes V = \bigcap_{\nu=0}^{h-3} V^{\otimes \nu} \otimes R \otimes V^{\otimes (h-2-\nu)}$. These are the conditions defining $\mathcal{K}_h$ with $\nu \leq h-3$, and the remaining condition is $\nu = h-2$, namely $V^{\otimes (h-2)} \otimes R$. This gives the first equality. For the second, $\mathcal{K}_{h-1} \otimes V \subseteq \mathcal{K}_{h-2} \otimes V \otimes V$, and $(\mathcal{K}_{h-2} \otimes V \otimes V) \cap (V^{\otimes (h-2)} \otimes R) = \mathcal{K}_{h-2} \otimes R$, as one sees by choosing an S-complement of $\mathcal{K}_{h-2}$ in $V^{\otimes (h-2)}$. Finally, $\mathcal{K}_{h+k}$ satisfies the conditions defining $\mathcal{K}_h$ on the first h factors and those defining $\mathcal{K}_k$ on the last k factors. □

Definition 6.13. *The Koszul complex of A is $K_\bullet = (K_\bullet, d_\bullet)$ with*

$$K_h := \mathcal{K}_h \otimes_S A(-h) \qquad (h \geq 0), \qquad K_0 = A,$$

where $\mathcal{K}_h$ is placed in degree 0 before the shift, so that K_h is generated in degree h. The differential $d_h : K_h \to K_{h-1}$ is the composition

$$\mathcal{K}_h \otimes_S A \xrightarrow{\iota_h \otimes \mathrm{id}_A} \mathcal{K}_{h-1} \otimes_S V \otimes_S A \xrightarrow{\mathrm{id}_{\mathcal{K}_{h-1}} \otimes \mu} \mathcal{K}_{h-1} \otimes_S A, \qquad (z \otimes v) \otimes a \longmapsto z \otimes va,$$

where $\iota_h : \mathcal{K}_h \hookrightarrow \mathcal{K}_{h-1} \otimes_S V$ is the inclusion of Lemma 6.12 and μ is the multiplication map.

The maps d_h are right A-linear and degree-preserving. Koszul complex is an honest complex $d_{h-1} d_h = 0$: for $h \geq 2$, by Lemma 6.12 we have $\mathcal{K}_h \subseteq \mathcal{K}_{h-2} \otimes_S R$, and $z \otimes r \otimes a \mapsto z \otimes \mu(ra) = 0$ for $z \in \mathcal{K}_{h-2}, r \in R, a \in A$.

We also have an exact sequence $K_1 \to K_0 \to S \to 0$ where $d_0 : K_0 = A \to A/A_+ = S$ is the projection; the exactness at K_0 follows from the definition that $VA \subseteq A_+$.

Theorem 6.14. *Let $P_\bullet \to S$ be a minimal graded projective resolution of a quadratic algebra A. Then $P_h^\Delta := P_{h,h} A \subseteq P_h$ is a subcomplex of $P_\bullet$, and $P_\bullet^\Delta \cong K_\bullet$ as complexes. In particular, $P_{h,h} \cong \mathcal{K}_h$ for all h.*

Thus the Koszul complex is the 'diagonal part' of the minimal graded projective resolution of S for a quadratic algebra.

Proof By Lemma 5.10, P_h lives in degrees $\geq h$, so $(P_h A_+)_h = 0$ and $P_{h,h} \cong (\operatorname{grtop} P_h)_h$. Choose a decomposition of P_h so that $P_h^{[i]}$ is the maximal direct summand generated in degree i. Hence, we have $P_h^{[h]} = P_{h,h} A \cong P_{h,h} \otimes_S A(-h)$ by Lemma 4.26.

Step 1: $P_\bullet^\Delta$ is a subcomplex of $P_\bullet$. It is enough to show that $d_h(P_{h,h}) \subseteq P_{h-1,h-1} \otimes_S V$. Indeed, by minimality (4.3) and that d_h is degree-preserving, we have $d_h(P_{h,h}) \subseteq (P_{h-1} A_+)_h$. Since $P_{h-1}^{[i]} A_+$ with $i \geq h$ contribute only in degrees $\geq h+1$, we have

$$(P_{h-1} A_+)_h = P_{h-1,h-1} A_1 \cong P_{h-1,h-1} \otimes_S V,$$

as claimed.

Step 2: $d_h : P_{h,h} \xrightarrow{\sim} (\operatorname{Ker} d_{h-1})_h$ is an isomorphism for $h \geq 1$. We have

$$\begin{aligned} (\operatorname{Ker} d_{h-1})_h &\overset{(*)}{=} (\operatorname{Im} d_h)_h \overset{(**)}{=} d_h(P_{h,h}) \overset{\text{Step 1}}{\subseteq} P^{\Delta}_{h-1}, \quad \text{and} \\ \operatorname{Ker}\big(d_h|_{P_{h,h}}\big) = (\operatorname{Ker} d_h)_h &\overset{(*)}{=} (\operatorname{Im} d_{h+1})_h \overset{(**)}{=} 0, \end{aligned}$$

where, $(*)$ is the exactness of $P_\bullet$, and $(**)$ is $P_h = P_{h,\geq h}$ for all $h \geq 1$ and d_h being degree-preserving.

Step 3: $d_h : K_{h,h} \xrightarrow{\sim} (\operatorname{Ker} d_{h-1})_h$ is an isomorphism for $h \geq 1$. Here $K_{h,h} = \mathcal{K}_h$, and d_h restricts to the inclusion $\iota_h : \mathcal{K}_h \hookrightarrow \mathcal{K}_{h-1} \otimes_S V = K_{h-1,h}$, so it is injective. For $h = 1$, we have $(\operatorname{Ker} d_0)_1 = A_1 = V = \iota_1(\mathcal{K}_1)$. For $h \geq 2$,

$$\begin{aligned} (\operatorname{Ker} d_{h-1})_h &\overset{(1)}{=} \operatorname{Ker}\Big(\mathcal{K}_{h-1} \otimes_S V \xrightarrow{\iota_{h-1}\otimes \mathrm{id}_V} \mathcal{K}_{h-2} \otimes_S V \otimes_S V \xrightarrow{\mathrm{id}\otimes \mu_2} \mathcal{K}_{h-2} \otimes_S A_2\Big) \\ &\overset{(2)}{=} (\mathcal{K}_{h-1} \otimes_S V) \cap (\mathcal{K}_{h-2} \otimes_S R) \\ &\overset{(3)}{=} \iota_h(\mathcal{K}_h), \end{aligned}$$

where $\mu_2 : V \otimes_S V \to A_2$ is the multiplication, and

(1) is the definition of d_{h-1} in degree h, as $K_{h-1,h} = \mathcal{K}_{h-1} \otimes_S A_1$ and $K_{h-2,h} = \mathcal{K}_{h-2} \otimes_S A_2$;

(2) holds because $\iota_{h-1} \otimes \mathrm{id}_V$ is injective, $\mathcal{K}_{h-2} \otimes_S -$ is exact, and $R = \operatorname{Ker} \mu_2$;

(3) is Lemma 6.12.

Step 4: $K_\bullet \cong P^{\Delta}_\bullet$ as complexes. Since $K_h \cong K_{h,h} \otimes_S A(-h)$ and $P^{\Delta}_h \cong P_{h,h} \otimes_S A(-h)$, an S-linear isomorphism $\varphi_h : K_{h,h} \to P_{h,h}$ extends uniquely to an isomorphism of graded right A-modules

$$\Phi_h : K_h \xrightarrow{\sim} P^{\Delta}_h, \qquad \Phi_h(za) = \varphi_h(z)a.$$

We construct φ_h by induction on h so that $d_h\Phi_h = \Phi_{h-1}d^K_h$. Start with $\Phi_0 = \mathrm{id}_A$, which commutes with the maps d_0 to S. For $h \geq 1$, define φ_h as the composite

$$\varphi_h : \; K_{h,h} \xrightarrow[\sim]{(1)} (\operatorname{Ker} d^K_{h-1})_h \xrightarrow[\sim]{(2)} (\operatorname{Ker} d_{h-1})_h \xrightarrow[\sim]{(3)} P_{h,h},$$

where

(1) is d^K_h, an isomorphism by Step 3;

(2) is Φ_{h-1}, an isomorphism because $\Phi_{h-1} : K_{h-1} \to P^{\Delta}_{h-1}$ is an isomorphism commuting with d_{h-1} by induction, so it maps the kernel onto the kernel (recall from Step 2 that $(\operatorname{Ker} d_{h-1})_h \subseteq P^{\Delta}_{h-1}$);

(3) is the inverse of $d_h|_{P_{h,h}}$, an isomorphism by Step 2.

By construction $d_h\varphi_h = \Phi_{h-1}d^K_h$ on $K_{h,h}$. Both sides are A-linear and $K_h = K_{h,h}A$, so $d_h\Phi_h = \Phi_{h-1}d^K_h$ on all of K_h. Hence $\Phi_\bullet : K_\bullet \to P^{\Delta}_\bullet$ is an isomorphism of complexes, and in particular $P_{h,h} \cong K_{h,h} = \mathcal{K}_h$. □

Corollary 6.15. *A quadratic algebra A is Koszul if and only if its Koszul complex $K_\bullet \to S$ is exact.*

Proof If $K_\bullet \to S$ is exact, it is a graded projective resolution with K_h generated in degree h, hence linear. Conversely, if A is Koszul, the minimal graded projective resolution is linear, so $P_h = P_{h,h}A = P^{\Delta}_h$ for all h. Thus $P_\bullet = P^{\Delta}_\bullet \cong K_\bullet$ is exact. □

The terms $\mathcal{K}_h$ are dual to the quadratic dual of Section 3.

Proposition 6.16. *For a quadratic algebra $A = T_S(V)/(R)$, the composite*

$$\rho_h : D(V)^{\otimes_S h} \xrightarrow{\theta} D(V^{\otimes_S h}) \xrightarrow{\text{res}} D(\mathcal{K}_h),$$

where θ is induced by the isomorphism $\theta_{V,V}$ of Proposition 3.3 and res *is the dual of canonical embedding $\mathcal{K}_h \hookrightarrow V^{\otimes_S h}$, is surjective with kernel $\operatorname{Ker}(\rho_h) = (A^!\theta^{-1}_{V,V}(R^\perp)A^!)_h$. Hence ρ_h induces $A^!_h \cong D(\mathcal{K}_h)$, and $K_h \cong D(A^!_h) \otimes_S A(-h)$.*

Proof By Definition 3.12, the degree h part of the ideal is $J_h = \sum_{p+q=h-2} D(V)^{\otimes p} \otimes \theta^{-1}(R^\perp) \otimes D(V)^{\otimes q}$. The iterated $\theta : D(V)^{\otimes h} \to D(V^{\otimes h})$ is an isomorphism that reverses the order of the factors, and it sends the (p,q) summand onto the annihilator of $V^{\otimes q} \otimes R \otimes V^{\otimes p}$. Since the annihilator of a sum of subspaces is the intersection of the annihilators, and conversely, $\theta(J_h)$ is the annihilator of $\bigcap_{p+q=h-2} V^{\otimes q} \otimes R \otimes V^{\otimes p} = \mathcal{K}_h$. Restriction $D(V^{\otimes h}) \to D(\mathcal{K}_h)$ is surjective with kernel $\mathcal{K}_h^\perp$, so ρ_h is surjective with kernel $\theta^{-1}(\mathcal{K}_h^\perp) = J_h$. The last claim follows by dualising, as $D(D(\mathcal{K}_h)) \cong \mathcal{K}_h$. □

The slogan of this proposition is:

$K_\bullet$ is "$D(A^!_\bullet) \otimes_S A$".
The quadratic dual $A^!$ records (the generators of)
the diagonal part of the minimal graded projective resolution of S over A.

6.5 Koszul complex of the polynomial ring

Let V be n-dimensional with basis $x_1, \dots, x_n$, and $A = \mathsf{S}(V) = \Bbbk[x_1, \dots, x_n] = T(V)/(R)$ with $R = \langle v \otimes w - w \otimes v \rangle$. We have a map

$$\bigwedge^h V \to V^{\otimes h}, \qquad v_1 \wedge \cdots \wedge v_h \mapsto \sum_\sigma \operatorname{sgn}(\sigma) v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(h)},$$

where σ varies over of all permutations of h letters. This is an injective map whose image lies in $\mathcal{K}_h$ ($\because$ pairing terms that differ by a transposition $(i+1, i+2)$ gives elements of $V^{\otimes \nu} \otimes R \otimes V^{\otimes(h-2-\nu)}$). By Proposition 6.16 and Proposition 3.20, we have

$$\dim \mathcal{K}_h = \dim \bigwedge^h D(V) = \dim \bigwedge^h V = \binom{n}{h}.$$

Hence,

$$\mathcal{K}_h \cong \bigwedge^h V, \qquad K_h \cong \bigwedge^h V \otimes_\Bbbk A(-h), \qquad K_0 = A.$$

To compute the differential, we express the inclusion $\iota_h : \mathcal{K}_h \hookrightarrow \mathcal{K}_{h-1} \otimes_S V$ of Lemma 6.12 under these identifications. In the antisymmetrisation of $v_1 \wedge \cdots \wedge v_h$, group the permutations σ according to the vector v_r sitting in the last tensor factor, that is, $\sigma(h) = r$. Moving v_r to the last position takes $h - r$ transpositions, and the remaining $h - 1$ tensor factors then run over the antisymmetrisation of $v_1, \dots, \widehat{v_r}, \dots, v_h$. Hence

$$\iota_h(v_1 \wedge \cdots \wedge v_h) = \sum_{r=1}^{h} (-1)^{h-r} \, (v_1 \wedge \cdots \widehat{v_r} \cdots \wedge v_h) \otimes v_r \in \bigwedge^{h-1} V \otimes V = \mathcal{K}_{h-1} \otimes V,$$

and composing with $\mathrm{id} \otimes \mu$ gives the differential

$$d_h\big((v_1 \wedge \cdots \wedge v_h) \otimes a\big) = \sum_{r=1}^{h} (-1)^{h-r} \, (v_1 \wedge \cdots \widehat{v_r} \cdots \wedge v_h) \otimes v_r a.$$

Proposition 6.17. *For $A = \mathsf{S}(V)$ the Koszul complex, with $d_0 : K_0 = A \to \Bbbk$ the projection,*

$$0 \longrightarrow K_n \xrightarrow{d_n} \cdots \xrightarrow{d_2} K_1 \xrightarrow{d_1} K_0 \xrightarrow{d_0} \Bbbk \longrightarrow 0$$

is exact.

Proof The idea is to split the complex into finite-dimensional pieces by a multigrading and to contract each piece by an explicit homotopy.

Step 1: multigrading. Give $A = \Bbbk[x_1, \dots, x_n]$ the $\mathbb{Z}^n_{\geq 0}$-grading with $\deg x_j$ the jth standard basis vector, and $\bigwedge V$ the one with $\deg(x_{j_1} \wedge \cdots \wedge x_{j_h}) = \mathbf{1}_J := \sum_r \mathbf{e}_{j_r}$ for $J = \{j_1 < \cdots < j_h\}$. Then $K_h = \bigwedge^h V \otimes_\Bbbk A$ is multigraded, and each d_h preserves the multidegree because it moves a factor x_{j_r} from the wedge to A. Hence $K_\bullet = \bigoplus_\alpha K_{\bullet,\alpha}$ as complexes, and it suffices to show that each multidegree piece $K_{\bullet,\alpha}$ is exact (in multidegree $\alpha = 0$ together with d_0).

Step 2: a basis and the differential in it. For $\alpha \in \mathbb{Z}^n_{\geq 0}$ put $\operatorname{supp}(\alpha) = \{j \mid \alpha_j > 0\}$. The space $K_{h,\alpha}$ has basis

$$b_J := (x_{j_1} \wedge \cdots \wedge x_{j_h}) \otimes x^{\alpha - \mathbf{1}_J}, \qquad J = \{j_1 < \cdots < j_h\} \subseteq \operatorname{supp}(\alpha),$$

since $x_J \otimes x^\beta$ has multidegree α exactly when $\beta = \alpha - \mathbf{1}_J$, which requires $J \subseteq \operatorname{supp}(\alpha)$. In particular $K_{h,\alpha} = 0$ for $h > |\operatorname{supp}(\alpha)|$. Since $x_{j_r}\, x^{\alpha - \mathbf{1}_J} = x^{\alpha - \mathbf{1}_{J\setminus\{j_r\}}}$, the formula for d_h gives

$$d_h(b_J) = \sum_{r=1}^{h} (-1)^{h-r}\, b_{J\setminus\{j_r\}}.$$

Step 3: normalising the sign. Replacing d_h by $(-1)^{h-1} d_h$ for all $h \geq 1$ does not affect exactness: multiplication by $(-1)^{h(h-1)/2}$ on K_h is an isomorphism from the complex with the old differentials to the one with the new differentials, and it is the identity on K_0 and K_1. So we may and do assume

$$d_h(b_J) = \sum_{r=1}^{h} (-1)^{r-1}\, b_{J\setminus\{j_r\}},$$

that is, the differential deletes one element of J at a time with alternating signs.

Step 4: a contracting homotopy for $\alpha \neq 0$. Let q be the least index in $\operatorname{supp}(\alpha)$, and define $s_h : K_{h,\alpha} \to K_{h+1,\alpha}$ by

$$s_h(b_J) = \begin{cases} b_{\{q\}\cup J}, & q \notin J, \\ 0, & q \in J, \end{cases}$$

that is, s_h inserts q in the first position of the wedge. We claim that

$$d_{h+1} s_h + s_{h-1} d_h = \mathrm{id}_{K_{h,\alpha}} \qquad (h \geq 0,\ s_{-1} := 0),$$

which we check on a basis vector b_J.

Case $q \notin J$. Since q is smaller than all elements of J, in $\{q\} \cup J$ it is the first element and j_r is the $(r+1)$st. So $d_{h+1}s_h(b_J) = b_J - \sum_{r=1}^{h} (-1)^{r-1} b_{\{q\}\cup(J\setminus\{j_r\})}$, where the first term comes from deleting q. On the other hand $q \notin J \setminus \{j_r\}$, so $s_{h-1}d_h(b_J) = \sum_{r=1}^{h} (-1)^{r-1} b_{\{q\}\cup(J\setminus\{j_r\})}$. The sum of the two is b_J. (For $h = 0$ we have $J = \varnothing$ and the second term is absent.)

Case $q \in J$. Then $q = j_1$, because $J \subseteq \operatorname{supp}(\alpha)$ and q is the least element of the support. Now $s_h(b_J) = 0$, and in $d_h(b_J)$ the only term on which s_{h-1} is non-zero is the one deleting q, that is $r = 1$, with sign $+1$; it gives $s_{h-1}(b_{J\setminus\{q\}}) = b_J$. Hence $d_{h+1}s_h(b_J) + s_{h-1}d_h(b_J) = b_J$ again.

Thus $K_{\bullet,\alpha}$ is contractible, in particular exact. (Here d_0 plays no role: its target $\Bbbk$ is concentrated in multidegree 0.)

Step 5: multidegree 0. For $\alpha = 0$ we have $\operatorname{supp}(\alpha) = \varnothing$, so only $K_{0,0} = \Bbbk \cdot 1$ is non-zero, and $d_0 : K_{0,0} \to \Bbbk$ is an isomorphism. Together with Step 4 this shows that $0 \to K_n \to \cdots \to K_0 \xrightarrow{d_0} \Bbbk \to 0$ is exact. □

A second proof: the Koszul complex as a tensor product

The proof above verifies exactness by brute force. Here is a conceptual explanation: the Koszul complex of $\mathsf{S}(V)$ is the tensor product of n one-variable complexes, and each one-variable complex is obviously exact. We first collect the three facts we need.

Lemma 6.18 (A polynomial ring is a tensor product). *Let $\Bbbk[x_j]$ be the polynomial ring in one variable, graded by $\deg x_j = 1$. Give $\Bbbk[x_1] \otimes_\Bbbk \cdots \otimes_\Bbbk \Bbbk[x_n]$ the ordinary tensor product algebra structure (Definition 3.21), that is, the componentwise product without signs. Then $p_1 \otimes \cdots \otimes p_n \mapsto p_1 \cdots p_n$ is an isomorphism of graded algebras onto $A = \Bbbk[x_1, \dots, x_n]$.*

Proof The map is well defined because the product $(p_1, \dots, p_n) \mapsto p_1 \cdots p_n$ is multilinear. A basis of a tensor product of vector spaces is given by the tensors of basis elements, so the pure tensors $x_1^{a_1} \otimes \cdots \otimes x_n^{a_n}$ form a basis of the left-hand side. They are sent to the monomials $x_1^{a_1} \cdots x_n^{a_n}$, which form a basis of A, so the map is bijective. It preserves degrees because degrees add on both sides, and it is multiplicative because A is commutative: both products multiply the monomials factor by factor. □

Under this isomorphism x_j corresponds to $1 \otimes \cdots \otimes x_j \otimes \cdots \otimes 1$. So if M_j is a graded $\Bbbk[x_j]$-module for each j, then $M_1 \otimes_\Bbbk \cdots \otimes_\Bbbk M_n$ is a graded A-module in which x_j acts on the jth factor only.

Lemma 6.19 (Tensor product of complexes). *Let C, C' be complexes of graded vector spaces, with differentials d, d' lowering the homological degree by 1 and preserving the Adams degree. Define $C \otimes_\Bbbk C'$ by $(C \otimes_\Bbbk C')_m = \bigoplus_{p+q=m} C_p \otimes_\Bbbk C'_q$ and $d(c \otimes c') = (-1)^q\, dc \otimes c' + c \otimes d'c'$ for $c \in C_p$ and $c' \in C'_q$. Then $d^2 = 0$.*

Proof Apply d twice. The terms containing $d^2 c$ or $d'^2 c'$ vanish, and the two remaining terms are $(-1)^q\, dc \otimes d'c'$ and $(-1)^{q-1}\, dc \otimes d'c'$, which cancel. □

The sign is there to make $d^2 = 0$. For a tensor product $C_1 \otimes_\Bbbk \cdots \otimes_\Bbbk C_n$ of n complexes, iterating the rule shows that the differential of a pure tensor $c_1 \otimes \cdots \otimes c_n$ is the sum over r of the terms obtained by applying d to the rth factor, each with the sign $(-1)^{m_r}$, where m_r is the sum of the homological degrees of the factors to the right of the rth one.

Lemma 6.20 (Künneth formula). *Let C, C' be complexes of vector spaces over a field. If $c \in C_p$ and $c' \in C'_q$ are cycles, then $c \otimes c'$ is a cycle of $C \otimes_\Bbbk C'$, and its homology class only depends on the classes of c and c'. The resulting map $\bigoplus_{p+q=m} H_p(C) \otimes_\Bbbk H_q(C') \to H_m(C \otimes_\Bbbk C')$, $[c] \otimes [c'] \mapsto [c \otimes c']$, is an isomorphism for every m.*

This is a standard result of homological algebra (see [Wei94, Section 3.6] or [ML95, Chapter V, Section 10], where the general version has an extra Tor term, which vanishes over a field); its proof totalises the double complex, and we omit it. Intuitively, over a field each complex is quasi-isomorphic to its homology, and tensor products preserve quasi-isomorphisms.

We now identify the Koszul complex. For $1 \le j \le n$ let $C^{(j)}$ be the complex of graded $\Bbbk[x_j]$-modules $0 \to \Bbbk[x_j]\, e_j \to \Bbbk[x_j] \to 0$, where $\Bbbk[x_j]$ is in homological degree 0, the free module on

e_j is in homological degree 1 with $\deg e_j = 1$, and the differential is $e_j p \mapsto x_j p$. Multiplication by x_j is injective with cokernel $\Bbbk$, so $H_1(C^{(j)}) = 0$ and $H_0(C^{(j)}) = \Bbbk$.

Lemma 6.21 (The Koszul complex is a tensor product). *Regard $C = C^{(1)} \otimes_\Bbbk \cdots \otimes_\Bbbk C^{(n)}$ as a complex of graded A-modules via Lemma 6.18. Then $C \cong (K_\bullet, d_\bullet)$ as complexes of graded A-modules (ignoring the augmentation d_0).*

Proof *The terms.* A pure tensor in C has, in each factor j, either an element of $\Bbbk[x_j]$ or an element of $\Bbbk[x_j]\, e_j$. Sort the pure tensors by the set $J = \{j_1 < \cdots < j_h\}$ of factors where e_j appears, which is exactly the homological degree $h = |J|$. For a fixed J, the corresponding summand is a tensor product of n copies of $\Bbbk[x_j]$ (one in each factor), so by Lemma 6.18 it is a free A-module of rank 1, generated by the pure tensor e_J with e_j in each factor $j \in J$ and 1 in the other factors. So $C_h = \bigoplus_{|J|=h} e_J A$, and $e_J \mapsto x_{j_1} \wedge \cdots \wedge x_{j_h}$ identifies this with $\bigwedge^h V \otimes_\Bbbk A(-h) = K_h$, since $\deg e_J = h$.

The differential. The differential on C is A-linear, since each $C^{(j)}$ has $\Bbbk[x_j]$-linear differential, so it is determined by its values on the e_J. Applying d to the factor j_r replaces e_{j_r} by x_{j_r}, so that the result is $x_{j_r} e_{J\setminus\{j_r\}}$. The factors to the right of it that have homological degree 1 are $e_{j_{r+1}}, \ldots, e_{j_h}$, which are $h-r$ in number, so the sign is $(-1)^{h-r}$. Hence $d(e_J) = \sum_{r=1}^{h} (-1)^{h-r}\, x_{j_r}\, e_{J\setminus\{j_r\}}$, which is the formula for d_h in the example above, now with x_{j_r} written on the right. □

For instance, for $n = 2$ we get $d(e_1) = x_1$, $d(e_2) = x_2$ and $d(e_1 e_2) = x_2 e_1 - x_1 e_2$, which matches $d_2((x_1 \wedge x_2) \otimes 1) = x_1 \otimes x_2 - x_2 \otimes x_1$ under $e_i \leftrightarrow x_i$.

Second proof of Proposition 6.17 By Lemma 6.21, $K_\bullet \cong C^{(1)} \otimes_\Bbbk \cdots \otimes_\Bbbk C^{(n)}$. Each $C^{(j)}$ has homology $\Bbbk$ in homological degree 0 and nothing else, so applying Lemma 6.20 repeatedly shows that $K_\bullet$ has homology $\Bbbk \otimes_\Bbbk \cdots \otimes_\Bbbk \Bbbk = \Bbbk$ in degree 0 and 0 in all other degrees. Thus $K_n \to \cdots \to K_1$ is exact at every K_h with $h \geq 1$, and $H_0 = K_0 / \operatorname{Im} d_1 = A/A_+ = \Bbbk$ (the image of d_1 is $\sum_j x_j A = A_+$), where d_0 is the projection to this quotient. This is the exactness of $0 \to K_n \to \cdots \to K_0 \xrightarrow{d_0} \Bbbk \to 0$. □

Corollary 6.22. *For every finite-dimensional vector space V, the symmetric algebra $\mathsf{S}(V)$ is Koszul, and $K_\bullet$ is its minimal graded projective resolution of $\Bbbk$.*

Here $\operatorname{grtop} K_h = \bigwedge^h V$ in degree h, so Proposition 5.6 gives $\operatorname{ext}^h_A(\Bbbk, \Bbbk(-h)) \cong D(\bigwedge^h V) \cong \bigwedge^h D(V)$, the degree h part of $\mathsf{S}(V)^! \cong \bigwedge D(V)$.

6.6 Exercises

Exercise 6.23. *Let $A = \mathsf{S}(V) = \Bbbk[x_1, \ldots, x_n]$ and $K_\bullet \to \Bbbk$ be the Koszul resolution of Proposition 6.17. In each degree, the alternating sum of the dimensions in an exact sequence of finite-dimensional vector spaces is zero. Use this to show*

$$(1-t)^n\, h_A(t) = 1,$$

and deduce the Hilbert series of $\Bbbk[x_1, \ldots, x_n]$.

Exercise 6.24. *Let $S = \bigoplus_{y=1}^{n} \Bbbk e_y$, let V be a finite-dimensional bimodule in degree 1, and let $A = T_S(V)/(V \otimes_S V) = S \oplus V$. Prove that the minimal graded projective resolution of S has $P_h \cong (V^{\otimes_S h} \otimes_S A)(-h)$ for all $h \geq 0$, and deduce that A is Koszul.*

Hint. *Show that* $\operatorname{Ker} d_0 = V$ *and that* $V A_+ = 0$*, so each kernel is killed by* A_+.

7 The Yoneda algebra

The groups $\operatorname{ext}^h_A(S, S(j))$ of Section 5 do not just sit side by side: they multiply, and the resulting algebra is intrinsic to A. We show that on the diagonal $j = -h$ this algebra is the quadratic dual $A^!$ of Section 3, for every quadratic algebra A, and that for Koszul algebras the diagonal is everything. Here A is as in Section 6, $P_\bullet \to S$ is its minimal graded projective resolution, and $S = \bigoplus_x \Bbbk e_x$ as in Assumption 1.50.

7.1 Comparing projective resolutions

To define the Yoneda product we must continue a map out of one term of a resolution to the whole resolution, and the continuation is unique only up to homotopy. The same comparison shows that ungraded Ext and graded ext do not depend on the chosen resolution, as used in Section 5.

Lemma 7.1 (Comparison lemma). *Let $P_\bullet \to M$ be a projective resolution with differentials d_h and with $d_0 : P_0 \to M$ the surjection onto M. Let $\cdots \to Q_1 \xrightarrow{d'_1} Q_0 \xrightarrow{d'_0} M' \to 0$ be an exact sequence of A-modules, and let $f : M \to M'$ be A-linear.*

(1) *There are A-linear maps $f_h : P_h \to Q_h$ for $h \geq 0$ making the following diagram commute:*

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & P_2 & \xrightarrow{d_2} & P_1 & \xrightarrow{d_1} & P_0 & \xrightarrow{d_0} & M & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_0} & & \downarrow{\scriptstyle f} & & \\
\cdots & \longrightarrow & Q_2 & \xrightarrow[d'_2]{} & Q_1 & \xrightarrow[d'_1]{} & Q_0 & \xrightarrow[d'_0]{} & M' & \longrightarrow & 0
\end{array}$$

(2) *If $(f'_h)_{h\geq 0}$ is another such family, there are A-linear maps $s_h : P_h \to Q_{h+1}$ for $h \geq 0$ (the diagonal arrows below) with $f_h - f'_h = d'_{h+1}s_h + s_{h-1}d_h$ for all $h \geq 0$, where $s_{-1} := 0$:*

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & P_2 & \xrightarrow{d_2} & P_1 & \xrightarrow{d_1} & P_0 \\
 & & \downarrow{\scriptstyle f_2-f'_2} & \swarrow{\scriptstyle s_1} & \downarrow{\scriptstyle f_1-f'_1} & \swarrow{\scriptstyle s_0} & \downarrow{\scriptstyle f_0-f'_0} \\
\cdots & \longrightarrow & Q_2 & \xrightarrow[d'_2]{} & Q_1 & \xrightarrow[d'_1]{} & Q_0
\end{array}$$

We say that $(f_h)_{h\geq 0}$ *lifts* f, and that two lifts are *homotopic* when maps s_h as in (2) exist. Intuitively, this is the algebraic version of a continuous deformation: (f_h) can be deformed into (f'_h), with the maps s_h recording the deformation. As for continuous maps, a deformation does not change the effect on (co)homology; this is checked in Proposition 7.2 below. For $M = M'$, $Q_\bullet = P_\bullet$ and $f = \mathrm{id}_M$ the lifts are the maps of resolutions in Theorem 4.29.

Proof of Lemma 7.1 (1) Since P_0 is projective and d'_0 is surjective, there is f_0 with $d'_0 f_0 = f d_0$. Suppose $f_0, \ldots, f_{h-1}$ are constructed, $h \geq 1$, and consider $f_{h-1}d_h : P_h \to Q_{h-1}$. For $h = 1$, we have $d'_0 f_0 d_1 = f d_0 d_1 = 0$, so $f_0 d_1$ has image in $\operatorname{Ker} d'_0 = \operatorname{Im} d'_1$. For $h \geq 2$, $d'_{h-1} f_{h-1} d_h = f_{h-2} d_{h-1} d_h = 0$, so $f_{h-1}d_h$ has image in $\operatorname{Ker} d'_{h-1} = \operatorname{Im} d'_h$. Since P_h is projective and $d'_h : Q_h \to \operatorname{Im} d'_h$ is surjective, there is $f_h : P_h \to Q_h$ with $d'_h f_h = f_{h-1} d_h$.

(2) Put $g_h = f_h - f'_h$, so $d'_0 g_0 = 0$ and $d'_h g_h = g_{h-1} d_h$ for $h \geq 1$. The image of g_0 lies in $\operatorname{Ker} d'_0 = \operatorname{Im} d'_1$, so projectivity of P_0 gives s_0 with $d'_1 s_0 = g_0$. Let $h \geq 1$ and suppose

$s_0, \dots, s_{h-1}$ are constructed. Then

$$d'_h(g_h - s_{h-1}d_h) = g_{h-1}d_h - (g_{h-1} - s_{h-2}d_{h-1})d_h = 0,$$

where $s_{-1} := 0$ and $d_{h-1}d_h = 0$. Thus $g_h - s_{h-1}d_h$ has image in $\operatorname{Ker} d'_h = \operatorname{Im} d'_{h+1}$, and projectivity of P_h gives s_h with $d'_{h+1}s_h = g_h - s_{h-1}d_h$. □

Proposition 7.2. *Let $P_\bullet \to M$ and $Q_\bullet \to M'$ be projective resolutions, $f : M \to M'$ be A-linear, (f_h) a lift of f, and N a module.*

(1) *The map $[\varphi] \mapsto [\varphi f_h]$ is a well-defined linear map $f^* : H^h(\operatorname{Hom}_A(Q_\bullet, N)) \to H^h(\operatorname{Hom}_A(P_\bullet, N))$ that does not depend on the choice of lift.*

(2) *If $g : M' \to M''$ is A-linear and $R_\bullet \to M''$ is a projective resolution, then $(gf)^* = f^*g^*$; also $(\mathrm{id}_M)^* = \mathrm{id}$.*

In particular, up to canonical isomorphism the ungraded $\operatorname{Ext}^h_A(M, N)$ does not depend on the projective resolution of M. It is contravariant in M through f^*. It is covariant in N: an A-linear map $\psi : N \to N'$ gives $\psi_*[\varphi] = [\psi\varphi]$.

Proof (1) Let $\varphi : Q_h \to N$ be a cocycle, so $\varphi d'_{h+1} = 0$. Then $\varphi f_h d_{h+1} = \varphi d'_{h+1} f_{h+1} = 0$, so φf_h is a cocycle. If $\varphi = \psi d'_h$ is a coboundary, then $\varphi f_h = \psi f_{h-1} d_h$ is a coboundary. For another lift (f'_h) with homotopy (s_h) as in Lemma 7.1(2),

$$\varphi(f_h - f'_h) = \varphi d'_{h+1}s_h + \varphi s_{h-1}d_h = (\varphi s_{h-1})d_h,$$

a coboundary; for $h = 0$ the same computation gives $\varphi d'_1 s_0 = 0$. (2) If (f_h) lifts f and (g_h) lifts g, then $(g_h f_h)$ lifts gf, and (id_{P_h}) lifts id_M; use (1). For two resolutions $P_\bullet, P'_\bullet$ of M, the lifts of id_M in each direction give maps id^* that are inverse to each other by (2). Finally $\psi\varphi$ is a cocycle (coboundary) if φ is, and ψ_* is compatible with f^*. □

7.2 The Yoneda product

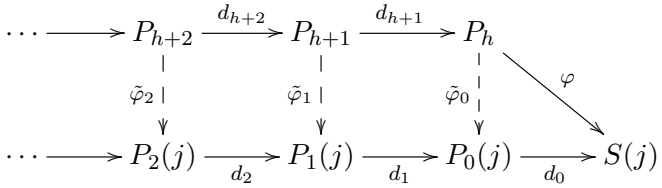
By Proposition 5.6, every cochain in $\operatorname{Hom}^{\mathbb{Z}}_A(P_\bullet, S(j))$ is a cocycle and no cochain is a coboundary, so

$$\operatorname{ext}^h_A(S, S(j)) = \operatorname{Hom}^{\mathbb{Z}}_A\big(P_h, S(j)\big) = \operatorname{Hom}_S\big((\operatorname{grtop} P_h)_{-j}, S\big).$$

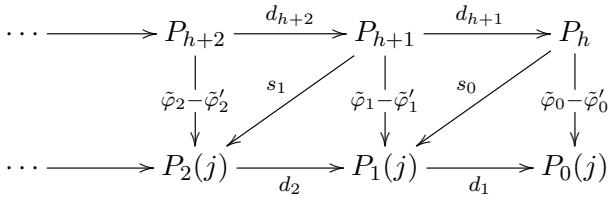
A class $\varphi \in \operatorname{ext}^h_A(S, S(j))$ is thus a degree-preserving A-linear map $\varphi : P_h \to S(j)$, which vanishes on $P_h A_+$.

The product is best understood by viewing φ as a kind of endomorphism of the resolution $P_\bullet$. Indeed, $S(j)$ is the end of the resolution $P_\bullet(j) \to S(j)$, so φ can be continued to a chain map $P_{h+\bullet} \to P_\bullet(j)$: it lowers the homological degree by h and raises the Adams degree by j. Composing two such maps is then the Yoneda product.

Lemma 7.3. *For $\varphi \in \operatorname{ext}^h_A(S, S(j))$ there are degree-preserving A-linear maps $\tilde\varphi_i : P_{h+i} \to P_i(j)$, $i \geq 0$, making the following diagram commute:*



Any two such families $\tilde\varphi, \tilde\varphi'$ are homotopic: there are maps $s_i : P_{h+i} \to P_{i+1}(j)$ (the diagonal arrows below) with $\tilde\varphi_i - \tilde\varphi'_i = d_{i+1}s_i + s_{i-1}d_{h+i}$, where $s_{-1} := 0$:



Proof Since φ kills $\operatorname{Im} d_{h+1} \subseteq P_h A_+$, it factors as $\varphi = \bar\varphi\, p_h$ with $p_h : P_h \twoheadrightarrow \operatorname{Im} d_h$ and $\bar\varphi : \operatorname{Im} d_h \to S(j)$. Both $P_{h+\bullet} \to \operatorname{Im} d_h$ (with p_h the map onto $\operatorname{Im} d_h$) and $P_\bullet(j) \to S(j)$ (with d_0 the map onto $S(j)$) are graded projective resolutions. Apply Lemma 7.1 (in its degree-preserving form, Subsection 5.2) to $\bar\varphi$. □

We call $\tilde\varphi = (\tilde\varphi_i)_i$ a *lift* of φ.

Definition 7.4. *For $\varphi \in \operatorname{ext}^h_A(S, S(j))$ and $\psi \in \operatorname{ext}^k_A(S, S(j'))$, the Yoneda product is*

$$\psi \cdot \varphi := \psi(j) \circ \tilde\varphi_k : \ P_{h+k} \longrightarrow S(j+j') \ \in \ \operatorname{ext}^{h+k}_A(S, S(j+j')).$$

The next statement says that on the level of lifts, the product is just composition.

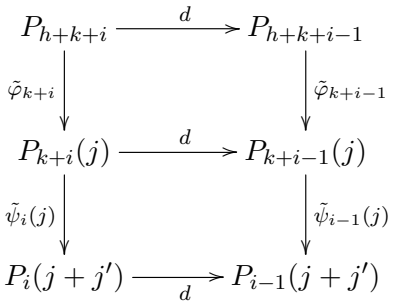
Proposition 7.5. *Let $\tilde\varphi$ and $\tilde\psi$ be lifts of $\varphi \in \operatorname{ext}^h_A(S, S(j))$ and $\psi \in \operatorname{ext}^k_A(S, S(j'))$.*

(1) *The maps $\tilde\psi_i(j) \circ \tilde\varphi_{k+i} : P_{h+k+i} \to P_i(j+j')$ form a lift of $\psi \cdot \varphi$.*

(2) *The Yoneda product is well defined, that is, does not depend on the choice of $\tilde\varphi$.*

(3) *The Yoneda product is bilinear and associative, with unit $\mathrm{id}_S \in \operatorname{ext}^0_A(S, S)$.*

Proof (1) The lift condition is the commutativity of



for $i \geq 1$, which holds because both squares commute, by Lemma 7.3. For $i = 0$ we need $d_0\tilde\psi_0(j)\tilde\varphi_k = \psi(j)\tilde\varphi_k$, which is $d_0\tilde\psi_0 = \psi$.

(2) Let $\tilde\varphi'$ be another lift, with $\tilde\varphi_k - \tilde\varphi'_k = d_{k+1}s_k + s_{k-1}d_{h+k}$ by Lemma 7.3. Then

$$\psi(\tilde\varphi_k - \tilde\varphi'_k) \overset{(*)}{=} \psi d_{k+1}s_k + \psi s_{k-1}d_{h+k} \overset{(**)}{=} 0,$$

where (*) is the homotopy formula, and (**) holds because $d_{k+1}s_k$ and $s_{k-1}d_{h+k}$ both map into P_kA_+ (by minimality, $\operatorname{Im} d_{k+1} \subseteq P_kA_+$ and $\operatorname{Im} d_{h+k} \subseteq P_{h+k-1}A_+$) while ψ kills P_kA_+.

(3) Bilinearity holds since sums of lifts are lifts. For $\chi \in \operatorname{ext}^l_A(S, S(j''))$, part (1) applied twice gives

$$\chi \cdot (\psi \cdot \varphi) \overset{(1)}{=} \chi\,\tilde\psi_l\,\tilde\varphi_{k+l} \overset{(2)}{=} (\chi \cdot \psi) \cdot \varphi,$$

where (1) uses that $(\tilde\psi_i\tilde\varphi_{k+i})_i$ is a lift of $\psi \cdot \varphi$, and (2) that $\chi\tilde\psi_l$ is $\chi \cdot \psi$. Finally $\mathrm{id}_S \in \operatorname{Hom}_S(S, S) = \operatorname{ext}^0_A(S, S)$ lifts to $\mathrm{id}_{P_\bullet}$, so $\mathrm{id}_S \cdot \varphi = \varphi$ and $\varphi \cdot \mathrm{id}_S = \varphi \circ \mathrm{id}_{P_h} = \varphi$. □

Definition 7.6. *The Yoneda algebra of A is the bigraded algebra*

$$E(A) := \bigoplus_{h\geq 0,\, j\in\mathbb{Z}} \operatorname{ext}_A^h(S, S(j)),$$

with $E(A)_j^h := \operatorname{ext}_A^h(S, S(j))$. The diagonal Yoneda algebra is

$$E^{\Delta}(A) := \bigoplus_{h\geq 0} E(A)_{-h}^{h} = \bigoplus_{h\geq 0} \operatorname{ext}_A^h(S, S(-h)).$$

By Remark 5.4, if the terms P_h are finitely generated then $E(A)$ is the ordinary (ungraded) Ext-algebra $\operatorname{Ext}_A^{\bullet}(S,S) = \bigoplus_h \operatorname{Ext}_A^h(S,S)$ of the simple modules, with the same construction in the ungraded setting; the extra index j remembers the grading of A. The degree 0 part is $E(A)_0^0 = \operatorname{Hom}_S(S,S) = S$. By Lemma 5.10, $E(A)_j^h = 0$ for $j > -h$. So the product of classes of types (h,j) and (k,j') lies in $j + j' \leq -(h+k)$, with equality only if both factors are diagonal. Hence $E^{\Delta}(A)$ is a subalgebra of $E(A)$, and the off-diagonal part $\bigoplus_{j<-h} E(A)_j^h$ is an ideal.

7.3 The Yoneda product as splicing of exact sequences

The Yoneda product has a second description that does not mention resolutions: it is the splicing of exact sequences. This is how it is usually introduced in the literature, and it explains the name; we only state the result and indicate why it agrees with Definition 7.4. For the general theory see [Wei94, Section 3.4] or [ML95, Chapter III, Sections 5–6].

From classes to exact sequences. Let $\varphi \in \operatorname{ext}_A^h(S, S(j))$ with $h \geq 1$, viewed as a map $\varphi : P_h \to S(j)$ vanishing on $\operatorname{Im} d_{h+1}$, so that $\varphi = \bar{\varphi}\, p_h$ with $\bar{\varphi} : \operatorname{Im} d_h \to S(j)$. Truncating the resolution gives an exact sequence

$$0 \to \operatorname{Im} d_h \hookrightarrow P_{h-1} \xrightarrow{d_{h-1}} \cdots \xrightarrow{d_1} P_0 \xrightarrow{d_0} S \to 0.$$

Replace $\operatorname{Im} d_h$ by $S(j)$ along $\bar{\varphi}$, that is, replace P_{h-1} by the *pushout*

$$E_h := \big(P_{h-1} \oplus S(j)\big) \big/ \{(z, -\bar{\varphi}(z)) : z \in \operatorname{Im} d_h\}.$$

The result is an exact sequence of graded modules with h middle terms

$$\xi_{\varphi} : \quad 0 \to S(j) \to E_h \to P_{h-2} \to \cdots \to P_0 \to S \to 0,$$

and a morphism of exact sequences from the truncated resolution to ξ_{φ} which is $\bar{\varphi}$ at the left end and the identity at the right end. For $h = 0$ a class is a map $S \to S(j)$ and we put $\xi_{\varphi} := \varphi$.

Yoneda's theorem. Call two exact sequences $0 \to S(j) \to E_h \to \cdots \to E_1 \to S \to 0$ equivalent if they are related by a chain of morphisms of sequences which are the identity at both ends. Then $\varphi \mapsto \xi_{\varphi}$ is a bijection from $\operatorname{ext}_A^h(S, S(j))$ to the set of equivalence classes of such sequences, and $\operatorname{ext}_A^1(S, S(j))$ is the set of extensions $0 \to S(j) \to E \to S \to 0$ of S by $S(j)$ in the usual sense; see [Wei94, Theorem 3.4.3]. Here all modules and maps are graded and degree-preserving.

The product is splicing. Let $\varphi \in \operatorname{ext}_A^h(S, S(j))$ and $\psi \in \operatorname{ext}_A^k(S, S(j'))$. Shifting ξ_{ψ} by (j) gives a sequence $0 \to S(j+j') \to F_k(j) \to \cdots \to F_1(j) \to S(j) \to 0$ ending in $S(j)$, and ξ_{φ} starts

with $S(j)$. Their *splice* is obtained by gluing along $S(j)$:

$$0 \longrightarrow S(j+j') \longrightarrow F_k(j) \longrightarrow \cdots \longrightarrow F_1(j) \searrow \quad \nearrow E_h \longrightarrow \cdots \longrightarrow E_1 \longrightarrow S \longrightarrow 0$$
$$S(j)$$

It is an exact sequence with $h+k$ middle terms from S to $S(j+j')$, hence represents a class in $\operatorname{ext}_A^{h+k}(S, S(j+j'))$.

Proposition 7.7. *$\xi_{\psi\cdot\varphi}$ is equivalent to the splice of $\xi_\psi(j)$ and ξ_φ.*

Sketch of proof Let $\tilde\varphi$ be a lift of φ. The maps $\tilde\varphi_i : P_{h+i} \to P_i(j)$ together with the identity maps on $P_{h-1}, \dots, P_0$ give a morphism from the truncated resolution $0 \to \operatorname{Im} d_{h+k} \to P_{h+k-1} \to \cdots \to P_0 \to S \to 0$ to the splice of $\xi_\psi(j)$ and ξ_φ, which is the identity at S and, at the left end, the map $\operatorname{Im} d_{h+k} \to S(j+j')$ induced by $\psi(j)\tilde\varphi_k$. A sequence admitting such a morphism from the truncated resolution is, by the construction of ξ, the pushout of the truncated resolution along this map, that is, $\xi_{\psi\cdot\varphi}$. □

Thus $E(A)$ is the algebra of exact sequences from S to shifts of S under splicing. In this language associativity is clear, since splicing is evidently associative; and the unit is the identity $S \to S$. It also shows that the product is intrinsic: it uses neither the resolution $P_\bullet$ nor the lifts $\tilde\varphi$ in its definition.

7.4 The diagonal part of a quadratic algebra

From now on $A = T_S(V)/(R)$ is quadratic. By Theorem 6.14, $P_{h,h} = T_h \cong \mathcal{K}_h$, so

$$E^\Delta(A)^h = \operatorname{Hom}_S(\mathcal{K}_h, S),$$

a space of right S-linear maps $\mathcal{K}_h \to S$. Recall that $\mathcal{K}_{h+k} \subseteq \mathcal{K}_h \otimes_S \mathcal{K}_k$ (Lemma 6.12). For $\varphi \in \operatorname{Hom}_S(\mathcal{K}_h, S)$ and $\psi \in \operatorname{Hom}_S(\mathcal{K}_k, S)$ put

$$\psi \star \varphi : \mathcal{K}_h \otimes_S \mathcal{K}_k \to S, \qquad x \otimes y \longmapsto \psi\big(\varphi(x)\, y\big),$$

which is the map $\theta_r(\psi \otimes \varphi)$ of Section 3 (advanced).

Proposition 7.8. *For $\varphi \in E^\Delta(A)^h$ and $\psi \in E^\Delta(A)^k$, the Yoneda product $\psi \cdot \varphi \in E^\Delta(A)^{h+k}$ is the restriction of $\psi \star \varphi$ to $\mathcal{K}_{h+k} \subseteq \mathcal{K}_h \otimes_S \mathcal{K}_k$.*

Proof The class $\psi \cdot \varphi$ is determined by its values on $T_{h+k} = \mathcal{K}_{h+k}$, and only the restriction of $\tilde\varphi_k$ to T_{h+k} matters: it maps T_{h+k} into $(P_k(-h))_{h+k} = T_k = \mathcal{K}_k$, where ψ is evaluated. We claim that $\tilde\varphi_i(z) = (\varphi \otimes \mathrm{id})(z)$ for $z \in \mathcal{K}_{h+i} \subseteq \mathcal{K}_h \otimes_S \mathcal{K}_i$, where $\varphi \otimes \mathrm{id} : \mathcal{K}_h \otimes_S \mathcal{K}_i \to S \otimes_S \mathcal{K}_i = \mathcal{K}_i$. For $i = 0$ this is $d_0\tilde\varphi_0 = \varphi$, as d_0 is the identity on S. For $i \geq 1$ write $z = \sum z' \otimes v$ with $z' \in \mathcal{K}_{h+i-1}$ and $v \in V$. Then $d_{h+i}(z) = \sum z'v$, so $\tilde\varphi_{i-1}d_{h+i}(z) = \sum \tilde\varphi_{i-1}(z')v = \sum(\varphi \otimes \mathrm{id})(z') \otimes v$ by the induction hypothesis. This is the image of $(\varphi \otimes \mathrm{id})(z) \in \mathcal{K}_i \subseteq \mathcal{K}_{i-1} \otimes_S V$ under d_i. Since d_i is injective on T_i (Step 2 of Theorem 6.14), the claim follows. Hence $(\psi \cdot \varphi)(z) = \psi((\varphi \otimes \mathrm{id})(z)) = (\psi \star \varphi)(z)$. □

We now compare with the quadratic dual. Under Assumption 1.50 there is the isomorphism $\alpha_X : \operatorname{Hom}_S(X, S) \to D(X)$, $f \mapsto \tau_S \circ f$, of Section 3, where $\tau_S(\sum_x c_x e_x) = \sum_x c_x$.

Theorem 7.9. *Let $A = T_S(V)/(R)$ be a quadratic algebra. Then there is an isomorphism of graded algebras*

$$A^! \xrightarrow{\sim} E^\Delta(A), \qquad A^!_h \xrightarrow{\sim} \operatorname{ext}^h_A(S, S(-h)) \cong \operatorname{Hom}_S(\mathcal{K}_h, S).$$

Proof Let $\Phi_h : D(V)^{\otimes h} \to E^\Delta(A)^h$ be $\alpha_{\mathcal{K}_h}^{-1} \circ \rho_h$, with ρ_h as in Proposition 6.16. By that proposition, Φ_h is surjective with kernel the degree h part of $(\theta^{-1}(R^\perp))$, so Φ induces a linear isomorphism $A^!_h \cong E^\Delta(A)^h$. It remains to show $\Phi_{h+k}(w' \otimes w) = \Phi_k(w') \cdot \Phi_h(w)$ for $w \in D(V)^{\otimes h}$ and $w' \in D(V)^{\otimes k}$.

For right S-modules X, Y and $\varphi \in \mathrm{Hom}_S(X, S)$, $\psi \in \mathrm{Hom}_S(Y, S)$ we have

$$\theta_{X,Y}\big(\alpha_Y \psi \otimes \alpha_X \varphi\big) = \alpha_{X \otimes_S Y}(\psi \star \varphi).$$

Indeed, writing $\varphi(x) = \sum_i c_i e_i$ with $c_i = \tau_S(\varphi(x)e_i) = \alpha_X \varphi(x e_i)$, both sides send $x \otimes y$ to $\sum_i c_i \, \tau_S(\psi(e_i y))$. By associativity and binaturality of θ (Proposition 3.3), $\rho_{h+k}(w' \otimes w)$ is the restriction to $\mathcal{K}_{h+k} \subseteq \mathcal{K}_h \otimes_S \mathcal{K}_k$ of $\theta_{\mathcal{K}_h, \mathcal{K}_k}(\rho_k(w') \otimes \rho_h(w))$. Applying α^{-1} and the identity above gives $\Phi_{h+k}(w' \otimes w) = \Phi_k(w') \star \Phi_h(w)|_{\mathcal{K}_{h+k}}$, which is $\Phi_k(w') \cdot \Phi_h(w)$ by Proposition 7.8. □

In particular $E^\Delta(A)$ is generated by $E(A)^1_{-1} = \mathrm{ext}^1_A(S, S(-1)) \cong D(V)$, with quadratic relations $\theta^{-1}(R^\perp)$.

> *Remark* 7.10 (General S). For arbitrary semisimple S, the proof of Proposition 7.8 works verbatim, and $E^\Delta(A)^h = \mathrm{Hom}_S(\mathcal{K}_h, S) = \mathcal{K}_h^*$ with product θ_r. Hence $E^\Delta(A) \cong T_S(V^*)/(R^\perp)$, the right quadratic dual of Definition 3.16, without choosing a symmetrising form.

7.5 Consequences for Koszul algebras

By Proposition 6.6, A is Koszul if and only if $E(A) = E^\Delta(A)$. Combining this with Theorem 7.9 gives the following.

Corollary 7.11. *Let A be a quadratic algebra, and let $A^! \cong E^\Delta(A) \hookrightarrow E(A)$ be the natural algebra map. The following are equivalent:*

(1) *A is Koszul;*

(2) *$E(A) = E^\Delta(A)$, that is, $\mathrm{ext}^h_A(S, S(j)) = 0$ for $j \neq -h$;*

(3) *$A^! \to E(A)$ is an isomorphism.*

In this case $\mathrm{ext}^h_A(S, S(-h)) \cong A^!_h$ for all h, the Yoneda algebra is generated by $\mathrm{ext}^1_A(S, S(-1))$, and $E(A)$ is the ordinary (ungraded) Ext-algebra of the simple A-modules.

Proof (1)⇔(2) is Proposition 6.6(2), and (2)⇔(3) is Theorem 7.9, since $E^\Delta(A) \subseteq E(A)$ is an equality if and only if the inclusion is bijective. If A is Koszul, each $P_h = K_h = \mathcal{K}_h \otimes_S A(-h)$ is finitely generated, so Remark 5.4 gives the last statement. □

Thus the Koszul property can be read as "the quadratic dual computes all of the graded Ext-groups $\mathrm{ext}^h_A(S, S(j))$"; for a quadratic algebra that is not Koszul, $A^!$ is a proper subalgebra of $E(A)$ (for an example of such an algebra, see Exercise 6.11).

Example 7.12. *(1) For $A = \mathsf{S}(V)$, Corollary 6.22 and Proposition 3.20 give $E(A) \cong \bigwedge D(V)$, concentrated on the diagonal.*

(2) For $A = \bigwedge W$, Proposition 3.20 gives $A^! \cong \mathsf{S}(D(W))$. If $\dim W = 1$, $A = \Bbbk[\epsilon]/(\epsilon^2)$ is Koszul (Example 4.32), so $E(A) \cong \Bbbk[t]$ with $t \in \mathrm{ext}^1_A(\Bbbk, \Bbbk(-1))$, in agreement with Example 5.9.

(3) For $A = \Bbbk A_2$ we have $R = 0$ and $V \otimes_S V = 0$, so $A^! = S \oplus D(V)$, of dimension $2 + 1$, matching Example 5.8.

7.6 The quadratic dual of a Koszul algebra is Koszul

Let $A = T_S(V)/(R)$ be quadratic and $B = A^!$, with $B_1 = D(V)$ and $B^! \cong A$ (Theorem 3.14); all A_i, B_j are finite-dimensional. By Corollary 6.15, A is Koszul if and only if $K_\bullet \to S$ is exact, where $K_h = D(B_h) \otimes_S A(-h)$ by Proposition 6.16. Fixing the internal degree n, this is the complex of vector spaces $K_{\bullet,n}$:

$$0 \to D(B_n) \otimes_S A_0 \to D(B_{n-1}) \otimes_S A_1 \to \cdots \to D(B_0) \otimes_S A_n \to 0,$$

which is exact for $n \geq 1$ (and equals S for $n = 0$). Transposing it produces a complex with terms $D(A_i) \otimes_S B_{n-i}$ that turns out to be a resolution of S over B.

Lemma 7.13. *Under $\mathcal{K}_h \cong D(B_h)$, the inclusion $\iota_h : \mathcal{K}_h \hookrightarrow \mathcal{K}_{h-1} \otimes_S V$ is the transpose of the multiplication $B_1 \otimes_S B_{h-1} \to B_h$, followed by $D(D(V)) \cong V$:*

$$\iota_h : D(B_h) \xrightarrow{D(m_B)} D(B_1 \otimes_S B_{h-1}) \cong D(B_{h-1}) \otimes_S D(B_1) \cong D(B_{h-1}) \otimes_S V.$$

Proof Let $\pi_h : D(V)^{\otimes h} \to B_h$ be the quotient map; the isomorphism $D(B_h) \cong \mathcal{K}_h$ of Proposition 6.16 is $D(\pi_h)$ followed by the evaluation identification $D(D(V)^{\otimes h}) \cong V^{\otimes h}$ (Lemma 3.6), which reverses the order of the factors. Since $\pi_h = m_B \circ (\mathrm{id} \otimes \pi_{h-1})$, we have $D(\pi_h) = D(\mathrm{id} \otimes \pi_{h-1}) \circ D(m_B)$. By Lemma 3.6 again, the last tensor factor of $V^{\otimes h}$ corresponds to the left factor B_1, so $D(\mathrm{id} \otimes \pi_{h-1}) \circ D(m_B)$ is the inclusion $\mathcal{K}_h \subseteq \mathcal{K}_{h-1} \otimes_S V \subseteq V^{\otimes(h-1)} \otimes_S V$. □

Theorem 7.14. *A quadratic algebra A is Koszul if and only if $A^!$ is Koszul.*

Proof Since $(A^!)^! \cong A$, it suffices to show that if A is Koszul then so is $B = A^!$. For $i \geq 0$ put

$$\Lambda_i := D(A_i) \otimes_S B(-i),$$

a graded projective right B-module generated in degree i, with $\Lambda_0 = B$. Let $\iota^A : D(A_{i+1}) \to D(A_i) \otimes_S D(A_1) = D(A_i) \otimes_S B_1$ be the transpose of the multiplication $A_1 \otimes_S A_i \to A_{i+1}$, and define the right B-linear map

$$\delta_{i+1} : \Lambda_{i+1} \to \Lambda_i, \qquad \lambda \otimes b \longmapsto \sum \mu \otimes \beta b \quad \text{where } \iota^A(\lambda) = \sum \mu \otimes \beta.$$

Let $\varepsilon : \Lambda_0 = B \to S$ be the projection. We claim that $\Lambda_\bullet \xrightarrow{\varepsilon} S$ is a linear projective resolution of S.

In internal degree n, the complex $\Lambda_{\bullet,n}$ is

$$0 \to D(A_n) \otimes_S B_0 \to D(A_{n-1}) \otimes_S B_1 \to \cdots \to D(A_0) \otimes_S B_n \to 0.$$

Dualising the complex above, $D\big(D(B_h) \otimes_S A_i\big) \cong D(A_i) \otimes_S D(D(B_h)) \cong D(A_i) \otimes_S B_h$ term by term. By Lemma 7.13, the differential of $K_\bullet$ is $\varphi \otimes a \mapsto (\mathrm{id} \otimes \mathrm{mult}_A)(\kappa(\varphi) \otimes a)$ with $\kappa = D(m_B)$, and its transpose is $(\mathrm{id} \otimes m_B) \circ (\iota^A \otimes \mathrm{id})$, which is δ. Hence $\Lambda_\bullet$ in internal degree n is the dual of $K_\bullet$ in internal degree n. Since D is exact on finite-dimensional spaces and $K_\bullet \to S$ is exact (Corollary 6.15), $\Lambda_\bullet \to S$ is exact in every internal degree $n \geq 1$, and for $n = 0$ it is $S \xrightarrow{\sim} S$. In particular $\delta\delta = 0$ and $\varepsilon\delta_1 = 0$. Thus $\Lambda_\bullet \to S$ is a resolution with Λ_i generated in degree i, so B is Koszul. □

Corollary 7.15. *If A is Koszul, then $E(E(A)) \cong A$.*

Proof By Corollary 7.11, $E(A) \cong A^!$, which is Koszul by Theorem 7.14. Applying Corollary 7.11 to $A^!$ gives $E(A^!) \cong (A^!)^! \cong A$. □

This is the algebraic core of Koszul duality: for a Koszul algebra A, the algebra $A^! = E(A)$ is again Koszul, and dualising twice returns A.

7.7 Exercises

Exercise 7.16. *Let V be a finite-dimensional bimodule over S. Verify Theorem 7.9 for $A = T_S(V)$ (no relations) and for $A = S \oplus V = T_S(V)/(V \otimes_S V)$, by computing $E(A)$ from the minimal resolutions: Exercise 6.24 for $S \oplus V$, and $0 \to V \otimes_S A(-1) \to A \to S \to 0$ for $T_S(V)$. Which of the two algebras has $A^! = S \oplus D(V)$?*

Exercise 7.17. *Let $A = \Bbbk[x]/(x^3)$ be as in Exercise 6.10. Show that $\operatorname{ext}_A^h(\Bbbk, \Bbbk(j)) \neq 0$ for some $j \neq -h$ when $h \geq 2$, and deduce that $E(A) \neq E^\Delta(A)$.*

Exercise 7.18. *Using the Koszulness of $\mathsf{S}(V)$, deduce that the exterior algebra $\bigwedge W$ of a finite-dimensional vector space W is Koszul, and that $E(\bigwedge W) \cong \mathsf{S}(D(W))$.*

List of Symbols

Page numbers link to the first defining passage. The final group lists notation used in advanced material.

Symbol	Meaning	Page
Algebras and categories		
$\Bbbk$	Base field.	1
$\mathrm{Mat}_n(\Bbbk)$	Algebra of square matrices over the base field.	2
$\Bbbk A_n$	Upper triangular matrix algebra, with paths composed from left to right.	2
e_{ij}	Matrix unit with its only non-zero entry in position (i, j).	2
$\mathrm{Hom}_{\mathcal{C}}(X, Y),\ \mathrm{id}_X$	Morphisms from X to Y; identity morphism of X.	2
$\mathrm{Mod}\, A,\ \mathrm{mod}\, A$	Right modules; finitely generated right modules.	2
$\mathrm{Hom}_A(M, N)$	Space of A-linear maps, without a degree restriction.	2
$\mathrm{add}\, M$	Direct summands of finite direct sums of M, in the right-module category unless specified.	3
$\mathcal{C}^{\mathrm{op}}$	Opposite category; morphisms have their directions reversed.	3
$\mathrm{rad}\, A,\ \mathrm{rad}\, M$	Jacobson radical of A; intersection of the maximal submodules of M.	4
$eA,\ eAe$	Right-module summand of A; corner algebra with identity e.	5
A^{op}	Opposite algebra; left A-modules are right A^{op}-modules.	7
$S = A_0$	Degree zero algebra; from the standing assumption, $S = \bigoplus_{i=1}^{n} \Bbbk e_i$.	15
$e_1, \ldots, e_n$	Coordinate idempotents of S under the standing assumption.	15
Gradings and graded modules		
$A_g,\ M_g$	Degree g components of a group-graded algebra and module.	6, 7
lfd	Locally finite-dimensional: every degree component is finite-dimensional.	6, 7
$M(d)$	Grading shift: $M(d)_j = M_{j+d}$.	7
$\mathrm{Hom}_A^{\mathbb{Z}}(M, N)$	Degree-preserving A-linear maps.	7
$\mathrm{Hom}_A^{\mathbb{Z}}(M, N(d))$	Homogeneous A-linear maps from M to N of degree d.	7
$\mathrm{GrMod}\, A,\ \mathrm{grmod}\, A$	Graded right modules; locally finite-dimensional graded right modules. Both use degree-preserving maps.	7
A_+	Positive degree ideal $\bigoplus_{i>0} A_i$.	9
Sums, bimodules, and tensor products		
$M^{\oplus r}$	Direct sum of r copies of M.	3
$U \otimes_{\Bbbk} V$	Tensor product of vector spaces.	9
$\bigoplus_{i \in I} M_i$	Direct sum: only finitely many coordinates of an element are non-zero.	10
$\prod_{i \in I} M_i$	Direct product: coordinates may be non-zero at infinitely many indices.	10
$\Bbbk[[x]]$	Formal power series ring; as a vector space, $\prod_{n \geq 0} \Bbbk x^n$.	10
${}_B M_C$	Bimodule with left B-action and right C-action.	11
$B\,\mathrm{Mod}\, C$	Category of bimodules with left B-action and right C-action; identified with $\mathrm{Mod}(C \otimes_{\Bbbk} B^{\mathrm{op}})$.	11
A^{en}	Enveloping algebra $A \otimes_{\Bbbk} A^{\mathrm{op}}$, acting on the right by $m(a \otimes b^{\mathrm{op}}) = bma$.	12

List of Symbols (continued)

Symbol	Meaning	Page
$M \otimes_A N$	Tensor product of a right module and a left module, with $ma \otimes n = m \otimes an$.	23
$V^{\otimes_A i}$	Tensor power of a bimodule over A; $V^{\otimes_A 0} = A$.	26
$(M \otimes_S N)_d$	Total grading: $\bigoplus_{p+q=d} M_p \otimes_S N_q$.	35
Linear duals		
$D(V),\ D(u)$	Ungraded linear dual $\mathrm{Hom}_{\Bbbk}(V, \Bbbk)$; dual map given by precomposition with u.	4
ev_V	Evaluation $V \to D(D(V))$, $v \mapsto (\lambda \mapsto \lambda(v))$.	4
v_i^*	Dual-basis functional, characterised by $v_i^*(v_j) = \delta_{ij}$.	4
$\mathbb{D}(V)$	Graded linear dual, with $\mathbb{D}(V)_i = D(V_{-i})$.	13
ev_M	Graded evaluation $M \to \mathbb{D}(\mathbb{D}(M))$; an isomorphism for locally finite-dimensional M.	14
$e_j D(M) e_i$	Dual idempotent component, identified with $D(e_i M e_j)$.	15
Presentations and Hilbert series		
$\mu_i : A_1^{\otimes_S i} \to A_i$	Multiplication of i elements of degree 1.	30
$T_S(V)$	Tensor algebra; degree zero part S and degree i part $V^{\otimes_S i}$ for $i \geq 1$.	30
$\mathsf{S}(V),\ \bigwedge W$	Symmetric algebra of V; exterior algebra of W.	31
$\mu : T_S(A_1) \to A$	Canonical multiplication map for an algebra generated in degree 1.	32
$I = \mathrm{Ker}\, \mu$	Homogeneous ideal of relations.	32
$A = T_S(V)/(R)$	Quadratic presentation: V consists of degree 1 generators, and $R \subset V \otimes_S V$ of degree 2 relations.	32
(R)	Two-sided ideal generated by the relation space R.	32
$L_j = e_j S$	Simple right S-module, viewed as $e_j A / e_j A_+$ in degree zero.	33
$[M_d : L_j]$	Multiplicity of the composition factor L_j in degree d.	33
$\mathbf{dim}\, M_d$	Row vector $([M_d : L_j])_{1 \leq j \leq n}$.	33
$h_M(t)$	Module Hilbert series $\sum_{d \in \mathbb{Z}} (\mathbf{dim}\, M_d) t^d$.	33
$h_A(t)$	Algebra Hilbert series; entry (i, j) is $\sum_{d \geq 0} [e_i A_d : L_j] t^d$.	33
Quadratic duality and algebra tensor products		
θ	Factor-reversing comparison $D(W) \otimes_S D(V) \to D(V \otimes_S W)$ for finite-dimensional bimodules over the standing S.	38
$U^\perp$	Annihilator of $U \subset M$ in $D(M)$.	41
$R^\perp$	Relation annihilator, viewed in $D(V) \otimes_S D(V)$ using θ.	43
$A^!$	Quadratic dual $T_S(D(V))/(R^\perp)$, with $D(V)$ placed in degree 1.	43
$A \otimes_{\Bbbk} B$	Ordinary algebra tensor product, with multiplication $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$.	36
Tops, resolutions, Ext, Tor, and the Koszul complex		
$\mathrm{top}\, M$	Ordinary top $M/\operatorname{rad} M$, for finite-dimensional A.	54
$p : P \to M$	Projective cover: no proper submodule of P maps onto M.	59
$C_\bullet,\ d_h$	Chain complex with differentials $d_h : C_h \to C_{h-1}$ and $d_h d_{h+1} = 0$.	52
$H_h(C_\bullet)$	Homology $\mathrm{Ker}\, d_h / \mathrm{Im}\, d_{h+1}$.	52
$P_\bullet \to M$	Projective resolution; $P_\bullet$ ends in $P_0 \to 0$, with the map $d_0 : P_0 \to M$.	52
$\mathrm{grtop}\, M$	Graded top M/MA_+.	53

List of Symbols (continued)

Symbol	Meaning	Page
$C_{h,i}$	Degree i part of the hth term of a complex of graded modules; h is the homological degree, i the internal (Adams) degree.	53
$C^\bullet,\ \delta^h,\ H^h(C^\bullet)$	Cochain complex with $\delta^h\delta^{h-1} = 0$; cohomology $\operatorname{Ker}\delta^h / \operatorname{Im}\delta^{h-1}$.	52
$\operatorname{Ext}^h_A(M,N)$	Ungraded Ext: cohomology of $\operatorname{Hom}_A(P_\bullet, N)$ for a projective resolution $P_\bullet \to M$.	61
$\operatorname{ext}^h_A(M,N)$	Graded Ext: cohomology of $\operatorname{Hom}^{\mathbb{Z}}_A(P_\bullet, N)$ for a graded projective resolution; $\operatorname{ext}^h_A(M,N(j))$ consists of maps raising the Adams degree by j.	62
$\operatorname{Tor}^A_h(M,N)$	Ungraded Tor: homology of $P_\bullet \otimes_A N$ for a projective resolution $P_\bullet \to M$.	63
$\operatorname{tor}^A_h(M,N)$	Graded Tor: homology of $(P_\bullet \otimes_A N)_0$ for a graded projective resolution; $\operatorname{tor}^A_h(M,N(i))$ is the Adams degree i part.	64
$\mathcal{K}_h,\ K_h$	Koszul terms of a quadratic algebra: $\mathcal{K}_h = \bigcap_\nu V^{\otimes\nu} \otimes R \otimes V^{\otimes(h-2-\nu)}$ and $K_h = \mathcal{K}_h \otimes_S A(-h)$; for $A = \mathsf{S}(V)$, $K_h = \bigwedge^h V \otimes_{\Bbbk} A(-h)$.	69
$K_\bullet$	Koszul complex, with degree-preserving differentials $d_h : K_h \to K_{h-1}$.	70
$E(A),\ E^\Delta(A)$	Yoneda algebra $\bigoplus_{h,j} \operatorname{ext}^h_A(S, S(j))$ with the Yoneda product; its diagonal part $\bigoplus_h \operatorname{ext}^h_A(S, S(-h))$.	79
Advanced notation		
$\operatorname{add}_{\operatorname{mod} A^{\mathrm{op}}}(M)$	Additive closure in the category of left A-modules.	26
$M^*,\ {}^*N$	Right and left S-duals: $\operatorname{Hom}_S(M,S)$ and $\operatorname{Hom}_{S^{\mathrm{op}}}(N,S)$.	16
$M^{(*)},\ {}^{(*)}N$	Graded right and left S-duals; degree i is the corresponding dual of degree $-i$.	16
$M^{*_{\mathrm{en}}}$	Two-sided dual $\operatorname{Hom}_{S^{\mathrm{en}}}(M, S^{\mathrm{en}})$.	17
$M^{(*_{\mathrm{en}})}$	Graded two-sided dual $\bigoplus_{i\in\mathbb{Z}} (M_{-i})^{*_{\mathrm{en}}}$.	17
$\tau : S \to \Bbbk$	Symmetrising form; $(s,t) \mapsto \tau(st)$ is symmetric and non-degenerate.	18
τ_S	Standard form on the standing S: $\sum_i a_i e_i \mapsto \sum_i a_i$.	20
$\alpha_M,\ \beta_N$	Comparison maps from one-sided S-duals to linear duals, given by composition with τ.	19
ω_M	Comparison $M^{*_{\mathrm{en}}} \to M^*$ obtained by applying τ to the second factor of the values.	20
$\theta_r,\ \theta_\ell$	Tensor–Hom comparison maps; specialise to factor reversal for right and left S-duals.	26
θ_D	Mixed comparison $D(W) \otimes_S V^* \to D(V \otimes_S W)$.	28
θ_{en}	Two-sided comparison $W^* \otimes_S V^{*_{\mathrm{en}}} \to (V \otimes_S W)^{*_{\mathrm{en}}}$.	28
qA	Quadratic part $T_S(V)/(I_2)$ of $A = T_S(V)/I$.	36
$U^\perp,\ {}^\perp U$	Annihilators in the right and left S-duals, respectively.	42
$A^!,\ {}^!A$	Right and left quadratic duals formed using the one-sided S-duals.	44
$A \otimes^{\mathrm{gr}}_{\Bbbk} B$	Graded algebra tensor product; multiplication includes the sign $(-1)^{\deg b \deg a'}$.	46
$\beta_{h,i,y}(M)$	Multiplicity of L_y in $(\operatorname{grtop} P_h)_i$ for a minimal resolution of M.	57
$\beta_{h,i}(M)$	Row vector $(\beta_{h,i,y}(M))_{1\le y\le n}$.	57

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